

Perturbation Theory

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In this chapter perturbation techniques are described which serve to generate approximate solutions to the Schrödinger equation. Such solutions appear in the form of an expansion away from known, unperturbed values. A special procedure is developed for systems with degenerate eigenenergies. Application is made to problems in atomic physics and the problem of an electron in a periodic potential, encountered previously in Chapter 8. Harmonic perturbation theory is applied in a rederivation of the Planck radiation formula and the theory of the laser. The chapter continues with a description of the interaction of a radiation field with an atom. Here we encounter the notion of oscillator strengths and the important Thomas-Reiche-Kuhn sum rule. The concluding section addresses the Hartree-Fock model for obtaining approximate atomic wavefunctions and eigenenergies.

13.1 ■ TIME-INDEPENDENT, NONDEGENERATE PERTURBATION THEORY

Approximate methods of solution to an assortment of problems in quantum mechanics were described previously in Chapter 7 (WKB analysis), Chapter 8 (LCAO approximation), Chapter 10 (Thomas-Fermi model), and Chapter 11 (interaction picture).

Our present concern lies in refinement of the approximation method described in Chapter 12, used in calculating both the ground-state wavefunction of the helium atom (Problem 12.28) and the fine-structure spectrum of hydrogen (12.13).

TABLE 13.1 Examples of perturbation Hamiltonians

Name	Description	Hamiltonian
L-S coupling	Coupling between orbital and spin angular momentum in a one-electron atom	$\hat{H} = \hat{H}_0 + f(r)\hat{\mathbf{L}} \cdot \hat{\mathbf{S}}$ $\hat{H}' = f(r)\hat{\mathbf{L}} \cdot \hat{\mathbf{S}}$ $\hat{H}_0 = \hat{p}^2/2m - e^2 Z/r$
Stark effect	One-electron atom in a constant, uniform electric field $\mathcal{E} = \mathbf{e}_z \mathcal{E}_0$	$\hat{H} = \hat{H}_0 + e\mathcal{E}_0 z$ $H' = e\mathcal{E}_0 z$ $\hat{H}_0 = (\hat{p}^2/2m) - e^2 Z/r$
Zeeman effect	One electron atom in a constant, uniform magnetic field $\mathcal{B}$	$\hat{H} = \hat{H}_0 + (e/2mc)\hat{\mathbf{J}} \cdot \mathcal{B}$ $\hat{H}' = (e/2mc)\hat{\mathbf{J}} \cdot \mathcal{B}$ $\hat{H}_0 = (\hat{p}^2/2m) - e^2 Z/r$
Anharmonic oscillator	Spring with nonlinear restoring force	$\hat{H} = \hat{H}_0 + K'x^4$ $\hat{H}' = K'x^4$ $\hat{H}_0 = (\hat{p}_x^2/2m) + \frac{1}{2}Kx^2$
Nearly free electron model	Electron in a periodic lattice	$\hat{H} = \hat{H}_0 + V(x)$ $V(x) = \sum_n V_n \exp[i(2\pi nx/a)]$ $\hat{H}_0 = \hat{p}_x^2/2m$

In both these problems the Hamiltonian encountered was of the form

$$\hat{H} = \hat{H}_0 + \hat{H}'$$

This breakup of a Hamiltonian into a part $\hat{H}_0$, whose eigenfunctions are known, and an addition term $\hat{H}'$, which is in some sense small compared to $\hat{H}_0$, is typical of many practical problems encountered in quantum mechanics. The theory that seeks approximate eigenstates of the *total Hamiltonian* $\hat{H}$ is called *perturbation theory*. In the expression above, the Hamiltonian, $\hat{H}_0$, is called the *unperturbed Hamiltonian*, while $\hat{H}'$ is called the *perturbation Hamiltonian*. Some typical perturbation problems are listed in Table 13.1.

The perturbation analysis we will develop in this chapter divides into three categories: (1) time-independent, nondegenerate; (2) time-independent, degenerate; (3) time-dependent. In the last category one investigates the time development of a system in a given state due to a perturbation on the system which is turned on at a given instant of time.

Smallness of the Perturbation

Perturbation theory begins with the assumption that the perturbation Hamiltonian, $\hat{H}'$, is in some sense small compared to the unperturbed Hamiltonian, $\hat{H}_0$. The criterion that establishes the smallness of $\hat{H}'$ compared to $\hat{H}_0$ will emerge in the course of the analysis. Another underlying assumption in perturbation theory

is that the eigenstates and eigenenergies of the total Hamiltonian $\hat{H}$, do not differ appreciably from those of the unperturbed Hamiltonian, $\hat{H}_0$. That is, suppose that $\{\varphi_n\}$ and $\{E_n\}$ are, respectively, the eigenstates and eigenenergies of the total Hamiltonian $\hat{H}$,

$$\hat{H}\varphi_n = (\hat{H}_0 + \hat{H}')\varphi_n = E_n\varphi_n$$

while $\{\varphi_n^{(0)}\}$ and $\{E_n^{(0)}\}$ are, respectively, the eigenstates and eigenenergies of the unperturbed Hamiltonian

$$\hat{H}_0\varphi_n^{(0)} = E_n^{(0)}\varphi_n^{(0)}$$

Then it is always possible to write

$$\begin{aligned}\varphi_n &= \varphi_n^{(0)} + \Delta\varphi_n \\ E_n &= E_n^{(0)} + \Delta E_n\end{aligned}$$

where, owing to the smallness of $\hat{H}'$, $\Delta\varphi_n$ is a small correction to $\varphi_n^{(0)}$ and ΔE_n is a small correction to $E_n^{(0)}$.

To keep the smallness of $\hat{H}'$ in mind, we rewrite it as $\lambda\hat{H}'$, where λ is an infinitesimal parameter and is introduced for “bookkeeping” purposes only. The equation to which we seek a solution is of the form

$$(\hat{H}_0 + \lambda\hat{H}')\varphi_n = E_n\varphi_n \quad (13.1)$$

The Perturbation Expansion

The eigenstates and eigenenergies of $\hat{H}_0$ are assumed known. Since $\varphi_n \rightarrow \varphi_n^{(0)}$ as $\lambda \rightarrow 0$, it is consistent to seek solution to (13.1) in the form of a series with $\varphi_n^{(0)}$ entering as the leading term. In a similar manner, E_n is expanded, with $E_n^{(0)}$ entering as the leading term.

$$\begin{aligned}\varphi_n &= \varphi_n^{(0)} + \lambda\varphi_n^{(1)} + \lambda^2\varphi_n^{(2)} + \dots \\ E_n &= E_n^{(0)} + \lambda E_n^{(1)} + \lambda^2 E_n^{(2)} + \dots\end{aligned} \quad (13.2)$$

Substituting these expansions into (13.1) and arranging terms according to powers in λ gives

$$\begin{aligned}[\hat{H}_0\varphi_n^{(0)} - E_n^{(0)}\varphi_n^{(0)}] + \lambda[\hat{H}_0\varphi_n^{(1)} + \hat{H}'\varphi_n^{(0)} - E_n^{(0)}\varphi_n^{(1)} - E_n^{(1)}\varphi_n^{(0)}] \\ + \lambda^2[\hat{H}_0\varphi_n^{(2)} + \hat{H}'\varphi_n^{(1)} - E_n^{(0)}\varphi_n^{(2)} - E_n^{(1)}\varphi_n^{(1)} - E_n^{(2)}\varphi_n^{(0)}] \\ + \dots = 0\end{aligned} \quad (13.3)$$

This equation is of the form

$$F^{(0)} + \lambda F^{(1)} + \lambda^2 F^{(2)} + \lambda^3 F^{(3)} + \dots = 0$$

If this equation is to be true for *arbitrarily* small values of λ , then

$$F^{(0)} = F^{(1)} = F^{(2)} = \dots = 0$$

In this manner (13.3) gives the coupled set of equations

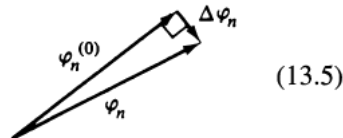
$$\begin{aligned} \text{(a)} \quad & \hat{H}_0 \varphi_n^{(0)} = E_n^{(0)} \varphi_n^{(0)} \\ \text{(b)} \quad & (\hat{H}_0 - E_n^{(0)}) \varphi_n^{(1)} = (E_n^{(1)} - \hat{H}') \varphi_n^{(0)} \\ \text{(c)} \quad & (\hat{H}_0 - E_n^{(0)}) \varphi_n^{(2)} = (E_n^{(1)} - \hat{H}') \varphi_n^{(1)} + E_n^{(2)} \varphi_n^{(0)} \\ \text{(d)} \quad & (\hat{H}_0 - E_n^{(0)}) \varphi_n^{(3)} = (E_n^{(1)} - \hat{H}') \varphi_n^{(2)} + E_n^{(2)} \varphi_n^{(1)} + E_n^{(3)} \varphi_n^{(0)} \\ & \vdots \end{aligned} \quad (13.4)$$

In the lowest approximation, (13.4a) returns the information that $\{\varphi_n^{(0)}\}$ and $\{E_n^{(0)}\}$ are, respectively, the eigenstates and eigenenergies of $\hat{H}_0$. The second (as well as all of the higher-order equations) has the following interesting property. The left-hand side of this equation remains the same under the replacement

$$\varphi_n^{(1)} \rightarrow \varphi_n^{(1)} + a \varphi_n^{(0)}$$

where a is an arbitrary constant. Suppose that one solves (13.4b) for $\varphi_n^{(1)}$ and $E_n^{(1)}$. Then $\varphi_n^{(1)} + a \varphi_n^{(0)}$; $E_n^{(1)}$ is also a solution. An extra constraint is needed to remove this arbitrary quality of solution. This constraint may be taken as follows.¹ We assume that all corrections to $\varphi_n^{(0)}$ in (13.2) are normal to $\varphi_n^{(0)}$.

$$\langle \varphi_n^{(s)} | \varphi_n^{(0)} \rangle = 0 \quad (\text{for } s > 0 \text{ and all } n)$$



(13.5)

In Hilbert space this relation indicates that $\Delta \varphi_n$ is normal to $\varphi_n^{(0)}$. This condition will aid us in the construction of $\varphi_n^{(s)}$.

Returning to (13.4b) we note that $\hat{H}_0$ operates on $\varphi_n^{(1)}$ in this equation, which suggests that the solution may be obtainable through expansion of $\varphi_n^{(1)}$ in a superposition of the eigenstates of $\hat{H}_0$.

$$|\varphi_n^{(1)}\rangle = \sum_i c_{ni} |\varphi_i^{(0)}\rangle \quad (13.6)$$

If this expansion is substituted into (13.4b), there results

$$(\hat{H}_0 - E_n^{(0)}) \sum_i c_{ni} |\varphi_i^{(0)}\rangle = (E_n^{(1)} - \hat{H}') |\varphi_n^{(0)}\rangle$$

¹ Another popular constraint is to construct $\varphi_n^{(s)}$ so that it is normalized. Both choices of constraint yield the same corrections to the energy, $\{E_n^{(s)}\}$, while the wavefunctions that emerge differ by at most a phase factor (see Section 4.1).

Multiplying from the left with $\langle \varphi_j^{(0)} |$ gives

$$(E_j^{(0)} - E_n^{(0)})c_{nj} + H'_{jn} = E_n^{(1)}\delta_{jn} \quad (13.7)$$

where H'_{jn} are the matrix elements of $\hat{H}'$ in the $\{\varphi_n^{(0)}\}$ representation

$$H'_{jn} \equiv \langle \varphi_j^{(0)} | \hat{H}' | \varphi_n^{(0)} \rangle$$

First-Order Corrections

With $j \neq n$, (13.7) gives the coefficients, $\{c_{nj}\}$, which when substituted into (13.6) gives the first-order correction to φ_n .

$$\begin{aligned} c_{ni} &= \frac{H'_{in}}{E_n^{(0)} - E_i^{(0)}} \\ \varphi_n^{(1)} &= \sum_{i \neq n} \frac{H'_{in}}{E_n^{(0)} - E_i^{(0)}} \varphi_i^{(0)} + c_{nn} \varphi_n^{(0)} \end{aligned} \quad (13.8)$$

Here one assumes that all corrections $\{\varphi_n^{(s)}\}$ lie in a Hilbert space that is spanned by the unperturbed wavefunctions, $\{\varphi_n^{(0)}\}$.

The coefficient c_{nn} is obtained from (13.5), which yields

$$c_{nn} = 0$$

With $j = n$, (13.7) gives the first-order corrections to the energy E_n .

$$E_n^{(1)} = H'_{nn} \quad (13.9)$$

These are the diagonal elements of $\hat{H}'$. Substituting (13.8) and (13.9) into (13.2) and setting $\lambda = 1$ gives

$$\begin{aligned} \varphi_n &= \varphi_n^{(0)} + \sum_{i \neq n} \frac{H'_{in}}{E_n^{(0)} - E_i^{(0)}} \varphi_i^{(0)} \\ E_n &= E_n^{(0)} + H'_{nn} \end{aligned} \quad (13.10)$$

The first of these equations tells us that in order for the expansion (13.2) to make sense, the coefficients of expansion should be less than 1.

$$|H'_{in}| \ll |E_n^{(0)} - E_i^{(0)}|$$

The matrix elements of $\hat{H}'$ should be small compared to the difference between the corresponding unperturbed energy levels. In similar manner, the second equa-

tion in (13.10) reveals that

$$|H'_{nn}| \ll E_n^{(0)}$$

The diagonal elements of the perturbation Hamiltonian should be small compared to the corresponding unperturbed energy level.

Second-Order Corrections

To find the second-order correction to φ_n and E_n , we must solve (13.4c). Again we note that $\hat{H}_0$ operates on $\varphi_n^{(2)}$, and it is again advantageous to expand $\varphi_n^{(2)}$ in the eigenstates of $\hat{H}_0$.

$$\varphi_n^{(2)} = \sum_i d_{ni} \varphi_i^{(0)} \quad (13.11)$$

Substitution into (13.4c) gives

$$\begin{aligned} \sum_i E_i^{(0)} d_{ni} |\varphi_i^{(0)}\rangle + \hat{H}' |\varphi_n^{(1)}\rangle \\ = E_n^{(0)} \sum_i d_{ni} |\varphi_i^{(0)}\rangle + E_n^{(1)} |\varphi_n^{(1)}\rangle + E_n^{(2)} |\varphi_n^{(0)}\rangle \end{aligned}$$

Multiplying from the left with $\langle \varphi_j^{(0)} |$ gives

$$(E_j^{(0)} - E_n^{(0)}) d_{nj} + \langle \varphi_j^{(0)} | \hat{H}' | \varphi_n^{(1)} \rangle = E_n^{(2)} \delta_{jn} + E_n^{(1)} \langle \varphi_j^{(0)} | \varphi_n^{(1)} \rangle \quad (13.12)$$

With $j = n$, this equation gives

$$\begin{aligned} E_n^{(2)} &= \langle \varphi_n^{(0)} | \hat{H}' | \varphi_n^{(1)} \rangle \\ &= \sum_{i \neq n} \langle \varphi_n^{(0)} | \frac{\hat{H}' H'_{in}}{E_n^{(0)} - E_i^{(0)}} | \varphi_i^{(0)} \rangle \\ &= \sum_{i \neq n} \frac{H'_{ni} H'_{in}}{E_n^{(0)} - E_i^{(0)}} \end{aligned}$$

Owing to the Hermiticity of $\hat{H}'$, this equation may be rewritten

$$E_n^{(2)} = \sum_{i \neq n} \frac{|H'_{ni}|^2}{E_n^{(0)} - E_i^{(0)}} \quad (13.13)$$

Note that in obtaining this result we have used the result that $c_{nn} = 0$. Substituting this expression for $E_n^{(2)}$ into (13.2) together with the expression for $E_n^{(1)}$ given by (13.9) gives the following second-order expression for E_n :

$$E_n = E_n^{(0)} + H'_{nn} + \sum_{i \neq n} \frac{|H'_{ni}|^2}{E_n^{(0)} - E_i^{(0)}} \quad (13.14)$$

To calculate the second-order corrections to the wavefunction φ_n , we must obtain the coefficients d_{ni} in (13.11). These are directly obtained from (13.12). With $n \neq j$ this equation gives

$$(E_n^{(0)} - E_j^{(0)})d_{nj} = \langle \varphi_j^{(0)} | \hat{H}' \sum_{k \neq n} \frac{H'_{kn}}{E_n^{(0)} - E_k^{(0)}} | \varphi_k^{(0)} \rangle - H'_{nn} \langle \varphi_j^{(0)} | \sum_{k \neq n} \frac{H'_{kn}}{E_n^{(0)} - E_k^{(0)}} | \varphi_k^{(0)} \rangle$$

In the second sum, only the $k = j$ term survives the $\langle \varphi_j^{(0)} | \varphi_k^{(0)} \rangle$ inner product. All terms in the first sum remain. There results

$$d_{nj} = \frac{1}{E_n^{(0)} - E_j^{(0)}} \left(\sum_{k \neq n} \frac{H'_{jk} H'_{kn}}{E_n^{(0)} - E_k^{(0)}} \right) - \frac{H'_{nn} H'_{jn}}{(E_n^{(0)} - E_j^{(0)})^2}$$

Again, with (13.5), one finds that

$$d_{nn} = 0$$

In this manner one obtains the following expression for φ_n , good to terms of second order in $\hat{H}'$.

$$\varphi_n = \varphi_n^{(0)} + \sum_{i \neq n} \left[\frac{H'_{in}}{E_n^{(0)} - E_i^{(0)}} - \frac{H'_{nn} H'_{in}}{(E_n^{(0)} - E_i^{(0)})^2} + \sum_{k \neq n} \frac{H'_{ik} H'_{kn}}{(E_n^{(0)} - E_i^{(0)})(E_n^{(0)} - E_k^{(0)})} \right] \varphi_i^{(0)} \quad (13.15)$$

PROBLEMS

13.1 Calculate the first-order correction to $E_3^{(0)}$ for a particle in a one-dimensional box with walls at $x = 0$ and $x = a$ due to the following perturbations:

- (a) $H' = 10^{-3} E_1 x/a$
- (b) $H' = 10^{-3} E_1 (x/a)^2$
- (c) $H' = 10^{-3} E_1 \sin(x/a)$

- 13.2** What is the eigenfunction φ_n for the same configuration as in Problem 13.1, to terms of second order, for the constant perturbation

$$H' = 10^{-3} E_1?$$

- 13.3** Calculate the eigenenergies of the anharmonic oscillator whose Hamiltonian is listed in Table 13.1, to first order in $\hat{H}'$.

Answer

In terms of raising and lowering operators, $(a^\dagger, a)$, the perturbation Hamiltonian appears as

$$\hat{H}' = Kx^4 = \frac{K}{4\beta^4} (\hat{a} + \hat{a}^\dagger)^4$$

The corrections to $E_n^{(0)}$ which we seek are given by

$$E_n^{(1)} = H'_{nn} = \langle n | \hat{H}' | n \rangle$$

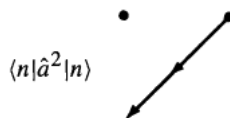
The only terms in the expansion of $(\hat{a} + \hat{a}^\dagger)^4$ which give nonzero contributions are those which maintain the eigenvector $|n\rangle$. All other terms vanish because of orthogonality with $\langle n|$. Of the 16 terms in the expansion of $(\hat{a} + \hat{a}^\dagger)^4$, only six survive this orthogonality condition. The energy $E_n^{(1)}$ may be determined by a graphical analysis, according to which a diagram is associated with each integral that contributes to $\langle n | \hat{H}' | n \rangle$. The eigenvector $|n\rangle$ is represented by a dot drawn at the right of the diagram. Another dot drawn at the left of the $|n\rangle$ dot, but on the same horizontal, represents the eigenbra $\langle n|$. The creation operator, $\hat{a}^\dagger$, is represented by a diagonal arrow from the right and inclined upward at $\pi/4$, while the annihilation operator, $\hat{a}$, is an arrow from the right at $-\pi/4$. Thus the diagram related to $\langle n | \hat{a} \hat{a}^\dagger | n \rangle$ is



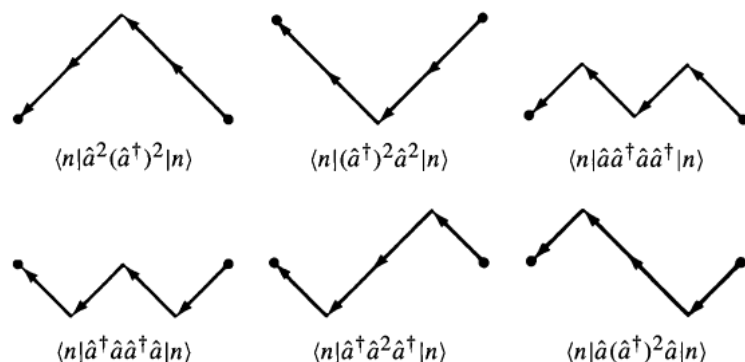
The diagram that represents the fourth-order term, $\langle n | \hat{a}^\dagger \hat{a} \hat{a}^\dagger \hat{a} | n \rangle$, is



while the second-order term, $\langle n | \hat{a}^2 | n \rangle$, is represented by



Any sequence of arrows that do not join the two horizontal dots represents a zero contribution. Continuing in this manner, we find that, in all, there are 16 fourth-order diagrams. Of these, only six gave nonzero contributions. These six diagrams are



Summing these contributions gives the desired result:

$$E_n^{(1)} = \frac{3K'}{4\beta^4} [2n(n+1) + 1]$$

13.4 Consider a particle of mass m in a potential well shown in Fig. 13.1a. Suppose that the bump at the bottom of the well can be considered a small perturbation.

- Calculate the corrected second eigenenergy and eigenfunction to first order in the perturbation. [Recall (6.100).]
- What dimensionless ratio must be small compared to 1 in order for your approximation to be valid?
- Is the parity of the unperturbed wavefunction maintained by this perturbation? Explain your answer.

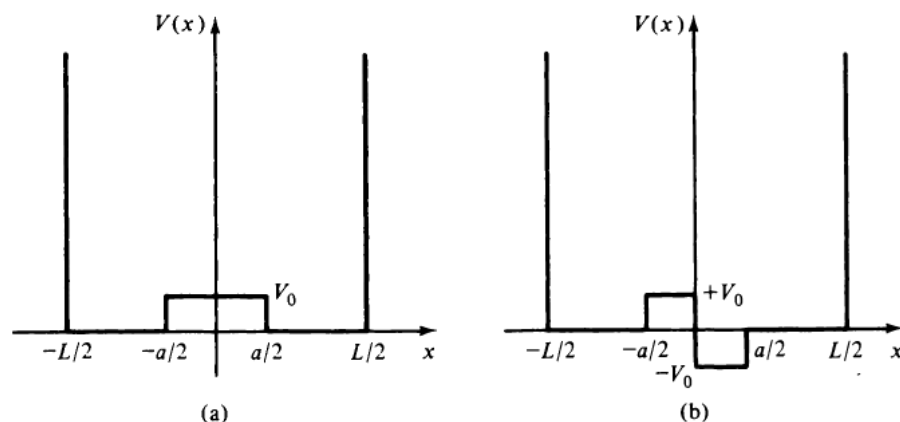


FIGURE 13.1 Potential configurations for Problems 13.4, 13.5, and 13.6.

- 13.5** (a) Consider the perturbation bump shown in Fig. 13.1b. What are the first-order corrected eigenenergies of a particle of mass m confined to this well?
- (b) Calculate the eigenenergies of this configuration, in the domain $E \gg V_0$, using the WKB formula (7.191) and compare your results with those obtained in part (a).
- 13.6** Again consider the potential shown in Fig. 13.1a. What is the unperturbed ground state for:
- (a) Two identical bosons of mass m confined to the box.
- (b) Two identical fermions of mass m confined to the box.
- (c) What are the unperturbed ground-state energies E_S and E_A for these two cases?
- (d) Use first-order perturbation theory to obtain the new ground-state energies for these two cases.

Answers (partial)

$$(a) \varphi_S(x_1, x_2) = \frac{2}{L} \cos\left(\frac{\pi x_1}{L}\right) \cos\left(\frac{\pi x_2}{L}\right)$$

$$(b) \varphi_A(x_1, x_2) = \frac{\sqrt{2}}{L} \left[\cos\left(\frac{\pi x_1}{L}\right) \sin\left(\frac{2\pi x_2}{L}\right) - \cos\left(\frac{\pi x_2}{L}\right) \sin\left(\frac{2\pi x_1}{L}\right) \right]$$

- 13.7** A particle of mass m is in an asymmetrical one-dimensional box, depicted in Fig. 13.2.
- (a) Use first-order perturbation theory to calculate the eigenenergies of the particle.
- (b) What are the first-order corrected wavefunctions?
- (c) If the particle is an electron, how do the frequencies emitted by the perturbed systems compare with those of the unperturbed system?
- (d) What smallness assumption is appropriate to V_0 ?

Answers

$$(a) E_n = E_n^{(0)} + E_n^{(1)} = n^2 E_1 + \frac{V_0}{2}$$

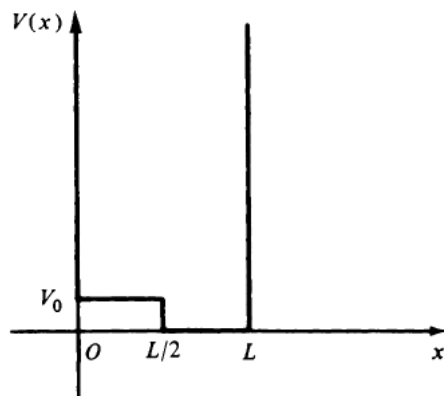


FIGURE 13.2 Potential configuration for Problem 13.7.

$$(b) \quad \varphi_n = \sqrt{\frac{2}{a}} \left[\sin\left(\frac{n\pi x}{a}\right) + \frac{V_0}{E_1} \sum_{l \neq n} \frac{l^2 n^2}{n^2 - l^2} \Lambda_{ln} \right]$$

$$\Lambda_{ln} \equiv \frac{1}{\pi} \left[\frac{\sin(n-l)\pi/2}{n-l} - \frac{\sin(n+l)\pi/2}{n+l} \right]$$

(c) They are the same.

(d) $V_0 \ll E_1$

13.8 Show that the matrix elements of the perturbation Hamiltonian $\hat{H}'$ obey the equality

$$\sum_m |H'_{nm}|^2 = (H'^2)_{nn}$$

Answer

$$\begin{aligned} \sum_m |H'_{nm}|^2 &= \sum_m \langle n | \hat{H}' | m \rangle \langle m | \hat{H}' | n \rangle \\ &= \langle n | \hat{H}' \hat{I} \hat{H}' | n \rangle = \langle n | \hat{H}'^2 | n \rangle = (H'^2)_{nn} \end{aligned}$$

Here we have recalled the relation

$$\hat{I} = \sum_m |m\rangle \langle m|$$

13.9 A hydrogen atom in its ground state is in a constant, uniform electric field that points in the z direction. The electric field polarizes the atom. Show that there is no change in the ground-state energy of the atom to terms of first order in the electric field. The interaction energy is

$$H' = e\mathcal{E}z = e\mathcal{E}r \cos \theta$$

13.10 The conditions are those of Problem 13.9. In calculating second-order corrections to the ground state, one must know the values of the matrix elements $\langle 100 | H' | nlm \rangle$, where $|nlm\rangle$ denotes a hydrogen eigenstate. Show that these matrix elements vanish if $l \neq 1$.

13.11 Consider again, as in Problem 13.9, that a hydrogen atom is in a constant, uniform electric field $\mathcal{E}$ that points in the z direction. If α is the *polarizability* of hydrogen, then the change in the ground-state energy is $\frac{1}{2}\alpha\mathcal{E}^2$, so we may write

$$\frac{1}{2}\alpha\mathcal{E}^2 = \sum_{n>1} \frac{|H'_{1n}|^2}{E_n^{(0)} - E_1^{(0)}}, \quad H'_{1n} \equiv \langle 100 | H' | n10 \rangle$$

(a) Use the result of Problem 13.8 to show that

$$\frac{1}{2}\alpha\mathcal{E}^2 < \frac{(H'^2)_{11}}{E_2^{(0)} - E_1^{(0)}}$$

(b) What maximum value for the polarizability of hydrogen does this inequality imply? How does this value compare with the correct value of α ?

Answers

$$(a) \quad [E_n^{(0)} - E_1^{(0)}]_{\min} = E_2^{(0)} - E_1^{(0)}$$

so

$$\frac{1}{2}\alpha\mathcal{E}^2 < \sum_{n>1} \frac{|H'_{1n}|^2}{E_2^{(0)} - E_1^{(0)}} = \frac{1}{E_2^{(0)} - E_1^{(0)}} \sum_{n>1} |H'_{1n}|^2$$

Using the results of Problem 13.8, we obtain

$$\sum_{n=1}^{\infty} |H'_{1n}|^2 = |H'_{11}|^2 + \sum_{n>1} |H'_{1n}|^2 = (H'^2)_{11}$$

so that

$$\sum_{n>1} |H'_{1n}|^2 = (H'^2)_{11} - |H'_{11}|^2 < (H'^2)_{11}$$

hence

$$\frac{1}{2}\alpha\mathcal{E}^2 < \frac{1}{E_2^{(0)} - E_1^{(0)}} \sum_{n>1} |H'_{1n}|^2 < \frac{(H'^2)_{11}}{E_2^{(0)} - E_1^{(0)}}$$

$$(b) \quad \begin{aligned} \frac{1}{2}\alpha\mathcal{E}_{\max}^2 &= \frac{4}{3\mathbb{R}} (H'^2)_{11} \\ &= \frac{4}{3\mathbb{R}} e^2 \mathcal{E}^2 \iiint |\varphi_{100}^*|^2 r^4 \cos^2 \theta \, d\cos \theta \, d\phi \, dr \\ &= \frac{8}{3} a_0^3 \mathcal{E}^2 \end{aligned}$$

which gives

$$\alpha_{\max} = \frac{16}{3} a_0^3$$

The more correct value is $\alpha = 9a_0^3/2$.

13.2 ■ TIME-INDEPENDENT, DEGENERATE PERTURBATION THEORY

Again we consider a system whose Hamiltonian has the form

$$\hat{H} = \hat{H}_0 + \hat{H}'$$

where $\hat{H}'$ is a small perturbation about the unperturbed Hamiltonian, $\hat{H}_0$. In the present case, however, $\hat{H}_0$ has degenerate eigenstates. We have found previously (see Section 8.5) that degeneracy in quantum mechanics stems from symmetries inherent to the system at hand. Any distortion of such symmetry should therefore tend to remove the related degeneracy.

Suppose, for example, that the ground state of $\hat{H}_0$ is q -fold-degenerate. If the symmetry producing this degeneracy is destroyed by the perturbation $\hat{H}'$, the

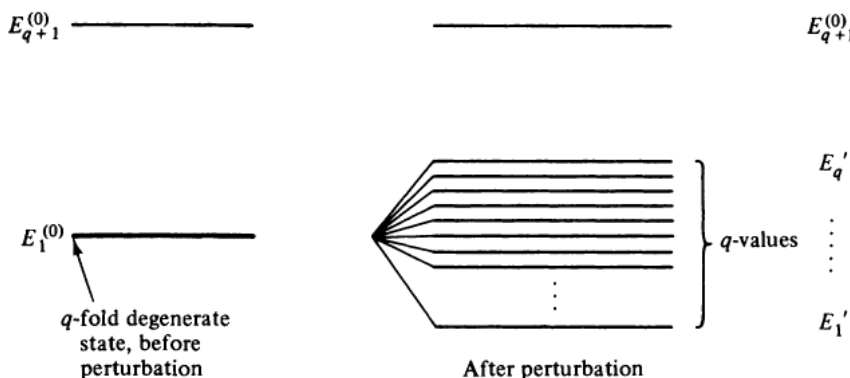


FIGURE 13.3 Perturbation causes a removal of degeneracy.

ground state $E_n^{(0)}$ separates into q distinct levels (Fig. 13.3). The primary aim of degenerate perturbation theory is to calculate these new energies. Suppose that we proceed as in the nondegenerate case described in the previous section and expand the first-order wavefunctions of $\hat{H}$ in the eigenstates of $\hat{H}_0$ (13.6).

$$\varphi_n^{(1)} = \sum_i c_{ni} \varphi_i^{(0)}$$

The formula that emerges for the coefficients $\{c_{ni}\}$ is given by (13.8):

$$c_{ni} = \frac{H'_{in}}{E_n^{(0)} - E_i^{(0)}}$$

If $E_1^{(0)}$ is q -fold-degenerate, then

$$E_1^{(0)} = E_2^{(0)} = \dots = E_q^{(0)}$$

and c_{ni} is infinite for $n, i \leq q$. This situation is remedied by constructing a new set of basis functions from the set $\{\varphi_n^{(0)}\}$ which diagonalize the submatrix, H'_{in} (for $n, i \leq q$). With the off-diagonal elements of H'_{in} vanishing, the corresponding singular c_{in} coefficients also vanish and we may proceed as in the nondegenerate case.

Diagonalization of the Submatrix

Thus the primary aim in degenerate perturbation theory is to diagonalize this submatrix of H'_{in} . As it turns out, the diagonal elements so constructed are the incremental energies, which when added to $E_1^{(0)}$ separate the q energies contained in the ground state.

Let the q functions that diagonalize H'_{in} ($i, n \leq q$) be labeled $\bar{\varphi}_n$:

$$\bar{\varphi}_n = \sum_{i=1}^q a_{ni} \varphi_i^{(0)} \quad (13.16)$$

These linear combinations of the degenerate eigenstates $\{\varphi_i^{(0)}\}$ diagonalize H'_{in} , so

$$\langle \bar{\varphi}_n | \hat{H}' | \bar{\varphi}_p \rangle = H'_{np} \delta_{np} \quad (n, p \leq q) \quad (13.17)$$

These functions, when joined with the complementary set of non-degenerate states, $\{\varphi_i^{(0)}, i > q\}$, give the basis²

$$B = \{\bar{\varphi}_1, \bar{\varphi}_2, \dots, \bar{\varphi}_q, \varphi_{q+1}^{(0)}, \varphi_{q+2}^{(0)}, \dots\} \quad (13.18)$$

The matrix of $\hat{H}'$ calculated in this basis appears as

$$\hat{H}' = \begin{pmatrix} H'_{11} & & 0 & H'_{1,q+1} & \cdots \\ & H'_{22} & & & \\ 0 & & \ddots & & \\ & & & H'_{qq} & \\ \hline H'_{q+1,1} & & & & \\ \vdots & & & & \end{pmatrix} \quad (13.19)$$

First-Order Energies

We will now show that the diagonal elements of the $q \times q$ submatrix of $\hat{H}'$ are the first-order energy corrections E'_n to $E_n^{(0)}$, $n \leq q$. That is,

$$E'_n = \langle \bar{\varphi}_n | \hat{H}' | \bar{\varphi}_n \rangle = H'_{nn} \quad (n \leq q) \quad (13.20)$$

If these diagonal elements are mutually distinct, then the q -fold degeneracy of $\hat{H}_0$ is removed by the perturbation $\hat{H}'$. To establish the equality (13.20), we proceed as follows.

The Schrödinger equation for the total Hamiltonian appears as

$$\hat{H}\varphi_n = (\hat{H}_0 + \hat{H}')\varphi_n = E_n\varphi_n$$

The ground state of $\hat{H}_0$ is a q -fold degenerate. Substituting

$$\left. \begin{aligned} \varphi_n &= \bar{\varphi}_n \\ E_n &= E_n^{(0)} + E'_n \end{aligned} \right\} (n \leq q) \quad (13.21)$$

into the Schrödinger equation gives

$$\hat{H}'\bar{\varphi}_n = E'_n\bar{\varphi}_n \quad (n \leq q) \quad (13.22)$$

Here we have recalled that $\bar{\varphi}_n$, being a linear combination of degenerate states, is itself a degenerate state (corresponding to the eigenvalue $E_1^{(0)}$). With the el-

²The sequence (13.16) spans the same subspace of Hilbert space spanned by the degenerate states $\{\varphi_n^{(0)}\}$, $n \leq q$. Thus the basis (13.18) spans the same Hilbert space spanned by the basis $\{\varphi_i^{(0)}\}$, $i \geq 1$.

ements of $\{\bar{\varphi}_n\}$ (as well as those of $\{\varphi_n^{(0)}\}$) taken to comprise an orthogonal sequence,³ one is able to identify (13.17) as being the matrix counterpart of the operator equation (13.22). This is so, provided that we set

$$E_n' = H_{nn}' \quad (n \leq q)$$

which again is the relation (13.20). This equality establishes the fact that the diagonal elements of the submatrix H_{np}' are the first-order corrections to the total Hamiltonian $\hat{H}$ (for $n \leq q$).

Let us now construct the new basis functions $\{\bar{\varphi}_n\}$ which diagonalize the said submatrix of $\hat{H}'$. These are given in terms of the a_{ni} coefficients in (13.16), which make $\{\bar{\varphi}_n\}$ obey the eigenvalue equation (13.22). Substituting the former into the latter gives

$$\hat{H}' \sum_{i=1}^q a_{ni} |\varphi_i^{(0)}\rangle = E_n' \sum_{i=1}^q a_{ni} |\varphi_i^{(0)}\rangle$$

Multiplying from the left with $\langle \varphi_p^{(0)} |$ gives

$$\sum_i a_{ni} H_{pi}' = E_n' \sum_i a_{ni} \delta_{pi} = E_n' a_{np} \quad (\text{for fixed } n, p \leq q)$$

This equation may be rewritten as

$$\sum_{i=1}^q (H_{pi}' - E_n' \delta_{pi}) a_{ni} = 0 \quad (n, p \leq q) \quad (13.23)$$

The coefficients $\{a_{ni}\}$ for a fixed value of n comprise the column vector representation of $\bar{\varphi}_n$ in the subbasis $\{\varphi_l^{(0)}, l \leq q\}$. Similarly,

$$H_{pi}' = \langle \varphi_p^{(0)} | \hat{H}' | \varphi_i^{(0)} \rangle$$

are the matrix elements of $\hat{H}'$ in this same basis.

The Secular Equation

For each value of n and p , (13.23) is one equation for E_n' and the q components $\{a_{ni}\}$. There are q such equations corresponding to the q values of p . For $n = 1$, for example, these equations appear as

$$\begin{pmatrix} H_{11}' - E_1' & H_{12}' & H_{13}' & \cdots & H_{1q}' \\ H_{21}' & H_{22}' - E_1' & H_{23}' & \cdots & H_{2q}' \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ H_{q1}' & & & & \end{pmatrix} \begin{pmatrix} a_{11} \\ a_{12} \\ \vdots \\ a_{1q} \end{pmatrix} = 0 \quad (13.24)$$

³One may always construct an orthogonal set of functions from a given sequence of degenerate functions through the so-called Schmidt orthogonalization procedure. See Problem 13.53.

This is the matrix equivalent of (13.22) in the basis $\{\varphi_n^{(0)}, n \leq q\}$. Setting $n = 2$ in (13.23) generates (13.24), with the modifications that E_1' is replaced with E_2' and the column vector $\{a_{1i}\}$ is replaced by $\{a_{2i}\}$. As n runs from 1 to q , one obtains q such equations. The condition that there be a nontrivial solution $\{a_{ni}\}$ for any one of these q matrix equations is that the determinant of the coefficient matrix vanish, which gives the *secular equation*

$$\det |H'_{pi} - E_n' \delta_{pi}| = 0 \quad (13.25)$$

This equation may be rewritten in a purely operational form,

$$\det |\hat{H}' - E_n' \hat{I}| = 0$$

The identity operator is $\hat{I}$ (or, equivalently, the $q \times q$ unit matrix). The q roots of the algebraic equation (13.25) are the eigenvalues of (13.22). They are the diagonal elements of the submatrix of $\hat{H}'$ depicted in (13.19). Substituting any value of E' so obtained, say E_1' , back into (13.24) permits one to solve for the coefficients $\{a_{1i}\}$. In similar manner, E_2' permits calculation of $\{a_{2i}\}$, and so on. These coefficients, in turn, give the new basis functions $\{\bar{\varphi}_n\}$ in (13.18).

Using this new basis (13.18), the ambiguities due to the degeneracy of $\hat{H}_0$ are removed⁴ and one may proceed with the analysis developed in the previous section for the nondegenerate case. For example, (13.2) appears as

$$\begin{aligned} \varphi_n &= \bar{\varphi}_n + \lambda \bar{\varphi}_n^{(1)} + \lambda^2 \bar{\varphi}_n^{(2)} + \cdots & n \leq q \\ \varphi_n &= \varphi_n^{(0)} + \lambda \varphi_n^{(1)} + \lambda^2 \varphi_n^{(2)} + \cdots & n > q \\ E_n &= E_n^{(0)} + \lambda E_n^{(1)} + \lambda^2 E_n^{(2)} + \cdots & n \leq q \quad (E_1^{(0)} = \cdots = E_q^{(0)}) \\ E_n &= E_n^{(0)} + \lambda E_n^{(1)} + \lambda^2 E_n^{(2)} + \cdots & n > q \\ E_n' &= \langle \bar{\varphi}_n | \hat{H}' | \bar{\varphi}_n \rangle & n \leq q \\ E_n^{(1)} &= \langle \varphi_n^{(0)} | \hat{H}' | \varphi_n^{(0)} \rangle & n > q \end{aligned}$$

An outline of this analysis is shown in Fig. 13.4. An important feature of degenerate perturbation theory is that solution of the matrix equation (13.24) gives (1) first-order corrections to the energy; and (2) corrected wavefunctions which, together with the nondegenerate states, serve as a proper basis for higher-order calculations.

Two-Dimensional Harmonic Oscillator

In this section we are primarily concerned with the degeneracy-removing property of the perturbation $\hat{H}'$. As a simple example of these procedures, consider the case

⁴It may be that first-order calculation does not remove the degeneracies of H_0 . For example, this occurs if the off-diagonal elements of $\hat{H}'$ are zero. For such cases it becomes necessary to include higher-order terms to remove the degeneracy. For a discussion and problems relating to such *second-order degenerate perturbation theory*, see L. I. Schiff, *Quantum Mechanics*, 3d ed., McGraw-Hill, New York, 1968.

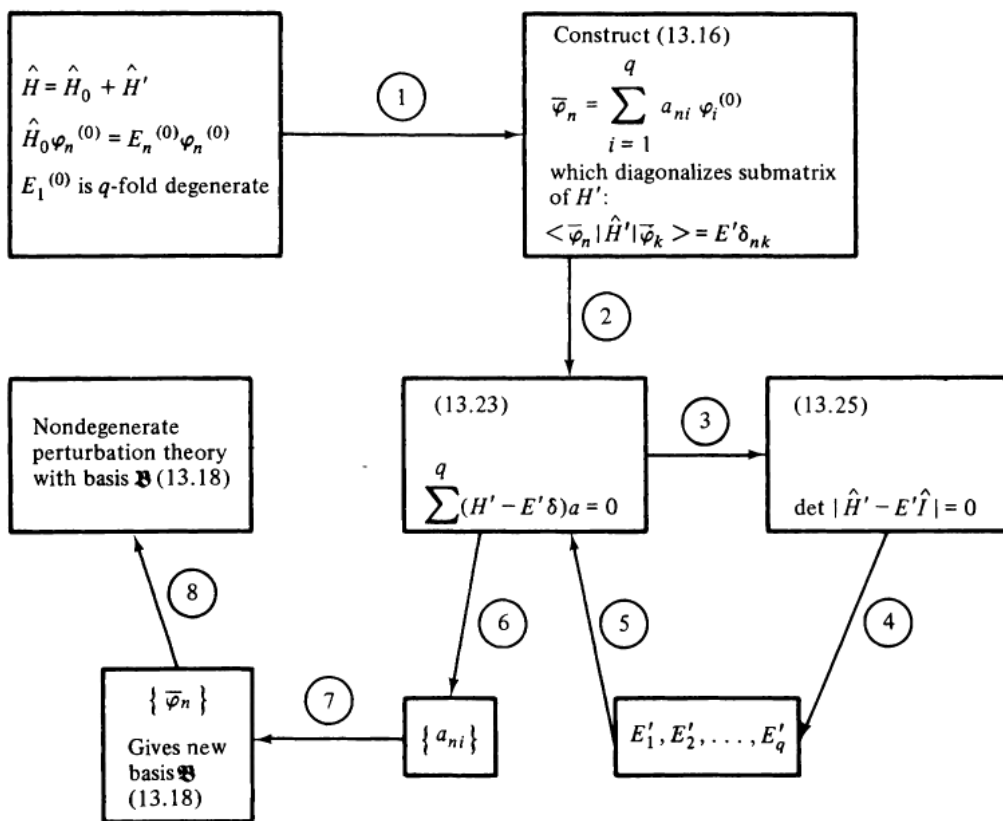


FIGURE 13.4 Elements of degenerate perturbation theory.

of the two-dimensional harmonic oscillator whose Hamiltonian is

$$\hat{H}_0 = \frac{\hat{p}_x^2 + \hat{p}_y^2}{2m} + \frac{K}{2}(x^2 + y^2)$$

or, equivalently,

$$\hat{H}_0 = \hbar\omega_0(\hat{a}^\dagger \hat{a} + \hat{b}^\dagger \hat{b} + 1)$$

where

$$x = \frac{1}{\sqrt{2}\beta}(\hat{a} + \hat{a}^\dagger) \quad y = \frac{1}{\sqrt{2}\beta}(\hat{b} + \hat{b}^\dagger) \quad \beta^2 = \frac{m\omega_0}{\hbar}$$

The eigenstates of $\hat{H}_0$ are the product forms (8.111)

$$\varphi_{np} \equiv \varphi_n(x)\varphi_p(y)$$

which we will label $|np\rangle$. The corresponding eigenenergy

$$E_{np} = \hbar\omega_0(n + p + 1)$$

is $(n + p + 1)$ -fold degenerate. It follows that the energy

$$E_{10} = E_{01} = 2\hbar\omega_0$$

is twofold degenerate, with corresponding eigenstates $|10\rangle$ and $|01\rangle$.

Let us apply the analysis developed above to determine how this energy separates due to the perturbing potential

$$H' = K'xy$$

Furthermore, we wish to find the two new wavefunctions that diagonalize $\hat{H}'$. These are given by the linear combinations (13.16)

$$\begin{aligned}\bar{\varphi}_1 &= \hat{a}\varphi_{10} + \hat{b}\varphi_{01} \\ \bar{\varphi}_2 &= \hat{a}'\varphi_{10} + \hat{b}'\varphi_{01}\end{aligned}$$

The submatrix of $\hat{H}'$ in the basis $\{\varphi_{10}, \varphi_{01}\}$ appears as

$$\begin{aligned}\hat{H}' &= K' \begin{pmatrix} \langle 10|xy|10\rangle & \langle 10|xy|01\rangle \\ \langle 01|xy|10\rangle & \langle 01|xy|01\rangle \end{pmatrix} = \mathbb{E} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\ \mathbb{E} &= \frac{K'}{2\beta^2}\end{aligned}$$

Consider, for example, the calculation of the $(1, 2)$ elements of H' .

$$\begin{aligned}\langle 10|\hat{x}\hat{y}|01\rangle &= \frac{1}{2\beta^2} \langle 10|\hat{a}\hat{b} + \hat{a}^\dagger\hat{b} + \hat{a}\hat{b}^\dagger + \hat{a}^\dagger\hat{b}^\dagger|01\rangle \\ &= \frac{1}{2\beta^2} \langle 10|\hat{a}^\dagger\hat{b}|01\rangle = \frac{1}{2\beta^2}\end{aligned}$$

With these values of the matrix elements of $\hat{H}'$, we are prepared to solve (13.25) for the incremental energies E' . This equation appears as

$$\begin{vmatrix} -E' & \mathbb{E} \\ \mathbb{E} & -E' \end{vmatrix} = 0$$

which has the solutions

$$E' = \pm\mathbb{E}$$

Thus we find that the perturbation separates the first excited state by the amount $2\mathbb{E}$.

$$\begin{array}{c}
 E_{10} \text{---} \begin{array}{l} \nearrow E_+ = E_{10} + \mathbb{E} \\ \searrow E_- = E_{10} - \mathbb{E} \end{array}
 \end{array} \quad (13.26)$$

The corresponding new wavefunctions are obtained by substituting these values into the matrix equation (13.23), which for the present case takes the form

$$\begin{pmatrix} -E' & \mathbb{E} \\ \mathbb{E} & -E' \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = 0$$

The two values of E' given above serve to determine two column vectors, (a, b) and (a', b') . The value $E' = \mathbb{E}$ gives $a = b$, while $E' = -\mathbb{E}$ gives $a' = -b'$. Thus we obtain for the new wavefunctions, $\bar{\varphi}_1$ and $\bar{\varphi}_2$, the values

$$\begin{aligned}
 E' = +\mathbb{E} &\rightarrow \bar{\varphi}_1 = \frac{1}{\sqrt{2}}(\varphi_{10} + \varphi_{01}) \\
 E' = -\mathbb{E} &\rightarrow \bar{\varphi}_2 = \frac{1}{\sqrt{2}}(\varphi_{10} - \varphi_{01})
 \end{aligned} \quad (13.27)$$

These new wavefunctions serve to diagonalize the perturbation Hamiltonian H' .

PROBLEMS

13.12 How does the threefold-degenerate energy

$$E = 3\hbar\omega_0$$

of the two-dimensional harmonic oscillator separate due to the perturbation

$$H' = K'xy?$$

13.13 Consider a particle confined in a two-dimensional square well with faces at $x = 0, a$; $y = 0, a$ (see Section 8.5). The doubly degenerate eigenstates appear as

$$\begin{aligned}
 \varphi_{np}(x, y) &= \frac{2}{a} \sin\left(\frac{n\pi x}{a}\right) \sin\left(\frac{p\pi y}{a}\right) \\
 E_{np} &= E_1(n^2 + p^2)
 \end{aligned}$$

What do these energies become under the perturbation

$$H' = 10^{-3} E_1 \sin\left(\frac{\pi x}{a}\right)?$$

13.14 The eigenstates of a rotating dumbbell, with moment of inertia I ,

$$E_l = \frac{\hbar^2 l(l+1)}{2I}$$

are $(2l+1)$ -fold degenerate. In the event that the dumbbell is equally and oppositely charged at its ends, it becomes a dipole. The interaction energy between such a

dipole and a constant, uniform electric field $\mathcal{E}$ is

$$\hat{H}' = -\mathbf{d} \cdot \mathcal{E} \quad (\hat{H} = \hat{H}_0 - \mathbf{d} \cdot \mathcal{E})$$

The dipole moment of the dumbbell is $\mathbf{d}$. Show that to terms of first order, this perturbing potential *does not separate* the degenerate E_l eigenstates.

- 13.15** Consider again the dipole moment described in Problem 13.14. If both ends are equally charged, the rotating dipole constitutes a magnetic dipole. If the dipole has angular momentum $\mathbf{L}$, the corresponding magnetic dipole moment is

$$\boldsymbol{\mu} = \frac{e}{2mc} \mathbf{L}$$

where e is the net charge of the dipole. The interaction energy between this magnetic dipole and a constant, uniform magnetic field $\mathcal{B}$ is

$$\hat{H}' = -\boldsymbol{\mu} \cdot \mathcal{B} = -\frac{e}{2mc} \hat{\mathbf{L}} \cdot \mathcal{B} \quad (\hat{H} = \hat{H}_0 - \boldsymbol{\mu} \cdot \mathcal{B})$$

- (a) If $\mathcal{B}$ points in the z direction, show that $\hat{H}'$ separates the $(2l + 1)$ -fold degenerate E_l energies of the rotating dipole.
- (b) Apply these results to one-electron atoms to find the splitting of the P states. (Neglect spin-orbit coupling.) (Note: This phenomenon is an example of the *Zeeman effect* discussed previously in Problems 12.15 et seq.)

13.3 ■ THE STARK EFFECT

In Problem 13.9 we found that to within a first-order calculation, an electric field does not remove m_l degeneracy of states of definite orbital number l . However, as we shall now see, a similar calculation reveals that such a field will induce a partial separation of the n^2 degeneracy of eigenenergies related to one-electron atoms. This effect was first noticed in 1913 by Stark. He observed the splitting of the Balmer lines in a field of 100,000 V/cm. (The more readily observed Zeeman effect was first observed in 1897.)

The Hamiltonian of a one-electron atom in a constant, uniform electric field $\mathcal{E}$ which points in the z direction, neglecting spin, is

$$\begin{aligned} \hat{H} &= \frac{\hat{p}_r^2}{2m} + \frac{\hat{L}^2}{2mr^2} - \frac{Ze^2}{r} - e\mathcal{E}z \\ &= \hat{H}_0 + \hat{H}' \\ \hat{H}' &= -e\mathcal{E}z = -e\mathcal{E}r \cos \theta \end{aligned}$$

The eigenstates of the unperturbed Hamiltonian are n^2 -fold degenerate. Let us consider how the perturbing electric field removes this degeneracy. Specifically, let us consider the fourfold degenerate $n = 2$ states. The related degenerate wave-

functions are, in the $|nlm\rangle$ notation,

$$|200\rangle, |211\rangle, |210\rangle, |21-1\rangle$$

To calculate the incremental changes in the energy E_2 , we must solve the determinantal equation

$$0 = \begin{vmatrix} \langle 200|\hat{H}'|200\rangle - E' & \langle 200|\hat{H}'|211\rangle & \langle 200|\hat{H}'|210\rangle & \langle 200|\hat{H}'|21-1\rangle \\ \langle 211|\hat{H}'|200\rangle & \langle 211|\hat{H}'|211\rangle - E' & \langle 211|\hat{H}'|210\rangle & \langle 211|\hat{H}'|21-1\rangle \\ \langle 210|\hat{H}'|200\rangle & \langle 210|\hat{H}'|211\rangle & \langle 210|\hat{H}'|210\rangle - E' & \langle 210|\hat{H}'|21-1\rangle \\ \langle 21-1|\hat{H}'|200\rangle & \langle 21-1|\hat{H}'|211\rangle & \langle 21-1|\hat{H}'|210\rangle & \langle 21-1|\hat{H}'|21-1\rangle - E' \end{vmatrix} \quad (13.28)$$

Only two elements survive integration. All elements with different m_l numbers vanish by orthogonality of the $|nlm_l\rangle$ states. Equivalently, one says, “ $\hat{H}'$ does not connect states of different m_l .” Integration gives⁵

$$\begin{aligned} \langle 210|\hat{H}'|200\rangle &= \langle 200|\hat{H}'|210\rangle \\ &= -\frac{ea_0\mathcal{E}}{32\pi} \int_0^\infty \rho^4(2-\rho)e^{-\rho} d\rho \int_{-1}^1 d\cos\theta \cos^2\theta \int_0^{2\pi} d\phi \\ &= \frac{e\mathcal{E}\hbar^2}{mZe} = -|e|3\mathcal{E}a_0 \equiv -\mathbb{E} \end{aligned}$$

With these values inserted into the determinant above, (13.25) becomes

$$\begin{vmatrix} -E' & 0 & -\mathbb{E} & 0 \\ 0 & -E' & 0 & 0 \\ -\mathbb{E} & 0 & -E' & 0 \\ 0 & 0 & 0 & -E' \end{vmatrix} = 0$$

which has the four roots

$$\begin{aligned} E' &= 0, 0, +\mathbb{E}, -\mathbb{E} \\ \mathbb{E} &= 3|e|\mathcal{E}a_0 \end{aligned} \quad (13.29)$$

Thus we find that to terms of lowest order in the electric field $\mathcal{E}$, the degenerate $n = 2$ state separates into three states:

$$E_2 \begin{cases} E_2^{(0)} + \mathbb{E} \\ E_2^{(0)} \\ E_2^{(0)} - \mathbb{E} \end{cases}$$

To calculate the new $n = 2$ wavefunctions

$$\varphi = a|200\rangle + b|211\rangle + c|210\rangle + d|21-1\rangle$$

⁵The nondimensional radius ρ is defined in Table 10.3. See also Tables 9.1 and 10.5.

we substitute the values (13.29) into the matrix equation

$$\begin{pmatrix} -E' & 0 & -\mathbb{E} & 0 \\ 0 & -E' & 0 & 0 \\ -\mathbb{E} & 0 & -E' & 0 \\ 0 & 0 & 0 & -E' \end{pmatrix} \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} = 0 \quad (13.30)$$

There results

$$\begin{aligned} E_2^+ &= E_2^{(0)} + \mathbb{E} \rightarrow \varphi_+ = \frac{1}{\sqrt{2}}(|200\rangle - |210\rangle) \\ E_2^- &= E_2^{(0)} - \mathbb{E} \rightarrow \varphi_- = \frac{1}{\sqrt{2}}(|200\rangle + |210\rangle) \\ E_2^0 &= E_2^{(0)} \begin{cases} \varphi = |211\rangle \\ \varphi = |21-1\rangle \end{cases} \end{aligned} \quad (13.31)$$

The perturbation mixes the $m = 0$ states, while the $m = 1, -1$ states are left degenerate. The values $\pm\mathbb{E}$ represent the average values of the interaction $\hat{H}'$ in the respective states, $\varphi_{\pm}$.

Symmetry Breaking and Removal of Degeneracy

Degeneracy, such as in the example of the two-dimensional harmonic oscillator considered above, is due to symmetry properties of $\hat{H}_0$. In this example the potential $x^2 + y^2$ is noted to be invariant under rotation about the origin in the plane. The perturbation xy destroys this symmetry, thereby removing the degeneracy. (Recall related discussions in Section 8.5.) It has been noted that an eigenenergy which is n -fold degenerate corresponds to n independent eigenfunctions. When the symmetry related to this degeneracy is broken, an n -fold degenerate eigenenergy may split into as many as n new eigenenergies. Another example addresses the case when a hydrogen atom is placed in an external field. This perturbation breaks rotational symmetry of the Hamiltonian about the nuclear center of the atom. In the problem of the Stark effect discussed immediately above, a uniform electric field breaks rotational symmetry and eigenenergies proliferate.⁶ In addition, as noted in (13.31), there is a decrease in the number of “good quantum numbers” from (n, l, m) to (n) . Consider the case in which the strength of the symmetry-breaking perturbation is represented by a parameter in the Hamiltonian, which we label g . If eigenenergies are plotted versus g , then at some value of g , eigenenergies bifurcate. Such behavior of eigenvalues is sometimes referred to as “level repelling.” The properties of repelling of levels and loss of good quantum numbers are often associated with nonintegrable systems.

⁶The order of degeneracy of a given system may exceed that implied by symmetry principles (i.e., group theory). Such, for example, is the case for the two-dimensional harmonic oscillator as well as for the hydrogen atom. For further discussion, see F. A. Cotten, *Chemical Applications of Group Theory*, 3d ed., Wiley, New York, 1993.

PROBLEMS

13.16 Show that the $n = 2$ matrix of the interaction Hamiltonian $\hat{H}'$, of the Stark effect, is diagonal in the basis (13.31).

13.17 What is the dipole moment of the hydrogen atom in the $\varphi_{\pm}$ states (13.31)?

Answer

The interaction energy between an electric dipole $\mathbf{d}$ and the electric field $\mathcal{E}$ is

$$H' = -\mathbf{d} \cdot \mathcal{E}$$

In the $\varphi_{\pm}$ states, the average value of H' is $E = \pm 3|e|a_0\mathcal{E}$. We may infer from this result that the magnitude of the dipole moment in the $\varphi_{\pm}$ states is $3|e|a_0$. The directions of these moments are parallel or antiparallel to the z axis (i.e., the direction of $\mathcal{E}$).

13.18 What is the charge density $q(r, \theta)$ of the hydrogen atom associated with the state φ_{-} (13.31)?

Answer

$$q(r, \theta) = e|\varphi_{-}|^2 = \frac{e}{16\pi a_0^3} \left[1 - \frac{r}{a_0} \sin^2 \left(\frac{\theta}{2} \right) \right]^2 e^{-r/a_0}$$

13.19 Of the two states $\varphi_{\pm}$ in (13.31), φ_{-} is said to be more *stable* than φ_{+} . Why? Discuss your answers in light of the interaction energy, $-\mathbf{d} \cdot \mathcal{E}$.

13.4 ■ THE NEARLY FREE ELECTRON MODEL

In this section we return to the problem of an electron in a periodic potential $V(x)$, discussed in Sections 8.2 and 12.9. Wavefunctions are in the Bloch form

$$\varphi(x) = u(x)e^{ikx}$$

where the periodic function $u(x)$ has the same period a as $V(x)$. Eigenenergies are functions of the crystal momentum wavenumber k . For a lattice of length L , the nearly continuous wavenumber k has the discrete values [see (8.42)]

$$k_j = \frac{j2\pi}{L}$$

We recall (see Problem 8.13) that in the high-energy domain ($E \gg V$), the energy spectrum reduces to the free-particle values $\hbar^2 k^2 / 2m$, or, equivalently,

$$E_{k_j} = \frac{\hbar^2 k_j^2}{2m} = \frac{j^2 \hbar^2}{2mL^2}$$

which is doubly degenerate: $E_{k_j} = E_{-k_j}$.

We now wish to obtain an expression for the energy gap δE_n at the n th band edge. A band edge, we recall, is a break in the energy spectrum which occurs at the k values $k_j a = n\pi$.

In the present analysis the periodic potential is considered a small perturbation to the free-particle Hamiltonian

$$\hat{H}_0 = \frac{\hat{p}^2}{2m}$$

The electron is “nearly free,” which is the same as saying that $E \gg V$.

Unperturbed eigenstates are normalized to the sample interval L .

$$\varphi_{k_j}^{(0)} = \frac{1}{\sqrt{L}} \exp(ik_j x)$$

With $k_j = j(2\pi/L)$ these functions comprise an orthonormal sequence, as may be seen as follows:

$$\begin{aligned} \langle k_q | k_j \rangle &= \frac{1}{L} \int_{-L/2}^{L/2} \exp[i(k_j - k_q)x] dx = \frac{\sin[(k_j - k_q)L/2]}{(k_j - k_q)L/2} \\ &= \frac{\sin(j - q)\pi}{(j - q)\pi} = \delta_{jq} \end{aligned} \quad (13.32)$$

The Perturbation Potential

The perturbing potential is periodic and may be expanded in the Fourier series (see Problem 13.58).

$$H' = V(x) = \sum_{n=-\infty}^{\infty} V_n \exp\left[i2\pi n \left(\frac{x}{a}\right)\right] \quad (13.33)$$

The zero energy line in the present analysis is set at the average of $V(x)$. This ensures that the dc component V_0 of V vanishes (see Fig. 13.5).

$$V_0 = \int_{-L/2}^{L/2} V(x) dx = 0$$

Application of the first-order perturbation theory necessitates calculation of the matrix elements of H' .

$$\begin{aligned} H'_{qj} &= \langle k_q | \sum_n V_n \exp\left[i2\pi n \left(\frac{x}{a}\right)\right] | k_j \rangle \\ &= \sum_n V_n \left\langle k_q \left| k_j + \left(\frac{2\pi n}{a}\right) \right. \right\rangle \\ &= \sum_n V_n \delta_{k_q, k_j + (2\pi n/a)} \end{aligned} \quad (13.34)$$

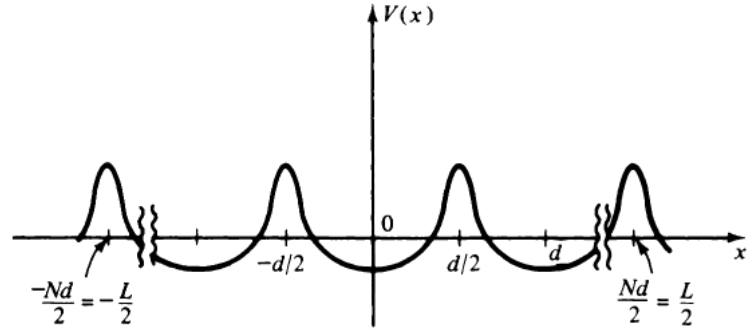


FIGURE 13.5 The zero in potential is chosen so as to eliminate V_0 .

$$V_0 = \int_{-L/2}^{L/2} V(x) dx = 0$$

Substituting these matrix elements into the first equation of (13.10) gives the first-order corrected wavefunctions

$$\varphi_{k_j} = \varphi_{k_j}^{(0)} + \frac{1}{\sqrt{L}} \sum_{n=-\infty}^{\infty} \frac{V_n \exp\{ix[k_j + (2\pi n/a)]\}}{E_{k_j}^{(0)} - E_{k_j + (2\pi n/a)}^{(0)}} \quad (13.35)$$

Calculation of the first-order corrected eigenenergies as given by the second equation in (13.10) gives

$$\begin{aligned} E_{k_j} &= E_{k_j}^{(0)} + \langle k_j | V | k_j \rangle = E_{k_j}^{(0)} + V_0 \\ &= E_{k_j}^{(0)} = \frac{\hbar^2 k_j^2}{2m} \end{aligned} \quad (13.36)$$

To first order, the energy remains unperturbed. This free-particle spectrum was found in Chapter 8 to maintain in the domain $E \gg V$. However, in the present analysis there is explicit evidence which indicates that this result is invalid at the band edges of the energy spectrum. Namely, the summation in (13.35) for the first-order corrected wavefunctions becomes singular if the denominator of any term vanishes.

$$E_{k_j}^{(0)} - E_{k_j + (2\pi n/a)}^{(0)} = -\frac{\hbar^2}{2m} \frac{4\pi n}{a^2} (k_j a + n\pi) = 0 \quad (13.37)$$

This singularity arises from the zeros of (13.37), corresponding to the degeneracy at the band edges, $k_j a + n\pi = 0$. To obtain correct energies at these values of k_j , one must use degenerate perturbation theory. As described in Section 13.2, the first step in this procedure is to construct a new basis $\{\bar{\varphi}\}$ that diagonalizes the relevant 2×2 submatrix of H' . This new subbasis $\{\bar{\varphi}\}$ is constructed from linear

combinations of the degenerate portion of the unperturbed basis and we may write

$$\bar{\varphi} = \bar{a}e^{ik_j x} + \bar{b}e^{-ik_j x} \quad (13.38)$$

Diagonalization of H' in the basis (13.38) yields the matrix equation

$$\begin{pmatrix} H'_{k_j k_j} - E_{k_j}' & H'_{k_j, -k_j} \\ H'_{-k_j, k_j} & H'_{-k_j, -k_j} - E_{k_j}' \end{pmatrix} \begin{pmatrix} \bar{a} \\ \bar{b} \end{pmatrix} = 0 \quad (13.39)$$

$$k_j = -\frac{n\pi}{a}$$

The matrix elements of H' are given by (13.34), from which we find

$$H'_{k_j k_j} = H'_{-k_j, -k_j} = V_0 = 0$$

Only the off-diagonal elements survive. Choosing the origin so that $V(x)$ is an even function gives

$$H'_{k_j, -k_j} = V_n = V_{-n} = H'_{-k_j, k_j}$$

The determinant equation (13.25) then becomes

$$\begin{vmatrix} -E_{k_j}' & V_n \\ V_n & -E_{k_j}' \end{vmatrix} = 0$$

which gives the roots

$$E_{k_j}' = \pm V_n$$

Resubstituting into the matrix equation (13.39) gives two nondegenerate eigenstates, which written together with the first-order corrected energies appear as

$$\begin{aligned} E_{k_j}^+ &= E_{k_j}^{(0)} + V_n & \bar{\varphi}_+ &= 2\bar{a} \sin k_j x \\ E_{k_j}^- &= E_{k_j}^{(0)} - V_n & \bar{\varphi}_- &= 2\bar{a} \cos k_j x \end{aligned}$$

These eigenstates, we recall, are the standing waves at the band edges previously obtained in Section 8.3 relevant to the Kronig-Penney potential. The present calculation affords an estimate (in the domain $E \gg V$) of the width of the energy gap at the band edges.

$$\delta E_n = E_{k_j}^+ - E_{k_j}^- = 2|V_n| \quad k_j a = \pm n\pi$$

Thus we find that the n th gap in the energy spectrum $E = E(k)$, at the values $k = \pm n\pi/a$, has width which is twice the n th Fourier coefficient of the potential $V(x)$ (Fig. 13.6).

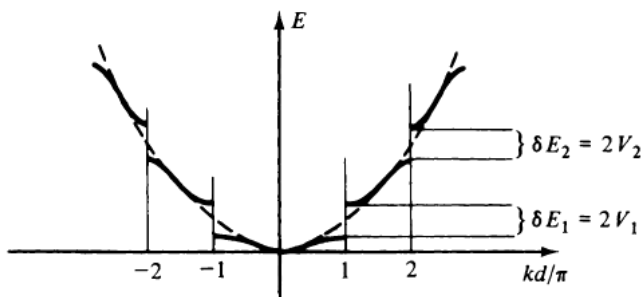


FIGURE 13.6 Energy gaps at the band edges in the nearly free electron model are given by twice the corresponding Fourier coefficient of the periodic potential.

In the nearly free electron model described above, conduction electrons are weakly bound to atoms of the lattice. This is opposite to the situation in the *tight-binding approximation* described in Problem 8.16 and in the LCAO analysis presented in Section 8.7.

PROBLEMS

- 13.20** Resketch the $E(k)$ curve shown in Fig. 13.6 for the case that the zeroth Fourier coefficient $V_0 > 0$.
- 13.21** A periodic potential has Fourier coefficients $V_n = V_1/n^2$. The width of the tenth energy gap is 0.031 eV. What is the value of V_1 ?
- 13.22** What are the energy gaps at the band edges for a particle in the periodic delta-function potential

$$V(x) = V_0 a \sum_{q=-\infty}^{\infty} \delta(x - qa)$$

defined over the entire x interval? [Hint: Consider the following delta-function representation

$$\delta(y) = \sum_{n=-\infty}^{\infty} \exp(i2\pi ny)$$

The right-hand side is periodic with period 1 and represents a delta function at $y = 0, \pm 1, \pm 2, \dots$. The left-hand side is a delta function only at the origin. The domain of validity of the equation may be extended to the whole y axis if one writes

$$\sum_{m=-\infty}^{\infty} \delta(y - m) = \sum_{n=-\infty}^{\infty} \exp(i2\pi ny).$$

- 13.23** (a) Show that the first-order corrected wavefunction φ_k as given by (13.35) may be cast in the form of a Bloch wavefunction $\varphi_k = u(x)e^{ikx}$.
- (b) Show that the expression you obtain for $u(x)$ has period a .

- 13.24** Estimate the energy gaps at the band edges for the Kronig–Penney potential of potential height V_0 , well width a , and barrier width b .

Answer

$$\delta E_n \simeq \frac{2V_0}{n\pi} \sin\left(\frac{n\pi a}{a+b}\right)$$

- 13.25** Estimate the band-gap widths for the potential

$$V(x) = 2V_0 \cos\left(\frac{2\pi x}{a}\right)$$

Answer

Rewriting the cos term in terms of exponentials,

$$V(x) = 2V_0 \cos\left(\frac{2\pi x}{a}\right) = V_0 \left[\exp\left(\frac{i2\pi x}{a}\right) + \exp\left(\frac{-i2\pi x}{a}\right) \right]$$

reveals that this potential has only two nonvanishing Fourier coefficients. First-order perturbation theory implies the existence of the gaps at $k = \pm\pi/a$, of width $2V_0$. Higher-order perturbation theory would uncover energy gaps at subsequent band edges as well. This may be concluded directly by writing the Schrödinger equation for the given potential,⁷

$$\varphi_{xx} + \frac{2m}{\hbar^2} \left[E - 2V \cos\left(\frac{2\pi x}{a}\right) \right] \varphi = 0$$

This is a well-known equation in mathematical physics and is called the *Mathieu equation*. As with most such equations, it stems from writing the “wave function,” $(\nabla^2 + k^2)\varphi = 0$, in a particular orthogonal coordinate frame. For the Mathieu equation these are elliptic cylinder coordinates. Solutions of the equation are called *Mathieu functions* and have been studied in detail. These analyses reveal a sequence of intervals on the E axis in which solutions to the equation are unstable.⁸

- 13.26** At 0 K a certain semiconductor has its valence band filled and conduction band empty. The potential “seen” by electrons may be approximated by the function

$$V(x) = 2V_0 \cos\left(\frac{2\pi x}{a}\right)$$

$$V_0 = 0.1 \text{ eV}, \quad a = 0.5 \text{ Å}$$

- (a) Assuming that the valence band is the band of lowest energy, estimate the width of the gap δE between the valence and conduction bands.

⁷This case has been studied in detail by P. M. Morse, *Phys. Rev.* **35**, 1310 (1930). Here it is also shown that the energy bands approach the line spectrum of an infinitely deep well as the amplitude V_0 of the periodic potential grows infinitely large.

⁸That is, series solutions in these domains do not converge. For further discussion of the properties of Mathieu functions, see P. M. Morse and H. Feshbach, *Methods of Theoretical Physics*, McGraw-Hill, New York, 1953, Chapter 5; also, E. T. Whittaker and G. N. Watson, *A Course of Modern Analysis*, Cambridge University Press, New York, 1952, Chapter 19.

- (b) Setting $\delta E = k_B T$, estimate the temperature at which the sample will begin to conduct appreciably. ($k_B = 8.6 \times 10^{-5}$ eV/K.)

Answers

(a) $\delta E \simeq 2V_0 = 0.20$ eV

(b) $T \simeq 2300$ K

13.5 ■ TIME-DEPENDENT PERTURBATION THEORY

Time-dependent perturbation theory addresses the following problem. Initially, the unperturbed system is in an eigenstate of $\hat{H}_0$. Then the perturbation, $\hat{H}'(t)$, is “turned on.” What is the probability, after a time t , that transition to another state (of $\hat{H}_0$) occurs?

The total Hamiltonian for these problems is of the form

$$\hat{H}(\mathbf{r}, t) = \hat{H}_0(\mathbf{r}) + \lambda \hat{H}'(\mathbf{r}, t) \quad (13.40)$$

where λ is again a parameter of smallness.

Let the time-dependent eigenstates of $\hat{H}_0$ be written

$$\begin{aligned} \psi_n(\mathbf{r}, t) &= \varphi_n(\mathbf{r}) e^{-i\omega_n t} \\ \hat{H}_0 \varphi_n &= E_n^{(0)} \varphi_n \equiv \hbar \omega_n \varphi_n \end{aligned} \quad (13.41)$$

Suppose that at time $t > 0$, the system is in the state

$$\psi(\mathbf{r}, t) = \sum_n c_n(t) \psi_n(\mathbf{r}, t) \quad (13.42)$$

Then, by the superposition principle, $|c_n|^2$ is the probability that measurement finds the system in the state ψ_n at the time t . Thus the primary aim of the present discussion is to calculate these coefficients. They are determined in the following manner. The wavefunction $\psi(\mathbf{r}, t)$ is a solution of

$$i\hbar \frac{\partial \psi}{\partial t} = (\hat{H}_0 + \lambda \hat{H}') \psi \quad (13.43)$$

Substituting the expansion (13.42) into this equation and then operating from the left with $\int d\mathbf{r} \psi_k^*(\mathbf{r}, t)$ gives

$$i\hbar \frac{dc_k}{dt} = \lambda \sum_n \langle k | H' | n \rangle c_n \quad (13.44)$$

This is an infinite sequence of coupled equations for the coefficients $\{c_k(t)\}$. In the limit that $\lambda \rightarrow 0$, the c_k coefficients are all constant. It is therefore consistent to seek solution in the form

$$c_k(t) = c_k^{(0)} + \lambda c_k^{(1)}(t) + \lambda^2 c_k^{(2)}(t) + \dots \quad (13.45)$$

Substituting this series into (13.44) and equating terms of equal powers in λ gives (with a dot denoting time differentiation and H'_{kn} written for the matrix elements of $\hat{H}'$)

$$\begin{aligned} i\hbar\dot{c}_k^{(0)} &= 0 \\ i\hbar\dot{c}_k^{(1)} &= \sum_n H'_{kn} c_n^{(0)} \\ i\hbar\dot{c}_k^{(2)} &= \sum_n H'_{kn} c_n^{(1)} \\ &\vdots \\ i\hbar\dot{c}_k^{(s+1)} &= \sum_n H'_{kn} c_n^{(s)} \end{aligned} \quad (13.46)$$

The lowest-order equations for $c_k^{(0)}$ indicate that these coefficients are all constant in time. They are the initial values of the coefficients $\{c_k(t)\}$.

We now specialize to the problem in which it is known that initially the system is in a definite eigenstate of $\hat{H}_0$, say $\psi_l(\mathbf{r}, t)$. With (13.42) this implies that as $t \rightarrow -\infty$,

$$\begin{aligned} \psi(\mathbf{r}, t) &\sim \psi_l(\mathbf{r}, t) = \sum_n \delta_{nl} \psi_n(\mathbf{r}, t) \\ c_n^{(0)}(-\infty) &= \delta_{nl} \end{aligned} \quad (13.47)$$

Note that we have taken “initially” to denote the time $t = -\infty$. Substituting this value into the second equation in (13.46) gives (dropping the superscripts 0 and 1)

$$i\hbar\dot{c}_k(t) = \sum_n H'_{kn} c_n(-\infty) = H'_{kl} \quad (13.48)$$

For $n \neq l$, $c_n(-\infty) = 0$, so the first-order solution for $c_k(t)$ is given by

$$c_k(t) = \frac{1}{i\hbar} \int_{-\infty}^t H'_{kl}(\mathbf{r}, t') dt' \quad (k \neq l) \quad (13.49)$$

If the time dependence of $\hat{H}'(\mathbf{r}, t)$ is factorable, then

$$\hat{H}'(\mathbf{r}, t) = \hat{\mathbb{H}}'(\mathbf{r}) f(t)$$

and the matrix elements of $\hat{H}$ becomes [with (13.41)]

$$\begin{aligned} H'_{kl}(t) &\equiv \langle \psi_k | H'(\mathbf{r}, t) | \psi_l \rangle = \langle \varphi_k | \mathbb{H}'(\mathbf{r}) | \varphi_l \rangle e^{i\omega_{kl}t} f(t) \\ &= \mathbb{H}'_{kl} e^{i\omega_{kl}t} f(t) \\ \hbar\omega_{kl} &\equiv \hbar(\omega_k - \omega_l) = E_k - E_l \end{aligned} \quad (13.50)$$

(Note that we have deleted the zero superscripts of E_k and E_l .) This gives the more explicit form of $c_k(t)$,

$$c_k(t) = \frac{\mathbb{H}'_{kl}}{i\hbar} \int_{-\infty}^t e^{i\omega_{kl}t'} f(t') dt' \quad (13.51)$$

These coefficients determine the effect of the perturbation on the initial state, ψ_l . The probability that the system has undergone a transition from this state to some other eigenstate of H_0 , ψ_k , at the time t , is

$$P_{l \rightarrow k} = |c_k|^2 = \left| \frac{\mathbb{H}'_{kl}}{\hbar} \right|^2 \left| \int_{-\infty}^t e^{i\omega_{kl}t'} f(t') dt' \right|^2 \quad (13.52)$$

Application of these results follows. The transition probability $P_{l \rightarrow k}$ is hereafter written P_{lk} .

Matrix elements in time-dependent perturbation theory are conventionally written with the initial state on the right and the final state on the left, as illustrated in the following symbolic form:

$$\langle \text{final state} | \text{interaction} | \text{initial state} \rangle$$

This convention is employed in the remainder of the text.

PROBLEMS

13.27 A system with discrete eigenstates $\{\varphi_n\}$ and eigenenergies $\{E_n\}$ is exposed to the perturbation

$$\hat{H}' = \hat{\mathbb{H}}'(\mathbf{r}) \frac{e^{-t^2/\tau^2}}{\tau\sqrt{\pi}}$$

The perturbation is turned on at $t = -\infty$, when the unperturbed system is in its ground state, ψ_0 . What is the probability that at $t = +\infty$ the system suffers a transition to the state ψ_k , $k > 0$?

Answer

To obtain the answer to this problem, we must calculate the time integral in (13.52). Writing ω for ω_{k0} , we have

$$\begin{aligned} I &= \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{i\omega t} e^{-t^2/\tau^2} d(t/\tau) \\ &= \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{i\bar{\omega}\xi - \xi^2} d\xi \quad (\xi \equiv t/\tau, \bar{\omega} \equiv \tau\omega) \\ &= \frac{1}{\sqrt{\pi}} e^{-\bar{\omega}^2/4} \int_{-\infty}^{\infty} \exp\left[-\left(\xi - \frac{i\bar{\omega}}{2}\right)^2\right] d\xi \\ &= e^{-\bar{\omega}^2/4} = e^{-(E_0 - E_k)^2 \tau^2 / 4\hbar^2} \end{aligned}$$

Substituting this into (13.52) gives the desired result

$$P_{0k} = \left| \frac{\mathbb{H}'_{k0}}{\hbar} \right|^2 e^{-(E_0 - E_k)^2 \tau^2 / 2\hbar^2}$$

13.28 Consider that the unperturbed system in Problem 13.27 is a particle of mass m confined to a one-dimensional box of width a . The spatial factor in $H'(x, t)$ is

$$\hat{\mathbb{H}}'(x) = \frac{10^{-4} \hat{p}_x^2}{2m}$$

What state does the perturbation leave the system in at $t = +\infty$?

13.6 ■ HARMONIC PERTURBATION

Stimulated Emission

As a first application of the analysis above, we consider a perturbation that is switched on at $t = 0$ and is subsequently monochromatically harmonic in time. The perturbation acts on a system whose Hamiltonian is $\hat{H}_0$. If it is definitely known that the unperturbed system was in one of its own stationary states before the perturbation was applied ($t < 0$), in what state will measurement find the system after the perturbation has been turned on for t seconds? This problem is appropriate, for example, to an atom that interacts with a (weak) electromagnetic field. The explicit form of the perturbation $\hat{H}'$ is (with $\omega > 0$)

$$\hat{H}'(\mathbf{r}, t) = \begin{cases} 0 & t < 0 \\ 2\hat{H}'(\mathbf{r}) \cos \omega t & t \geq 0 \end{cases} \quad (13.53)$$

(see Fig. 13.7). Substituting this form into (13.51), with $f(t) = 2 \cos \omega t$, gives

$$H' = \hat{\mathbb{H}}'(\mathbf{r}) f(t)$$

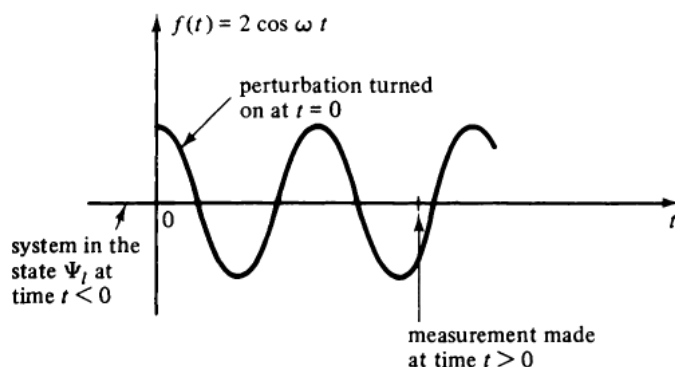


FIGURE 13.7 Harmonic perturbation.

$$c_k(t) = \frac{\mathbb{H}'_{kl}}{i\hbar} \int_0^t e^{i\omega_{kl}t'} (e^{-i\omega t'} + e^{i\omega t'}) dt'$$

$$= -\frac{\mathbb{H}'_{kl}}{\hbar} \left[\frac{e^{i(\omega_{kl}-\omega)t} - 1}{\omega_{kl} - \omega} + \frac{e^{i(\omega_{kl}+\omega)t} - 1}{\omega_{kl} + \omega} \right]$$

Employing the relation (see Problem 1.21)

$$e^{i\theta} - 1 = 2ie^{i\theta/2} \sin(\theta/2)$$

permits the equation above to be rewritten

$$c_k(t) = -\frac{i2\mathbb{H}'_{kl}}{\hbar} \left[\frac{e^{i(\omega_{kl}-\omega)t/2} \sin(\omega_{kl}-\omega)t/2}{\omega_{kl} - \omega} + \frac{e^{i(\omega_{kl}+\omega)t/2} \sin(\omega_{kl}+\omega)t/2}{\omega_{kl} + \omega} \right] \quad (13.54)$$

The dominant contributions to c_k come from the values $\omega \simeq \pm\omega_{kl}$. At these values, the moduli of the two bracketed terms in c_k , respectively, assume their maximum value, $t/2$. These *resonant* frequencies correspond to the energies

$$\omega \simeq +\omega_{kl} \longrightarrow E_k > E_l$$

$$\omega \simeq -\omega_{kl} \longrightarrow E_l > E_k$$

In the first case, the “final” energy E_k is larger than the “initial” energy E_l . The system absorbs energy and jumps to a higher energy level.

$$E_k = E_l + \hbar\omega$$

The energy absorbed, $\hbar\omega$, is that of a photon in the incident radiation field (Fig. 13.8).

For the second case the perturbation induces a decay in energy

$$E_k = E_l - \hbar\omega$$

A photon of energy $\hbar\omega$ is radiated away from the system. This decay process is *stimulated* by a photon of the same frequency in the perturbation field.

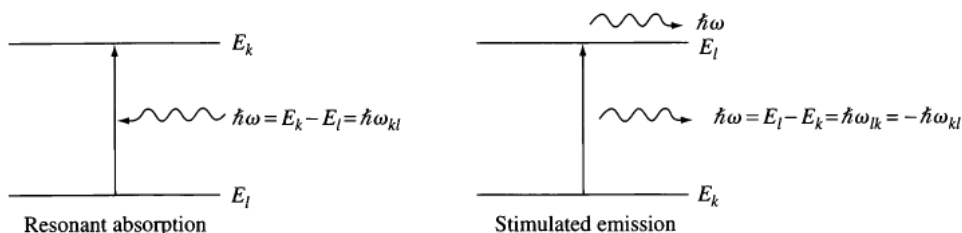


FIGURE 13.8 Dominant transition processes due to harmonic perturbation.

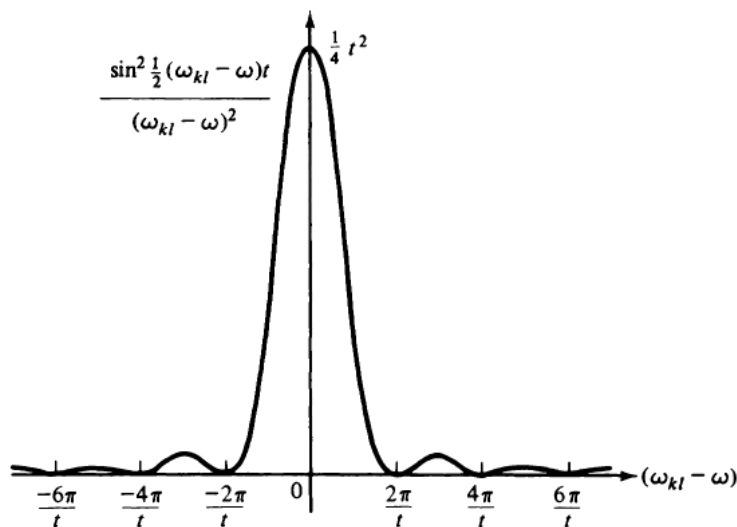


FIGURE 13.9 Frequency dependence of probability of transition from the l th to the k th state at the time t due to harmonic perturbation of frequency ω .

Let us consider the case that the incident radiation field excites only higher energies, so that $\omega_{kl} > 0$. Under such conditions, the first term in (13.54) dominates and the probability that the perturbing field causes a transition to the k th state becomes

$$P_{lk} = |c_k|^2 = \frac{4|\mathbb{H}'_{kl}|^2}{\hbar^2(\omega_{kl} - \omega)^2} \sin^2 \left[\frac{1}{2}(\omega_{kl} - \omega)t \right] \quad (13.55)$$

The frequency dependence of this function is plotted in Fig. 13.9.

Energy-Time Uncertainty

From this sketch it is evident that states falling in the interval

$$|\hbar\omega_{kl} - \hbar\omega| = |E_k - (E_l + \hbar\omega)| \lesssim \frac{2\pi\hbar}{t} \simeq \Delta E \quad (13.56)$$

have the greatest probability of being excited, after the perturbing field has acted for t seconds. If this perturbation is applied many times to an ensemble of independent, identical systems, all initially in the state ψ_l , then after a time t the energies excited among the members of the ensemble will lie primarily in the interval ΔE . Thus (13.56) gives the uncertainty in the values of energy observed. The final energies E_k are spread throughout the interval

$$\Delta E \simeq \frac{\hbar}{t}$$

$$E_k \simeq (E_l + \hbar\omega) \pm \Delta E \quad (13.57)$$

$$E_{\text{final}} \simeq E_{\text{initial}} \pm \Delta E$$

Note that E_{initial} refers to the energy of the incident photon plus that of the system in its initial state. Our perturbation analysis returns the principle of conservation of energy, properly modified by the uncertainty relation, $\Delta E \simeq \hbar/t$.

Long-Time Evolution

The transition probability formula for absorption (13.55) and its counterpart for stimulated emission may be written together as

$$P_{lk} = \frac{4|\mathbb{H}'_{kl}|^2}{\hbar^2(\omega_{kl} \mp \omega)^2} \sin^2 \left[\frac{1}{2}(\omega_{kl} \mp \omega)t \right] \quad (13.58)$$

where the $\mp$ signs refer to absorption and stimulated emission, respectively (see Fig. 13.8). This expression takes a convenient form in the long-time or, equivalently, high-frequency limit. This form follows from the delta-function representation

$$\delta(\omega) = \frac{2}{\pi} \lim_{t \rightarrow \infty} \frac{\sin^2(\omega t/2)}{t\omega^2} \quad (13.59)$$

so that in the same limit,

$$P_{lk} \rightarrow \frac{2\pi t}{\hbar^2} |\mathbb{H}'_{kl}|^2 \delta(\omega_{kl} \mp \omega) \quad (13.60)$$

The corresponding transition probability *rate* appears as

$$w_{lk} = \frac{2\pi}{\hbar^2} |\mathbb{H}'_{kl}|^2 \delta(\omega_{kl} \mp \omega) \quad (13.61)$$

In this or the formula above, the delta function expresses the fact that in the long-time limit, the Fourier transform of a monochromatic perturbation becomes sharply peaked about the frequency of perturbation. The system “sees” only a single frequency. Since the uncertainty in energy $\hbar/t$ vanishes in this limit, the argument of the delta function is also an expression of conservation of energy.

Short-Time Approximation

If a harmonic perturbation is applied to a system for a short-time interval such that $(\omega_{kl} - \omega)t \ll 1$, (13.55) may be expanded to yield

$$P_{lk} = \frac{t^2 |\mathbb{H}'_{kl}|^2}{\hbar^2} \quad (13.62)$$

The related transition probability rate is

$$w_{lk} = \frac{t |\mathbb{H}'_{kl}|^2}{\hbar^2} \quad (13.63)$$

At early times, the rate at which transitions to the k th state occur grows linearly with time.

The Golden Rule

In many problems of practical interest, the final excited states lie in a band of energies (Fig. 13.10). Such is the case, for example, for ionization or free-particle scattering states. Such states comprise a continuum. If the density of final states is $g(E_k)$, then

$$dN = g(E_k) dE_k$$

is the number of energy states in the interval E_k to $E_k + dE_k$. The probability that a transition occurs to a state in a band of width 2Δ centered at E_k is

$$\overline{P}_{lk} = \int_{E_k - \Delta}^{E_k + \Delta} P_{lk} g(E'_k) dE'_k$$

Inserting (13.55) gives

$$\overline{P}_{lk} = \int_{E_k - \Delta}^{E_k + \Delta} dE'_k g(E'_k) \left| \frac{\mathbb{H}'_{kl}}{\hbar} \right|^2 \frac{\sin^2 \beta}{\beta^2 / t^2}$$

$$2\hbar\beta \equiv \hbar(\omega_{kl} - \omega)t = (E'_k - E_l - \hbar\omega)t$$

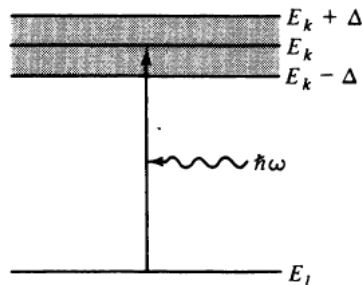


FIGURE 13.10 Resonant absorption to a band of energies. If the density of states about E_k is $g(E_k)$, then the probability rate of transition to the band is

$$\overline{w}_{lk} = \frac{2\pi}{\hbar} g(E_k) |\mathbb{H}'_{kl}|^2$$

For fixed E_l , t and ω

$$dE'_k = \frac{2\hbar d\beta}{t}$$

and

$$\overline{P}_{lk} = \frac{2t}{\hbar} \int_{-\delta}^{+\delta} g(E'_k) |\mathbb{H}'_{kl}|^2 \frac{\sin^2 \beta}{\beta^2} d\beta$$

where 2δ is the corresponding spread in β values. Owing to the rapid decay of the $\sin^2 \beta / \beta^2$ function (see Fig. 13.9), only a slight error is introduced in the expression above if we replace the interval $(-\delta, +\delta)$ by $(-\infty, +\infty)$. Furthermore, if we assume that g and $\mathbb{H}'_{kl}$ are slowly varying functions of E_k , they may be taken outside the integral. There results

$$\begin{aligned} \overline{P}_{lk} &= \frac{2t}{\hbar} g(E_k) |\mathbb{H}'_{kl}|^2 \int_{-\infty}^{\infty} \frac{\sin^2 \beta}{\beta^2} d\beta \\ &= t \left[\frac{2\pi}{\hbar} g(E_k) |\mathbb{H}'_{kl}|^2 \right] \end{aligned}$$

The related transition probability rate is

$$\overline{w}_{lk} = \frac{2\pi}{\hbar} g(E_k) |\mathbb{H}'_{kl}|^2 \quad (13.64)$$

This formula was found to have such widespread application that Fermi⁹ dubbed it "Golden Rule No. 2." It will be applied in Section 13.9 in study of atom-radiation interaction and in Chapter 14 in study of the Born approximation in the theory of scattering.

PROBLEMS

13.29 Show that if the perturbation field

$$\hat{H}' = 2\hat{\mathbb{H}}'(\mathbf{r}) \cos \omega t$$

acts on a system initially in the l th state of a very long time, and $\omega \simeq \omega_{kl}$, then the only state that will be excited is the k th state. Interpret this result in terms of the Fourier decomposition (in time) of $\hat{H}'$.

13.30 The expression in the text obtained for $\overline{P}_{lk}$, the probability of transition to a band of states, is seen to be independent of the frequency ω of the perturbing field. In what approximation is this result valid?

13.31 What does the transition probability P_{lk} (13.55) become if the perturbing field is precisely "on resonance"; that is, if $\omega = \omega_{kl}$?

⁹E. Fermi, *Nuclear Physics*, University of Chicago Press, Chicago, 1950.

- 13.32** A polarized beam of current, $J \text{ cm}^{-2}/\text{s}$, contains electrons with spins aligned with a steady magnetic field of magnitude $\mathcal{B}$ which points in the z direction. The beam propagates in the x direction. The wavefunction for an electron in the beam is

$$\psi = \frac{1}{(2\pi)^{1/2}} e^{i(kx - \omega t)} \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

A monochromatic electromagnetic field of frequency

$$\hbar\omega = 2\mu_b \mathcal{B}$$

extends over a length of beam path, L cm long. A Stern–Gerlach analyzer is in the path of the beam at a point beyond the domain of the electromagnetic field. Its orientation is such that spins aligned with $\mathcal{B}$ are not deflected from the beam. If the beam moves with the speed v , what is the current of electrons scattered out of the beam by the S-G analyzer? Assume that the interaction between electrons in the beam and the electromagnetic wave is

$$\hat{H}' = -\boldsymbol{\mu} \cdot \mathcal{B}' \cos \omega t$$

where $\mathcal{B}'$ is the amplitude of magnetic field of the wave. The component of $\mathcal{B}'$ in the z direction is small compared to $\mathcal{B}$.

- 13.33** A one-dimensional harmonic oscillator of charge to mass ratio e/m and spring constant K is in its ground state. An oscillating uniform electric field

$$\mathcal{E}(t) = 2\mathcal{E} \cos \omega_0 t, \quad \omega_0^2 = K/m$$

is applied for t seconds, parallel to the motion of the oscillator. What is the probability that the oscillator is excited to the n th state given that $(\omega_{n0} - \omega_0)t \ll 1$?

Answer

$$P_{0n} = \delta_{n,1} \left| \frac{e\mathcal{E}}{\sqrt{2}\hbar\beta} \right|^2 t^2, \quad \beta^2 \equiv \frac{m\omega_0}{\hbar}$$

13.7 ■ APPLICATION OF HARMONIC PERTURBATION THEORY

In this section we apply the transition probability formula (13.61) found above to Einstein's derivation of the Planck radiation formula (2.3) and to a brief qualitative description of the laser.

Einstein considered the equilibrium state between the walls of an enclosed cavity and the radiation field interior to the cavity. Atoms in the walls constantly exchange energy with the radiation field. The excited states of these atoms are very closely spaced and essentially comprise a continuum. Consider that the two energies E_l and E_k are representative of two states in the continuum (Fig. 13.11). Photons with energy $h\nu = E_k - E_l$ can be absorbed by atoms in the E_l state raising them to the E_k state. Atoms may decay from the E_k to the E_l state by *stimulated* or *spontaneous* emission. Stimulated emission was discussed in the preceding section. Spontaneous emission is related to the natural lifetime of the excited state and is more dependent on internal properties of the radiating system.

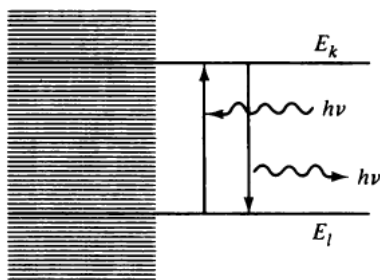
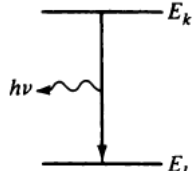


FIGURE 13.11 Energy states of closely packed atoms in the walls of a radiation cavity.

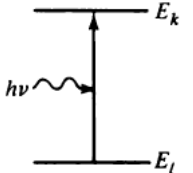
Einstein A and B Coefficients

The rate at which atoms in the E_k state decay by stimulated emission is proportional to the number of such atoms (N_k) and number of photons of frequency ν in the radiation field. This number of photons is proportional to photon energy density, $u(\nu)$. The rate at which the atoms in the E_k state decay by spontaneous emission, on the other hand, is proportional only to the number of atoms N_k in the E_k state. Thus the total transmission rate of decay of atoms with energy E_k to energies E_l is

$$W_{kl} = [A_{kl} + B_{kl}u(\nu)]N_k = w_{kl}N_k \quad (13.65)$$


The transition probability rate per atom is written w_{kl} . The proportionality constants A and B are called *Einstein A and B coefficients*.

Atoms in the E_l state can be excited to the E_k state only by absorption of a photon of frequency $(E_k - E_l)/h$. Thus the rate of elevation of atoms in the E_l state to the E_k state is

$$W_{lk} = B_{lk}u(\nu)N_l = w_{lk}N_l \quad (13.66)$$


Planck Radiation Formula

In equilibrium, these rates must equal:

$$W_{lk} = W_{kl}$$

Equivalently, we may write

$$\frac{N_l}{N_k} = \frac{A_{kl} + B_{kl}u(v)}{B_{lk}u(v)} \quad (13.67)$$

The ratio N_l/N_k may be obtained from elementary statistical mechanics. If the energy of an aggregate of atoms at the temperature T is partitioned such that N_1 of the atoms have energy E_1 , N_2 have energy E_2 , and so on, and the total energy of the aggregate is constant, then the most probable distribution of energies is given by the *Boltzmann distribution*.¹⁰ In this distribution the number of atoms N_i with energy E_i is proportional to the Boltzmann factor, $\exp(-E_i/k_B T)$, where k_B is Boltzmann's constant. It follows that the ratio N_l/N_k has the value

$$\frac{N_l}{N_k} = e^{(E_k - E_l)/k_B T} = e^{h\nu/k_B T}$$

Substituting this into (13.67) gives

$$u(v) = \frac{A_{kl}}{B_{lk}e^{h\nu/k_B T} - B_{kl}} \quad (13.68)$$

The basic structure of the Planck radiation formula (2.3) follows if $B_{lk} = B_{kl}$. This equality may be obtained from the harmonic perturbation theory developed above. To these ends we consider the interaction between a wave mode in the radiation field,

$$\mathcal{E} = \mathcal{E}_0 \cos \omega t$$

and an atom in the wall of the cavity. If $\mathbf{d}$ is the dipole moment of the atom, this interaction is

$$H' = -\mathbf{d} \cdot \mathcal{E} = -\mathcal{E}_0 \cdot \mathbf{d} \cos \omega t \quad (13.69)$$

which is identical to the perturbation (13.53). Substituting into (13.61) gives the transition probability rate for radiative absorption,

$$w_{lk} = \frac{2\pi}{\hbar^2} |\langle k | \mathcal{E}_0 \cdot \mathbf{d} | l \rangle|^2 \delta(\omega_{kl} - \omega)$$

If the electric field is isotropic ($\mathcal{E}_0$ is randomly oriented), then

$$|\langle \mathbf{d} \cdot \mathcal{E}_0 \rangle|^2 = \frac{1}{3} \langle \mathcal{E}_0^2 \rangle |\mathbf{d}_{lk}|^2$$

Furthermore, the energy density associated with this mode is

$$U(\omega) = \frac{1}{2\pi} \langle \mathcal{E}_0^2 \rangle$$

¹⁰This distribution was employed previously in Problem 2.36. For further discussion, see F. Reif, *Fundamentals of Statistical and Thermal Physics*, McGraw-Hill, New York, 1965.

It follows that

$$w_{lk} = \frac{4\pi^2}{3\hbar^2} U(\omega) |\mathbf{d}_{lk}|^2 \delta(\omega_{kl} - \omega)$$

With $u(\omega) = U(\omega)\delta(\omega_{kl} - \omega)$, (13.66) may be rewritten

$$w_{lk} = B_{lk} U(\omega) \delta(\omega_{kl} - \omega)$$

Equating this value to the preceding expression for w_{lk} gives the desired result,

$$B_{lk} = \frac{4\pi^2}{3\hbar^2} |\langle l | \mathbf{d} | k \rangle|^2 \quad (13.70)$$

The square moduli of the matrix elements of $\mathbf{d}$ obey the relation

$$\begin{aligned} |\langle l | \mathbf{d} | k \rangle|^2 &= \langle l | \mathbf{d} | k \rangle \cdot \langle k | \mathbf{d} | l \rangle \\ &= \langle k | \mathbf{d} | l \rangle \cdot \langle l | \mathbf{d} | k \rangle = |\langle k | \mathbf{d} | l \rangle|^2 \end{aligned}$$

It follows that

$$B_{lk} = B_{kl}$$

and (13.68) reduces to the desired form,

$$u(\nu) = \frac{A/B}{e^{h\nu/k_B T} - 1} \quad (13.71)$$

To obtain the ratio A/B we will use the correspondence principle. This rule stipulates that (13.71) should reduce to the classical Rayleigh–Jeans law (discussed in Section 2.2) in the limit $h \rightarrow 0$.

$$u_{RJ} = \frac{8\pi\nu^2}{c^3} k_B T$$

Expanding the exponential in (13.71) in this limit gives

$$\frac{A/B}{h\nu/k_B T} = \frac{8\pi\nu^2}{c^3} k_B T$$

from which we obtain the desired result (see Problem 13.52)

$$\frac{A}{B} = \frac{8\pi h\nu^3}{c^3}$$

Substituting this value into (13.71) gives Planck's formula, (2.3).

The Laser

The concepts of stimulated and spontaneous emission play an important role in the theory of the *laser*. A laser is a device for producing an intense beam of coherent monochromatic light. A coherent beam may be defined as follows. Two or more collinear, unidirectional, monochromatic beams of electromagnetic radiation which propagate in the same region of space, and are in phase, form a coherent beam. In 1954, C. H. Townes¹¹ conceived of a process for the generation and amplification of such coherent radiation in the microwave domain. The device was called a *maser*. The term is an acronym for the words *microwave amplification by the stimulated emission of radiation*. Shortly after, these concepts were extended to the optical region.¹² In this domain the corresponding device is called a *laser*.

Coherent Photons

The central principle in the realization of the laser is as follows. In constructing formula (13.65) for the transition rate from the E_k state to the E_l state, decay due to *stimulated* emission was taken to be proportional to the number of resonant photons present in the radiation field. Consider that a number of atoms in a gas are in the excited state E_k . Then when a photon of frequency

$$h\nu = E_k - E_l$$

falls on one of these atoms it stimulates the emission of another photon of the same frequency. These two photons, the emitted and incident, travel in phase in the same direction and combine coherently. A second atom, in the vicinity of the first and also excited to the E_k state, suddenly “sees” a duplication of resonant photons and is stimulated to emit another coherent photon, thereby adding to the intensity of the coherent radiation and amplifying it.

We found above that the matrix elements for radiative excitation, B_{lk} , and stimulated emission, B_{kl} , are equal [(13.70) et seq.]. To ensure that resonant photons stimulate decay to the E_l state more than they are absorbed in exciting the atom to the higher E_k state, the number of atoms, N_k , in the E_k state must outweigh the number of those in the lower E_l state, N_l . At any finite temperature, T , the ratio of these numbers of atoms, N_k/N_l , is given by the Boltzmann formula [preceding (13.68)].

$$\frac{N_k}{N_l} = e^{-h\nu/k_B T}$$

¹¹J. P. Gordon, H. J. Zeiger, and C. H. Townes, *Phys. Rev.* **99**, 1264 (1955).

¹²A. L. Schawlow and C. H. Townes, *Phys. Rev.* **112**, 1940 (1958). For further discussion and reference, see B. L. Lengyel, *Introduction to Laser Physics*, Wiley, New York, 1966; T. B. Melia, *An Introduction to Masers and Lasers*, Chapman & Hall, London, 1967.

Population Inversion and Optical Pumping

The number of atoms in the higher E_k state decreases exponentially with energy difference, $h\nu$. To effect a *population inversion*, so that $N_k > N_l$, an outside source must be brought into play.¹³ In *optical pumping*, N_k is increased by irradiation with light of frequency $\nu \geq (E_k - E_0)/h$, where E_0 is the ground state of the atom. Another technique for effecting a population inversion is by bombardment of electrons with energy $E > E_k - E_0$, thereby exciting atoms in the ground state to higher energy states. In a third technique, inelastic collisions between atoms in the ground state with those of a foreign gas which are in an excited state $E > E_k - E_0$ serve to populate the higher-energy states.

Consider three atomic states: E_0 , E_1 , and E_2 . The ground state is E_0 , while E_2 is a short-lived state with a lifetime of the order of 10 to 100 ns. The more stable ("metastable") E_1 state has a lifetime of the order of μs to ms. The upper E_2 state replenishes the metastable state through *spontaneous decay* [the A coefficient in (13.65)]. These randomly emitted photons comprise an incoherent radiation field. In Fig. 13.12 a scheme is depicted where the E_2 state is populated through pumping with an external source. When an atom in the metastable E_1 state decays to the E_0 state, neighboring atoms in the populated E_1 state decay, through stimulated emission to the ground state, and a coherent beam is generated.

Such a device is realized in an optical cavity of cylindrical shape with end mirrors positioned so that the cavity is tuned to the frequency mode $(E_1 - E_0)/h$. Photons propagate parallel to the axis of the tube. The radiation field is coherently amplified by a stimulated emission ($E_1 \rightarrow E_0$) and may be tapped through a small aperture on the axis, in one of the mirrors. The spontaneously emitted photons ($E_2 \rightarrow E_1$) propagate in random directions and are dissipated in collisions with the walls.

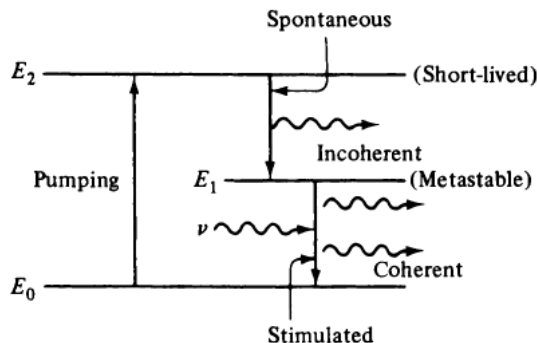


FIGURE 13.12 Schematic for the three-level laser.

¹³Including the effects of degeneracy, the number of atoms in the n th state is given by $\bar{N}_n = g_n N_n$. The condition for growth of radiation then becomes $\bar{N}_k/g_k > \bar{N}_j/g_l$, which returns the inequality stated in the text. For further discussion, see A. Yariv, *Quantum Electronics*, Wiley, New York, 1968.

PROBLEMS

- 13.34** Obtain a relation between the spontaneous emission coefficient A_{kl} and the dipole matrix element $|d_{kl}|$, analogous to (13.70).
- 13.35** The electric dipole moment of the ammonia molecule, NH_3 , has magnitude $d = 1.47 \times 10^{-18}$ esu-cm. A beam of these molecules with dipoles polarized in the $+z$ direction enters a domain of electric field of strength 1.62×10^4 V/cm, which points in the $-z$ direction. The resulting interaction between the molecules and the field gives rise to coherent radiation. What is the frequency of this radiation (Hz)? (Note: Units of voltage in cgs are statV; 1 statV = 300 V.)

Answer

$$\nu = 24 \text{ GHz}$$

- 13.36** In deriving Planck's formula (see Problem 2.36 et seq.) for the density of photons in frequency interval $\nu, \nu + d\nu$, in a radiation field in equilibrium at the temperature T , in cubic volume $V = a^3$,

$$n(\nu) d\nu = \frac{8\pi\nu^2 d\nu}{c^3} \frac{1}{e^{h\nu/k_B T} - 1}$$

one assumes the relation

$$n(\nu) d\nu = f_{\text{BE}}(\nu) g(\nu) d\nu$$

The term $g(\nu) d\nu$ is the density of available states in the said frequency interval, while f_{BE} is the *Bose-Einstein* factor, giving the average number of photons per state,

$$f_{\text{BE}}(\nu) = \frac{1}{e^{h\nu/k_B T} - 1}$$

Using the momentum-position uncertainty relation (appropriate to a three-dimensional box), derive the following expression for the *density of states* g , relevant to a radiation field:

$$g(\nu) d\nu = \frac{8\pi\nu^2 d\nu}{c^3}$$

Answer

With $p = h\nu/c$, we look at Cartesian $\mathbf{p}$ space and note the volume of a spherical shell of radius p and thickness dp .

$$\text{Volume of shell} = 4\pi p^2 dp$$

All momenta in this shell correspond to nearly the same energy—and therefore the same frequency. Owing to the uncertainty principle for a particle confined to a three-dimensional box (see Problem 5.42), the smallest volume (a cell in $\mathbf{p}$ space) that may be specified with certainty to contain a state is

$$(\Delta p)_{\min}^3 = \frac{(h/2)^3}{a^3} = \frac{(h/2)^3}{V}$$

It follows that the number of states in the dp shell is

$$\frac{\text{Volume of shell}}{\text{Volume of cell}} = \frac{4\pi p^2 dp V}{(h/2)^3}$$

Finally, we note that one counts states only in the first quadrant of $\mathbf{p}$ space insofar as photons have only positive frequencies. Furthermore, to each such $\mathbf{p}$ state there are two photon states, corresponding to two possible polarizations. This gives

$$V g(\nu) d\nu = \frac{1}{8} \times 2 \times \frac{4\pi V p^2 dp}{h^3/8} = \frac{8\pi V p^2 dp}{h^3}$$

The desired answer follows using the given relation between p and ν for a photon.

- 13.37** (a) What are the Hamiltonian $\hat{H}_0$, eigenstates and eigenvalues of a collection of harmonic oscillators with natural frequencies, $\omega_1, \omega_2, \dots$, in the number operator representation?
- (b) The Hamiltonian constructed in part (a) represents a radiation field in *second quantization*. In this representation the radiation field is viewed as a collection of harmonic oscillators. A perturbation $\hat{H}'$ is applied to the radiation field such that

$$[\hat{H}', \hat{N}_1] = \frac{i\hbar}{\tau} \hat{N}_1$$

$$\hat{N}_1 \equiv \hat{a}_1^\dagger \hat{a}_1$$

The subscript 1 denotes the frequency ω_1 . Show that $[\hat{H}_0, \hat{N}_1] = 0$. Then show that the perturbation $\hat{H}'$ represents a sink that diminishes the number of photons of frequency ω_1 exponentially with an e -folding time τ .

Answers

(a)
$$\hat{H}_0 = \sum_i \hbar \omega_i \left(\hat{a}_i^\dagger \hat{a}_i + \frac{1}{2} \right)$$

Eigenstates take the form

$$|\psi_{n_1 n_2 \dots}\rangle = |n_1, n_2, n_3, \dots\rangle$$

The number of photons in the ω_i mode is n_i . These wavefunctions have the properties

$$\hat{a}_i |\dots, n_i, \dots\rangle = \sqrt{n_i} |\dots, n_i - 1, \dots\rangle$$

$$\hat{a}_i^\dagger |\dots, n_i, \dots\rangle = \sqrt{n_i + 1} |\dots, n_i + 1, \dots\rangle$$

It follows that

$$\hat{H}_0 |n_1, n_2, \dots\rangle = \sum_i \hbar \omega_i \left(n_i + \frac{1}{2} \right) |n_1, n_2, \dots\rangle$$

(b) The average number of photons of frequency ω_1 is

$$\langle N_1 \rangle = \langle n_1, n_2, \dots | \hat{N}_1 | n_1, n_2, \dots \rangle = n_1$$

The equation of motion for $\langle N_1 \rangle$ is [recall (6.68)]

$$\begin{aligned} \frac{d\langle N_1 \rangle}{dt} &= \frac{i}{\hbar} \langle [\hat{H}, \hat{N}_1] \rangle = \frac{i}{\hbar} \langle [\hat{H}', \hat{N}_1] \rangle = -\frac{1}{\tau} \langle N_1 \rangle \\ \frac{dn_1}{dt} &= -\frac{1}{\tau} n_1 \quad n_1 = n_1(0) e^{-t/\tau} \end{aligned}$$

13.38 In Problem 13.37 we found that the Hamiltonian of an electromagnetic radiation field may be written¹⁴

$$\hat{H}_R = \sum_j \hbar \omega_j \hat{a}_j^\dagger \hat{a}_j$$

Recalling that the frequency of a photon of momentum $\hbar \mathbf{k}$ is $\omega = ck$ [see (2.28)], it follows that the operator corresponding to the total momentum of the radiation field is

$$\hat{\mathbf{P}} = \sum_j \hbar \mathbf{k}_j \hat{a}_j^\dagger \hat{a}_j$$

Here we are assuming that the field is contained in a large cubical box with perfectly reflecting walls. Boundary conditions then imply a discrete sequence of wavenumber vectors $\{\mathbf{k}_i\}$. (See Problem 2.37.) Consider that a charged particle bound to a point within the radiation field vibrates along the z axis. Its Hamiltonian is

$$\hat{H}_P = \hbar \omega_0 \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right)$$

- (a) If the oscillator and the field are uncoupled from each other, what are the Hamiltonian and eigenstates of the composite system?
- (b) Consider now that a small coupling exists between the particle and the field whose interaction energy is proportional to the scalar product of the displacement of the particle and the total momentum of the field, with coupling constant $\alpha \hbar \omega_0^2 / mc^2$, where α is the fine-structure constant. Assuming (13.61) to be appropriate, calculate the probability rate that as a result of this interaction, the oscillator emits a photon of energy $\hbar \omega_0$.

Answers

$$(a) \quad \hat{H}_{PF} = \hat{H}_P + \hat{H}_F = \hbar \omega_0 \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right) + \sum_j \hbar \omega_j \hat{a}_j^\dagger \hat{a}_j$$

$$|\psi\rangle = |\psi_P\rangle |\psi_F\rangle = |n; n_1, n_2, \dots\rangle$$

$$\hat{H}_{PF} |\psi\rangle = \left[\hbar \omega_0 \left(n + \frac{1}{2} \right) + \sum_j \hbar \omega_j n_j \right] |\psi\rangle$$

¹⁴Here we are omitting the infinite zero-point energy $\sum \frac{1}{2} \hbar \omega_j$.

The notation is such that n_j denotes the number of photons with momentum $\hbar \mathbf{k}_j$.

(b) The interaction Hamiltonian is

$$\hat{H}' = \frac{\alpha \hbar \omega_0^2}{mc^2} \frac{1}{\sqrt{2} \beta} (\hat{a} + \hat{a}^\dagger) \sum_j \hbar k_{jz} \hat{a}_j^\dagger \hat{a}_j$$

where $\beta^2 \equiv m\omega_0/\hbar$ [recall (7.26)] and k_{jz} denotes the z component of $\mathbf{k}_j$. To apply (13.61) we must first calculate the matrix element of $\hat{H}'$ between the initial state

$$|\psi_i\rangle = |n; n_1, \dots, n_k, \dots\rangle, \quad E_i = \hbar\omega_0 \left(n + \frac{1}{2}\right) + \hbar\omega_0 n_k + \sum_{j \neq k} \hbar\omega_j n_j$$

and the final state

$$\langle\psi_f| = \langle n-1; n_1, \dots, n_k+1, \dots|, \\ E_f = \hbar\omega_0 \left(n - \frac{1}{2}\right) + \hbar\omega_0 (n_k + 1) + \sum_{j \neq k} \hbar\omega_j n_j$$

Here n_k represents the number of photons with wavenumber ω_0/c . This choice of $\langle\psi_f|$ guarantees conservation of energy in the transition, as prescribed by the delta-function factor $\delta(E_i - E_f)$ in (13.61). Since $|\psi_i\rangle$ (as well as $\langle\psi_f|$) is an eigenvector of $\sum \hat{a}_j^\dagger \hat{a}_j$, it may be brought through the field component of $\hat{H}'$. When completed with the bra vector $\langle\psi_f|$, the inner product of the field component factors of the wavefunctions vanishes and we conclude that for the given perturbation,

$$\langle\psi_f|\hat{H}'|\psi_i\rangle = H'_{if} = 0$$

The hypothetical field-particle coupling does not induce a transition of the state of the harmonic oscillator.

13.8 ■ SELECTIVE PERTURBATIONS IN TIME

The Adiabatic Theorem

Let us now consider the case that $\hat{H}'$ is *adiabatically* turned on, which means that $\hat{H}'$ changes slowly in time (Fig. 13.13). Consequently, at any instant of time, the Hamiltonian may be treated as constant and an approximate solution may be obtained by regarding the Schrödinger equation as time-independent. This will be shown below.

The slowly changing quality of $\hat{H}'$ may be incorporated into the analysis through a parts integration of (13.51).

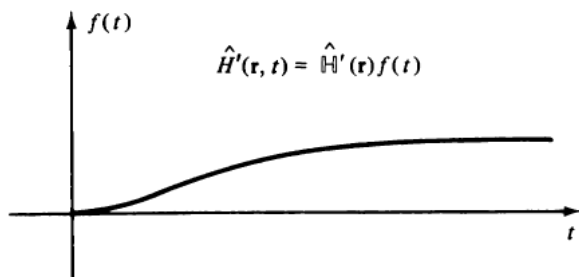


FIGURE 13.13 Adiabatic perturbation. The rise time, ω^{-1} , of the perturbation obeys the inequality

$$\omega \ll \omega_{kl}$$

where ω_{kl} are the natural frequencies of the unperturbed system.

$$\begin{aligned} c_k(t) &= \frac{1}{i\hbar} \int_{-\infty}^t H'_{kl}(t') e^{i\omega_{kl}t'} dt' = -\frac{1}{\hbar\omega_{kl}} \int_{-\infty}^t H'_{kl}(t') \frac{\partial}{\partial t'} e^{i\omega_{kl}t'} dt' \quad (13.72) \\ &= -\frac{1}{\hbar\omega_{kl}} \left\{ H'_{kl}(t) e^{i\omega_{kl}t} - \int_{-\infty}^t e^{i\omega_{kl}t'} \frac{\partial}{\partial t'} H'_{kl}(t') dt' \right\} \end{aligned}$$

where we have set $H'_{kl}(t) \equiv \mathbb{H}'_{kl} f(t)$. If $\hat{H}'(t)$ is slowly varying, the second term is small compared to the first and $c_k(t)$ is well approximated by

$$c_k(t) \simeq -\frac{1}{\hbar\omega_{kl}} H'_{kl}(t) e^{i\omega_{kl}t} = -\frac{\langle k | \hat{H}' | l \rangle e^{i\omega_{kl}t}}{E_k^{(0)} - E_l^{(0)}} \quad (13.73)$$

Let us recall that the analysis leading to (13.49) presumes that the system is in the stationary state ψ_l at $t < 0$. To terms of first order in the perturbation H' , the series (13.42) then appears as

$$\psi(\mathbf{r}, t) = \psi_l + \sum_{k \neq l} c_k \psi_k$$

As the perturbation is slowly varying, we write $\psi(\mathbf{r}, t) \simeq \exp(-i\omega_l t) \varphi(\mathbf{r})$. Substituting this expression into the preceding and employing (13.73) gives

$$\varphi(\mathbf{r}) = \varphi_l + \sum_{k \neq l} \frac{H'_{kl} \varphi_k}{E_l^{(0)} - E_k^{(0)}} \quad (13.74)$$

This is precisely the first-order result which stationary (time-independent) perturbation theory gives, (13.10). But the solution (13.10) represents an eigenfunction of the new Hamiltonian, which in the present case is $\hat{H} = \hat{H}_0 + \hat{H}'(t)$. That is, the wavefunction (13.74) represents the l th eigenstate of $\hat{H}_0 + \hat{H}'(t)$, to first order in H' . [Note that we are writing $\hat{H}'(t)$ for the operator $\hat{H}'$ evaluated at the specific

time t .] Since the system was originally in the l th state of $\hat{H}_0$, we may conclude the following. A system originally in the l th state of an unperturbed Hamiltonian will at the end of an adiabatic perturbation be found in the l th state of the new Hamiltonian. One says that the system “remains in the l th state.”

Having established that φ as given by (13.74) is an eigenstate of the new Hamiltonian, it follows that expectation of energy in this state is the same as the eigenenergy of the state. The expectation of energy to first order in the perturbation Hamiltonian is easily calculated with the explicit form of φ as given by (13.74). There results

$$E_l = E_l^{(0)} + H'_{ll}(t) \quad (13.75)$$

This is the l th eigenenergy of the Hamiltonian $\hat{H}_0 + \hat{H}'(t)$. So under an adiabatic perturbation, a system originally in the l th eigenstate of an unperturbed Hamiltonian will at time t be found in the l th eigenstate of the new Hamiltonian with the l th eigenenergy of the new Hamiltonian. With $\varphi_l(t)$ written for the time-independent wavefunction evaluated at the time t , we may write that under an adiabatic perturbation

$$\begin{aligned} \varphi_l(0) &\longrightarrow \varphi_l(t) \\ E_l(0) &\longrightarrow E_l(t) \end{aligned} \quad (13.76a)$$

where

$$[\hat{H}_0 + \hat{H}'(t)]\varphi_l(t) = E_l(t)\varphi_l(t) \quad (13.76b)$$

This equation is *not* a time-dependent equation. One first evaluates $\hat{H}'(t)$ and then finds the solution $\varphi_l(t)$. These results constitute the *adiabatic theorem*, first proved by Born and Fock in 1928.¹⁵

As an example of the use of this theorem, consider that a particle of mass m is in a one-dimensional box of edge length a . The width of the box is slowly increased to αa over a long interval lasting t seconds, where $\alpha > 1$. Let us calculate the new wavefunction and new energy of the system given that the particle initially was in the ground state. The adiabatic theorem tells us that the new state is the ground state of the new Hamiltonian.

$$\varphi_1 \rightarrow \varphi_1(t) = \sqrt{\frac{2}{\alpha a}} \sin\left(\frac{\pi x}{\alpha a}\right)$$

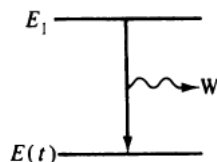
Here, as above, we are writing $\varphi(t)$ for the time-independent wavefunction evaluated at the time t . The new energy is the ground state of the new Hamiltonian.

$$E_1 \rightarrow E_1(t) = \frac{h^2}{8m(\alpha a)^2}$$

¹⁵M. Born and V. Fock, *Z. Physik*, **51**, 165 (1928).

Note that in slowly expanding (as in the classical case), the particle loses energy to the receding walls in slowing down. The amount of work absorbed by the walls in the present example is

$$W = E_1 - E(t) = \frac{\hbar^2}{8ma^2} \left(1 - \frac{1}{\alpha^2}\right)$$



Domain of Validity

Our conclusions regarding an adiabatic perturbation rest on neglecting the integral term in (13.72). Let us obtain a quantitative criterion which allows this term to be discarded. Let the perturbation be gradually turned on at $t = 0$. If $\hat{H}'$ changes slowly, then $\partial \hat{H}' / \partial t$ may be taken outside the integral term in (13.72) and we obtain

$$\left| \int_0^t e^{i\omega_{kl}t'} \frac{\partial H'_{kl}}{\partial t'} dt' \right| \simeq \left| \frac{\partial H'_{kl}}{\partial t} \right| \left| \int_0^t e^{i\omega_{kl}t'} dt' \right| \simeq \left| \frac{2}{\omega_{kl}} \frac{\partial H'_{kl}}{\partial t} \right| \left| \sin \left(\frac{\omega_{kl}t}{2} \right) \right|$$

It follows that the nonintegral term in (13.72) dominates provided that

$$|\omega_{kl} H'_{kl}| \gg 2 \left| \frac{\partial H'_{kl}}{\partial t} \right| \quad (13.77)$$

Thus a perturbation that undergoes a small fractional change in a typical period of the unperturbed system may be termed adiabatic.

Transition Probability

We have found that in the adiabatic limit, neglecting the integrated term in (13.72) leads to the system remaining in the initial state of the unperturbed Hamiltonian. What is the probability in this same limit that there is a transition out of the initial state? Suppose, for example, that an adiabatic perturbation is turned on at $t = 0$ and that the unperturbed system is originally in the l th state of the unperturbed Hamiltonian. The probability for a transition from the l to the k state is given by $|c_k|^2$. With (13.72) in the adiabatic limit, we obtain

$$|c_k|^2 = \frac{1}{\hbar^2(\omega_{kl})^2} \left[|H'_{kl}|^2 + \frac{4}{(\omega_{kl})^2} \left| \frac{\partial}{\partial t} H'_{kl} \right|^2 \sin^2 \left(\frac{\omega_{kl}t}{2} \right) - \frac{2}{\omega_{kl}} \frac{\partial (H'_{kl})^2}{\partial t} \sin \left(\frac{\omega_{kl}t}{2} \right) \cos \left(\frac{\omega_{kl}t}{2} \right) \right]$$

Over a long time interval, the last term averages to zero, whereas the $\sin^2$ term in the middle expression averages to $\frac{1}{2}$. As we have found above, the first term represents the probability that there is no transition out of the l th state. It follows that the probability that there is a transition out of the initial l th state to some k th state ($k \neq l$) in the adiabatic limit is

$$P_{lk} = \frac{2}{\hbar^2(\omega_{kl})^4} \left| \frac{\partial}{\partial t} H'_{kl} \right|^2 \quad (13.78)$$

Once more we find that there is vanishingly small probability of transition out of the l th state for sufficiently slowly changing perturbation.

An instructive laboratory example of the adiabatic theorem is given by elastic collision of molecules in a gas. During the collision, electron states are acted upon by intermolecular forces. This collisional interaction varies roughly as the intermolecular velocity, which, typically, is far smaller than mean atomic electron speeds. Thus, collisional forces which act on atomic states may be viewed as being adiabatically switched on. Consistent with this observation, one finds that whereas final rotational and vibrational molecular states are altered in the collision, atomic states suffer minimal change.¹⁶

Sudden Perturbation

The next type of perturbation problem we wish to examine involves a sudden change in a parameter in $\hat{H}_0$. For example, suppose that the spring constant of a simple harmonic oscillator is suddenly doubled. If the oscillator is in its ground state before the perturbation, in what state is it after perturbation? For such problems it is presumed that the eigenstates of both Hamiltonians (i.e., before and after perturbation) are known. This, together with the assumption of an instantaneous change of $\hat{H}_0$, are basic to the *sudden approximation*.

Let us call the Hamiltonian before the change in parameter, $\hat{H}$, and the Hamiltonian after the change in parameter, $\hat{H}'$, so that

$$\begin{aligned} \hat{H}\varphi_n &= E_n\varphi_n & t < 0 \\ \hat{H}'\varphi_n' &= E_n'\varphi_n' & t \geq 0 \end{aligned}$$

(Note that $\hat{H}'$ is now a total Hamiltonian.) Initially, the system is in an eigenstate φ_l of $\hat{H}$.

$$\psi'(\mathbf{r}, 0) = \varphi_l(\mathbf{r})$$

At later times the system is in a superposition state of $\hat{H}'$.

$$\psi'(\mathbf{r}, t) = \sum_n c_n \varphi_n' e^{-i\omega_n' t}$$

¹⁶For further discussion, see R. D. Levine and R. B. Bernstein, *Molecular Reaction Dynamics and Chemical Reactivity*, Oxford, New York, 1987.

Equating this function to its initial value at $t = 0$ gives

$$\varphi_l = \sum_n c_n \varphi_n'$$

so that

$$c_s = \langle \varphi_s' | \varphi_l \rangle$$

The probability that the sudden change from $\hat{H}$ to $\hat{H}'$ causes a transition from the l th state of $\hat{H}$ to the s th state of $\hat{H}'$ is

$$P_{ls} = |c_s|^2 = |\langle \varphi_s' | \varphi_l \rangle|^2 \quad (13.79)$$

A criterion for the validity of this approximation scheme may be drawn from the preceding relation. As described above, let the sudden change in Hamiltonian occur at $t = 0$. For the relation (13.79) to have meaning it is necessary that the initial wavefunction $\varphi_l(\mathbf{r})$: (1) maintain its form at $t = 0^+$ and (2) lie in the Hilbert space spanned by the eigenstates of the new Hamiltonian. Thus, for a particle confined to a rigid-walled box, we may conclude that the approximation is inappropriate to sudden compression of the box but is appropriate to sudden expansion of the box.

To further demonstrate this approximation, let us apply the formalism to the latter case. Thus we consider a particle of mass m confined to a one-dimensional box which suffers a sudden expansion. With no loss in generality we take the box to be of unit width ($a = 1$). The particle is in the energy eigenstate φ_l at $t < 0$. At $t = 0$ the box is suddenly expanded to the edge length $\alpha > 1$ (recall Problem 5.11). With the preceding formalism we may establish the somewhat surprising result that although no work is done on the particle in the expansion, energy is not necessarily conserved.

The new Hamiltonian $\hat{H}'$ (after expansion) is given by (4.23) with a replaced by α . New basis wavefunctions are the eigenfunctions

$$\varphi_s' = \sqrt{\frac{2}{\alpha}} \sin\left(\frac{s\pi x}{\alpha}\right)$$

The probability for transition is given by

$$\begin{aligned} \sqrt{P_{ls}} &= \langle \varphi_l | \varphi_s' \rangle = \frac{2}{\sqrt{\alpha}} \int_0^1 \sin l\pi x \sin\left(\frac{s\pi x}{\alpha}\right) dx \\ &= \frac{1}{\sqrt{\alpha}} \left\{ \frac{\sin \pi[l - (s/\alpha)]}{\pi[l - (s/\alpha)]} - \frac{\sin \pi[l + (s/\alpha)]}{\pi[l + (s/\alpha)]} \right\} \end{aligned} \quad (13.80)$$

Note that the integral in the preceding expression goes over $(0, 1)$ and not $(0, \alpha)$ because $\varphi_l(x)$ is zero over $(1, \alpha)$.

The transition which conserves energy occurs for the s value $\bar{s} = \alpha l$. With $A \equiv \hbar^2 \pi^2 / 2m$, we obtain

$$E'_s \Big|_{s=\bar{s}} = A \frac{s^2}{\alpha^2} \Big|_{s=\bar{s}} = A l^2 = E_l \quad (13.81)$$

At this value of s , (13.80) gives

$$P_{l\bar{s}} = \frac{1}{\alpha} < 1 \quad (13.81a)$$

Thus, as stated, it is not certain that the transition conserves energy. However, it is simply shown that expectation of energy is conserved. To show this, we note that at $t = 0^+$ the wavefunction for the system at hand is given by

$$\begin{aligned} \bar{\varphi}_l &= \varphi_l & 0 \leq x \leq 1 \\ \bar{\varphi}_l &= 0 & 1 < x \leq \alpha \end{aligned}$$

It follows that

$$E_l = \langle E \rangle = \langle \varphi_l | \hat{H} | \varphi_l \rangle = \langle \bar{\varphi}_l | \hat{H}' | \bar{\varphi}_l \rangle \quad (13.82)$$

Note that the third term is defined over $(0, 1)$ and is the value of $\langle E \rangle$ at $t < 0$. The last term is defined over $(0, \alpha)$ and is appropriate to $t = 0^+$. We conclude that $\langle E \rangle$ is conserved in the expansion. Further developing (13.82) gives

$$\begin{aligned} E_l &= \sum_k \langle \bar{\varphi}_l | \hat{H}' | \varphi'_k \rangle \langle \varphi'_k | \bar{\varphi}_l \rangle \\ E_l &= \sum_k P_{lk} E'_k \end{aligned} \quad (13.83)$$

or, equivalently,

$$l^2 = \frac{1}{\alpha^2} \sum_{k=1}^{\infty} P_{lk} k^2 \quad (13.84)$$

with P_{lk} given by (13.80). This result is valid for arbitrary $\alpha > 1$, i.e., for irrational as well as rational values. In particular, note that if $k/\alpha \neq l$ for all k , then no direct transition conserves energy. However, with (13.83), expectation of energy is still conserved.

Expressions obtained above for the transition probability in various limiting cases are listed in Table 13.2.

PROBLEMS

13.39 A neutron in a rigid spherical well of radius $a = 0.1 \text{ \AA}$ is in the ground state. The radius of the well is slowly decreased to $0.9a$.

TABLE 13.2 Transition probabilities for time-dependent perturbations

1. $\hat{H}'$ is separable. $\hat{H}'(\mathbf{r}, t) = \mathbb{H}'(\mathbf{r})f(t)$

$$P_{ik} = \frac{|\mathbb{H}'_{kl}|^2}{\hbar^2} \left| \int_{-\infty}^{\infty} e^{i\omega_{kl}t'} f(t') dt' \right|^2$$

Harmonic perturbation, $f = 2 \cos \omega t$,
turned on at $t = 0$:

$$P_{lk} = \frac{|\mathbb{H}'_{kl}|^2}{[\hbar(\omega_{kl} - \omega)/2]^2} \sin^2[(\omega_{kl} - \omega)t/2]$$

DC perturbation ($\omega = 0$, $f = 1$)
turned on at $t = 0$:

$$P_{lk} = \frac{|\mathbb{H}'_{kl}|^2 \sin^2(\omega_{kl}t/2)}{(\hbar\omega_{kl}/2)^2}$$

Long-time or high-frequency limit:

$$P_{lk} = \frac{2\pi t |\mathbb{H}'_{kl}|^2}{\hbar^2} \delta(\omega_{kl} - \omega)$$

$$w_{kl} = \frac{2\pi |\mathbb{H}'_{kl}|^2}{\hbar^2} \delta(\omega_{kl} - \omega)$$

Short-time or low-frequency limit:

$$P_{lk} = \frac{t^2 |\mathbb{H}'_{kl}|^2}{\hbar^2}$$

$$w_{kl} = \frac{t |\mathbb{H}'_{kl}|^2}{\hbar^2}$$

Probability for transition to a band centered at E_k with $g|\mathbb{H}'|$ slowly varying in energy:

$$\bar{P}_{lk} = \frac{2\pi t}{\hbar} g(E_k) |\mathbb{H}'_{kl}|^2$$

$$\bar{w}_{kl} = \frac{2\pi}{\hbar} g(E_k) |\mathbb{H}'_{kl}|^2$$

2. $\hat{H}' = \hat{H}'(\mathbf{r}, t)$ is slowly changing

Probability for transition out of the initial l state:

$$P_{lk} \simeq \frac{2}{\hbar^2(\omega_{kl})^4} \left| \frac{\partial}{\partial t} H'_{kl} \right|^2$$

Adiabatic theorem:

$$P_{kl} \simeq 0 \quad (k \neq l)$$

3. $\hat{H}$ changes suddenly to $\hat{H}'$

Eigenstates of both $\hat{H}$ and $\hat{H}'$ are known.

$$P_{lk} = |\langle \varphi'_k | \varphi_l \rangle|^2$$

(a) What is the energy and wavefunction of the neutron after the decrease in the well radius? (For normalization, see Table 10.2.)

(b) How much work (in eV) is performed during the compression of the well?

13.40 A collection of $N_0 = 10^{13}$ independent electrons have spins polarized parallel to a uniform magnetic field that points in the z direction, of magnitude $\mathcal{B}_0$. A perturbation field is applied in the x direction of magnitude

$$\mathcal{B}'(t) = 10^{-3} \mathcal{B}_0 (1 - e^{-t/\tau}) \quad (t \geq 0)$$

- (a) Obtain a criterion involving τ which ensures that the perturbation is adiabatic.
 (b) Given that $\Omega\tau = 10^2$ and $\mathcal{B}_0 = 10^4$ gauss, estimate the number of electrons ΔN that are thrown out of the ground spin state at $t = 10\tau$.

Answers

- (a) $\tau \gg \Omega^{-1} = mc/e\mathcal{B}_0$
 (b) A rough estimate is obtained from (13.78).

$$\Delta N \simeq 2N_0 \left| \frac{10^{-3}\mu_B\mathcal{B}_0}{\hbar\Omega} \right|^2 \left| \frac{e^{-t/\tau}}{\Omega\tau} \right|^2$$

- 13.41** A one-dimensional harmonic oscillator has its spring constant suddenly reduced by a factor of $\frac{1}{2}$. The oscillator is initially in its ground state. Show that the probability that the oscillator remains in the ground state is $P = 0.986$.
13.42 A particle of mass m in a one-dimensional box of width a is in the third excited state (i.e., $n = 3$). The width of the box is suddenly doubled. What is the probability that the particle drops to the ground state?
13.43 A one-dimensional harmonic oscillator in the ground state is acted upon by a uniform electric field

$$\mathcal{E}(t) = \frac{\mathcal{E}_0}{\sqrt{\pi}} \exp \left[-\left(\frac{t}{\tau} \right)^2 \right]$$

switched on at $t = -\infty$. The field is parallel to the axis of the oscillator. What is the probability that the oscillator suffers a transition to its first excited state at $t = +\infty$ in the limits: (a) $\omega_0\tau \gg 1$, (b) $\omega_0\tau \approx 1$. For case (b), are any other transitions possible? [Hint: See Problem 13.27.]

- 13.44** Radioactive tritium, H^3 , decays to light helium, (He^{3+}), with the emission of an electron. (This electron quickly leaves the atoms and may be ignored in the following calculation.) The effect of the β decay is to change the nuclear charge at $t = 0$ without effecting any change in the orbital electron. If the atom is initially in the ground state, what is the probability that the He^+ ion is left in the ground state after the decay?
13.45 A hydrogen atom in the ground state is placed in a uniform electric field in the z direction,

$$\mathcal{E} = \mathcal{E}_0 e^{-t/\tau}$$

which is turned on at $t = 0$. What is the probability that the atom is excited to the $2P$ state at $t \gg \tau$?

- 13.46** The perturbation

$$H' = \frac{A}{\tau\sqrt{\pi}} e^{i(\pi x/a)} e^{-t^2/\tau^2}$$

is applied to a particle of mass m in a one-dimensional box of width a at $t = -\infty$. At this time the particle is in the ground state. If $\hbar/\tau \ll E_1$, in what state is it most probable that the particle will be at $t = +\infty$?

13.47 An electron in a one-dimensional potential well

$$V = \frac{1}{2}Kx^2$$

is immersed in a constant, uniform electric field of magnitude $\mathcal{E}$ which points in the x direction. The corresponding perturbation to the Hamiltonian is

$$H' = e\mathcal{E}x$$

(a) Find the exact eigenenergies of the total Hamiltonian,

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}Kx^2 + e\mathcal{E}x$$

(see Problem 7.16). Discuss your findings with respect to the corresponding classical motion.

(b) Show that the first-order corrections to the energy vanish. Then calculate the second-order corrections. Show that these agree with your answer to part (a), so that the second-order corrections give the complete solution for this problem.

Answers

(a) Setting

$$\chi \equiv \frac{e\mathcal{E}}{K}$$

together with the transformation of variables

$$x' \equiv x + \chi$$

permits $\hat{H}$ to be rewritten

$$\begin{aligned}\hat{H} &= \frac{p^2}{2m} + \frac{1}{2}K(x^2 + 2\chi x) \\ &= \frac{p^2}{2m} + \frac{1}{2}Kx'^2 - \frac{1}{2}K\chi^2 \\ &= \hat{H}_0 - \mathbb{E}\end{aligned}$$

Since

$$\mathbb{E} \equiv \frac{1}{2}K\chi^2 = \frac{e^2\mathcal{E}^2}{2K}$$

is constant, the eigenenergies of $\hat{H}$ are simply

$$E_n = E_n^0 - \mathbb{E} = \hbar\omega_0\left(n + \frac{1}{2}\right) - \frac{e^2\mathcal{E}^2}{2K}$$

All levels are equally depressed by the constant energy $\mathbb{E}$. The new wavefunctions are

$$\varphi_n = \varphi_n(x') = \varphi_n(x + \chi)$$

The center of symmetry of these wavefunctions is at $x = -\chi$.

In the corresponding classical problem, the potential of the electron in the presence of the uniform electric field is

$$V = \frac{1}{2}K(x + \chi)^2 - \mathbb{E}$$

This parabola is congruent to the original potential $Kx^2/2$. The new equilibrium at $x = -\chi$ occurs where the electric force is balanced by the spring force. This new potential minimum is lower than the original minimum by the amount $\mathbb{E}$. (Work must be done to move the electron, quasi-statically, from $x = -\chi$ to $x = 0$.) The classical analog of the quantum mechanical problem considered above involves an electric field that is very slowly (adiabatically) turned on. If the energy of the oscillator initially is $E^{(0)}$, what is it after the electric field is established? In classical mechanics, for such adiabatically changing harmonic motion, the ratio A/ω is constant (it is an *adiabatic invariant*). The amplitude of oscillation is A . Since the new potential well is congruent to the old well, ω is the same for both wells. This means that the amplitude of oscillation must also be preserved (during the adiabatic change). Owing again to congruency of the parabolas, this is ensured (only) if the distance between $E^{(0)}$ and the bottom of the new well is the same as the distance between $E^{(0)}$ and the bottom of the original well. That is, if

$$E^{(0)} = E + \mathbb{E}$$

It follows that the new energy of the oscillator

$$E = E^{(0)} - \mathbb{E}$$

is depressed from the initial value, $E^{(0)}$, by the amount $\mathbb{E}$. This is identical to the quantum mechanical result (Fig. 13.14).

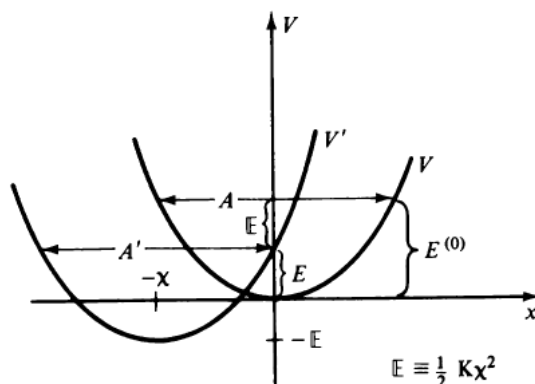


FIGURE 13.14 The classical adiabatic change from a parabolic potential, V , to a congruent ($\omega_0 = \omega'_0$) parabolic potential, V' , with a new center of symmetry, preserves amplitude A . We see that the amplitude of oscillation is preserved, $A = A'$, provided that $E = E^{(0)} - \mathbb{E}$. This result for the new energy of oscillation, E , is identical to the quantum mechanical result. (See Problem 13.47.)

(b) The perturbation Hamiltonian may be rewritten

$$\hat{H} = \frac{e\mathcal{E}}{\sqrt{2}\beta}(\hat{a}^\dagger + \hat{a}) \equiv \mathcal{E}'(\hat{a}^\dagger + \hat{a})$$

$$\beta^2 \equiv \frac{m\omega_0}{\hbar}, \quad \mathcal{E}' \equiv \frac{e\mathcal{E}}{\sqrt{2}\beta}$$

It follows immediately that

$$\langle n | \hat{H}' | n \rangle = 0$$

so there is no first-order correction to $E_n^{(0)}$. To calculate the second-order corrections, we must evaluate the off-diagonal matrix elements of $\hat{H}'$.

$$\begin{aligned} \langle n | \hat{H}' | l \rangle &= \mathcal{E}' \langle n | \hat{a}^\dagger + \hat{a} | l \rangle \\ &= \mathcal{E}'(\sqrt{l+1} \delta_{n,l+1} + \sqrt{l} \delta_{n,l-1}) \end{aligned}$$

Substituting this expression into (13.14) gives the desired result.

13.48 A system with discrete energy states and Hamiltonian $\hat{H}_0$ has the density operator $\hat{\rho}_0$, which is diagonal. Furthermore, $[\hat{\rho}_0, \hat{H}_0] = 0$, so $\hat{\rho}_0$ is constant in time. Show that after a perturbation $\hat{H}'$ is applied, the diagonal elements of $\hat{\rho}$ change according to the *Pauli equation*

$$\frac{\partial \rho_{nn}}{\partial t} = \sum_k w_{nk}(\rho_{kk} - \rho_{nn})$$

The transition rates w_{nk} are given by (13.63).

$$w_{nk} = \lim_{\Delta t \rightarrow 0} \frac{\Delta t}{\hbar^2} |H'_{nk}|^2$$

The density operator was discussed in Section 11.11.

Answer

For a short-time interval after the perturbation $\hat{H}'$ is applied, we may expand $\hat{\rho}$ to obtain

$$\hat{\rho}(\Delta t) - \hat{\rho}(0) = \left(\frac{\partial \hat{\rho}}{\partial t} \right)_0 \Delta t + \left(\frac{\partial^2 \hat{\rho}}{\partial t^2} \right)_0 \frac{(\Delta t)^2}{2} + \dots$$

Using the equation of motion (11.122) for $\hat{\rho}$ permits the last equation to be written, with $\hat{\rho}(0) = \hat{\rho}_0$,

$$\frac{\partial \hat{\rho}}{\partial t} = \lim_{\Delta t \rightarrow 0} \left[\frac{\hat{\rho}(\Delta t) - \hat{\rho}_0}{\Delta t} \right] = \lim_{\Delta t \rightarrow 0} \left\{ \frac{1}{i\hbar} [\hat{H}, \hat{\rho}]_0 - \frac{\Delta t}{2\hbar^2} [\hat{H}, [\hat{H}, \hat{\rho}]_0] \right\}$$

The diagonal form of $\hat{\rho}_0$ leads to the following properties:

$$\begin{aligned} \langle n | [\hat{H}_0 + \hat{H}', \hat{\rho}_0] | n \rangle &= 0 \\ \langle n | [\hat{H}_0, [\hat{H}', \hat{\rho}_0]] | n \rangle &= 0 \end{aligned}$$

Forming the diagonal elements of $\partial \hat{\rho} / \partial t$ then gives

$$\frac{\partial \rho_{nn}}{\partial t} = - \lim_{\Delta t \rightarrow 0} \frac{\Delta t}{2\hbar^2} \langle n | [\hat{H}', [\hat{H}', \hat{\rho}_0]] | n \rangle$$

The diagonal element on the right-hand side reduces to

$$\langle n | [\hat{H}', [\hat{H}', \hat{\rho}_0]] | n \rangle = 2 \sum_k \{ |H'_{nk}|^2 \rho_{nn}(0) - |H'_{nk}|^2 \rho_{kk}(0) \}$$

which when substituted into the preceding equation gives the desired result.

13.9 ■ ATOM-RADIATION INTERACTION

Our first description of the interaction between an atom and a radiation field was given in terms of Einstein's derivation of the Planck radiation law (Section 13.7). We now wish to present a Hamiltonian formulation of this problem. The Hamiltonian of an electron in an electromagnetic field with vector potential $\mathbf{A}$ (previously introduced in Section 10.4) and electric potential $V(\mathbf{r})$ is given by

$$\hat{H} = \frac{1}{2m} \left[\hat{\mathbf{p}} - \frac{e}{c} \mathbf{A}(\mathbf{r}, t) \right]^2 + V(\mathbf{r}) \quad (13.85)$$

In the full quantum electrodynamic analysis of this problem, electrodynamic fields are quantized. The present analysis is termed a "semiclassical" description in that the vector potential $\mathbf{A}$ in (13.85) is taken as a classical field.¹⁷

Expanding the kinetic energy term in (13.85) gives

$$\begin{aligned} \hat{H} &= \hat{H}^{(0)} + \hat{H}' + \hat{H}'' \\ \hat{H}^{(0)} &\equiv \frac{\hat{p}^2}{2m} + V; \quad \hat{H}' = -\frac{e}{mc} [\hat{\mathbf{p}} \cdot \mathbf{A} + \mathbf{A} \cdot \hat{\mathbf{p}}] \\ \hat{H}'' &= \frac{e^2}{2mc^2} A^2 \end{aligned} \quad (13.86)$$

We neglect $\hat{H}''$ and consider $\hat{H}'$ a perturbation term to the unperturbed atomic Hamiltonian $\hat{H}^{(0)}$.

The matrix element of $\hat{H}'$ between initial $|n\rangle$ and final $|n'\rangle$ states of $\hat{H}^{(0)}$ is given by

$$\langle n' | \hat{H}' | n \rangle = -\frac{e}{mc} \int \psi_{n'}^* (\hat{\mathbf{p}} \cdot \mathbf{A} + \mathbf{A} \cdot \hat{\mathbf{p}}) \psi_n d\mathbf{r} \quad (13.87)$$

Note that the index n denotes a quantum state, not the principal quantum number.

¹⁷For a more complete quantum analysis, see E. G. Harris, *A Pedestrian Approach to Quantum Field Theory*, Wiley, New York, 1972; A. S. Davydov, *Quantum Mechanics*, 2d ed., Pergamon, Elmsford, N.Y., 1973.

The first term in this integral may be written

$$\hat{\mathbf{p}} \cdot \mathbf{A} \psi_n = \frac{\hbar}{i} \nabla \cdot (\mathbf{A} \psi_n) = \frac{\hbar}{i} [\psi_n (\nabla \cdot \mathbf{A}) + (\mathbf{A} \cdot \nabla) \psi_n] \quad (13.88)$$

The gradient and curl qualities of the potentials which enter electrodynamics permit certain gauge conditions to be imposed. In the so-called *Coulomb gauge* one sets

$$\nabla \cdot \mathbf{A} = 0$$

so that (13.88) reduces to

$$\hat{\mathbf{p}} \cdot \mathbf{A} \psi_n = \frac{\hbar}{i} \mathbf{A} \cdot \nabla \psi_n$$

Thus we may write (13.87) in the equivalent forms

$$\langle n' | H' | n \rangle = -\frac{e}{mc} \int \psi_n^* \mathbf{A} \cdot \hat{\mathbf{p}} \psi_n d\mathbf{r} \quad (13.89a)$$

$$\langle n' | H' | n \rangle = -\frac{e}{mc} \int \psi_n^* \hat{\mathbf{p}} \cdot \mathbf{A} \psi_n d\mathbf{r} \quad (13.89b)$$

A revealing insight into this analysis is gained if one associates the vector potential $\mathbf{A}$ with the wavefunction of a photon of energy $\hbar\omega$. This permits the preceding two matrix elements to be symbolically written

$$\langle n' | \hat{H}' | n \rangle \rightarrow -\frac{e}{mc} \langle n' | \hbar\omega | \hat{\mathbf{p}} | n \rangle \quad (\text{emission}) \quad (13.90a)$$

$$\langle n' | \hat{H}' | n \rangle \rightarrow -\frac{e}{mc} \langle n' | \hat{\mathbf{p}} | \hbar\omega; n \rangle \quad (\text{absorption}) \quad (13.90b)$$

Thus, (13.90a) corresponds to the case where a photon is emitted by the system, whereas (13.90b) corresponds to the case of the absorption of a photon. Since these two integrals are equal, we conclude that probabilities of emission and absorption, at the same frequency, are equal. This is an example of the principle of *microscopic reversibility*.

To obtain a more explicit representation of the matrix elements (13.89), we consider the atom in interaction with a plane electromagnetic wave whose vector potential is of the form

$$\mathbf{A}(\mathbf{r}, t) = \mathbf{a} A_0 \cos(\mathbf{k} \cdot \mathbf{r} - \omega t) \quad (13.91)$$

where $\mathbf{a}$ is a unit polarization vector ($|\mathbf{a}|^2 = 1$) which is normal to the propagation vector $\mathbf{k}$. We wish to construct the amplitude A_0 so that the corresponding wave carries one photon per unit volume. The time-averaged energy density carried in

a plane electromagnetic wave is

$$\langle U \rangle = \frac{1}{4\pi} \langle \mathcal{B}^2 \rangle = \frac{1}{4\pi} \langle \mathcal{E}^2 \rangle \quad (13.92)$$

With

$$\mathcal{B} = \nabla \times \mathbf{A} \quad (13.93)$$

and given the form (13.91), we find that

$$\langle U \rangle = \frac{1}{8\pi} k^2 A_0^2 \quad (13.94)$$

For one photon per unit volume we set

$$\frac{k^2 A_0^2}{8\pi} = \frac{\hbar\omega}{V} \quad (13.95)$$

which, with $\omega = ck$, gives

$$A_0^2 = \frac{8\pi\hbar c^2}{\omega V} \quad (13.96)$$

Henceforth we set the volume $V = 1$.

In what follows, we will employ Fermi's golden rule (13.64). In formulating this relation it was found that for the time-dependent perturbation, $\cos \omega t$, the $\exp(-i\omega t)$ term was responsible for resonant absorption and the $\exp(+i\omega t)$ was responsible for decay. For consistent application of Fermi's rule, these observations must be incorporated into the present analysis. Thus, we first write (13.91) in the form

$$\mathbf{A} = \frac{A_0}{2} \mathbf{a} [e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} + e^{-i(\mathbf{k} \cdot \mathbf{r} - \omega t)}] \quad (13.97)$$

which permits the identification

$$\mathbf{A}_{\pm} = \frac{A_0}{2} \mathbf{a} e^{\pm i(\mathbf{k} \cdot \mathbf{r} - \omega t)} \quad (13.98)$$

where $\mathbf{A}_+$ corresponds to photon absorption and $\mathbf{A}_-$ to photon emission. So we may write

$$\mathbf{A}_{\pm} = c \left(\frac{2\pi\hbar}{\omega} \right)^{1/2} \mathbf{a} e^{\pm i(\mathbf{k} \cdot \mathbf{r} - \omega t)} \quad (13.99)$$

With

$$\hat{H}' = \hat{\mathbb{H}}_{\pm}(r) e^{\mp i\omega t} \quad (13.100)$$

and (13.89) we obtain

$$\langle n' | \mathbb{H}_{\pm} | n \rangle = -\frac{e}{m} \left(\frac{2\pi \hbar}{\omega} \right)^{1/2} \langle n' | \mathbf{a} \cdot \hat{\mathbf{p}} e^{\pm i \mathbf{k} \cdot \mathbf{r}} | n \rangle \quad (13.101)$$

The Dipole Approximation

For typical atomic transitions $\lambda \gg a_0$, where the Bohr radius a_0 is a length characteristic of atomic dimensions. Thus, over the domain of integration in the preceding matrix element, $\mathbf{k} \cdot \mathbf{r} \ll 1$, and we may write

$$e^{i \mathbf{k} \cdot \mathbf{r}} = 1 + i \mathbf{k} \cdot \mathbf{r} + \dots$$

In the *dipole approximation*, one sets $e^{i \mathbf{k} \cdot \mathbf{r}} = 1$. There results

$$\langle n' | \mathbb{H}_{\pm} | n \rangle = -\frac{e}{m} \left(\frac{2\pi \hbar}{\omega} \right)^{1/2} \langle n' | \mathbf{a} \cdot \hat{\mathbf{p}} | n \rangle \quad (13.102)$$

With the aid of the commutator relation (5.62) this matrix element may be transformed to one only involving $\mathbf{r}$ (see Problem 13.60):

$$\hat{\mathbf{p}} = \frac{im}{\hbar} [\hat{H}^{(0)}, \mathbf{r}] \quad (13.103)$$

It follows that

$$\langle n' | \mathbf{p} | n \rangle = \frac{im}{\hbar} \langle n' | H^{(0)} \mathbf{r} - \mathbf{r} H^{(0)} | n \rangle \quad (13.104)$$

Recalling that $|n'\rangle$ and $|n\rangle$ are eigenfunctions of the unperturbed Hamiltonian $\hat{H}^{(0)}$, we obtain

$$\begin{aligned} \langle n' | \mathbf{p} | n \rangle &= -\frac{im}{\hbar} (E_n^{(0)} - E_{n'}^{(0)}) \langle n' | \mathbf{r} | n \rangle \\ \langle n' | \mathbf{p} | n \rangle &= -im\omega \langle n' | \mathbf{r} | n \rangle \end{aligned} \quad (13.105)$$

Note that we are writing ω for $\omega_{nn'} > 0$. With these relations at hand, (13.102) may be rewritten more explicitly as

$$\begin{aligned} \langle n' | \mathbb{H}_{+} | n, \mathbf{k} \rangle &= \langle n', \mathbf{k} | \mathbb{H}_{-} | n \rangle \\ &= ie(2\pi \hbar \omega)^{1/2} \langle n' | \mathbf{a} \cdot \mathbf{r} | n \rangle \end{aligned} \quad (13.106)$$

Here we have specified that in the absorption process, the initial state contains a photon of wavevector $\mathbf{k}$, whereas in the emission process, the final state contains a photon of wavevector $\mathbf{k}$. Again as in (13.90), we find that the related probabilities of these two processes are equal. Spontaneous decay may be associated with the matrix element $\langle n', \mathbf{k} | \mathbb{H}_{-} | n \rangle$.

Spontaneous Decay

As described previously, the Einstein A coefficient represents the probability rate for spontaneous decay. [See, for example, (13.65).] Ordinarily, one would suspect that such spontaneous decay occurs in the absence of any perturbation. However, in our earlier description of hydrogen (Section 10.6) we found atomic states to be stationary. So in reality, spontaneous decay must have a triggering mechanism. What is this mechanism?

The answer to this question stems from the observation that (as described in Problem 13.37) an electromagnetic field may be represented as a collection of harmonic oscillators. We have found previously that a harmonic oscillator always has a residual energy which is called its “zero-point energy.” In like manner, no region of space is ever free of electromagnetic energy. It is such *vacuum fluctuations of electrodynamic fields* which are responsible for spontaneous decay.

Golden Rule Revisited

In describing spontaneous decay we concentrate on the emission matrix element $\langle n', \mathbf{k} | \mathbb{H}_- | n \rangle$, and consider that the emitted photon lies in the differential of solid angle $d\Omega$. The corresponding transition probability rate is given by Fermi's golden rule (13.64), which, in the present case, assumes the form

$$dw_{nn'} = \sum_{\mathbf{a}_i} \frac{2\pi}{\hbar} |\mathbb{H}_{n'n}|^2 \bar{g}(E) d\Omega \quad (13.107)$$

Here $\mathbf{a}_i$ denotes possible photon polarization and $\bar{g}(E) d\Omega$ is the density of states of photons emitted in the solid angle $d\Omega$. This value of $\bar{g}(E)$ may be obtained from the value of $g(\nu)$ given in Problem 2.37 by setting

$$V g(\nu) d\nu = V g(\omega) d\omega = 2 \left[V \int \bar{g}(E) d\Omega \right] dE \quad (13.108)$$

The factor 2 in this equality accounts for photon polarizations. Setting the volume $V = 1$, we obtain

$$\bar{g}(E) = \frac{E^2}{(2\pi\hbar c)^3} = \frac{\omega^2}{\hbar(2\pi c)^3} \quad (13.109)$$

Combining these results gives the following probability rate for spontaneous decay:

$$dw_{nn'} = \sum_{\mathbf{a}_i} \frac{e^2 \omega^3}{2\pi\hbar c^3} |\langle n', \mathbf{k} | \mathbf{a}_i \cdot \mathbf{r} | n \rangle|^2 d\Omega \quad (13.110)$$

The summation over polarizations in (13.110) is performed as follows. First, we note that

$$\langle n', \mathbf{k} | \mathbf{a} \cdot \mathbf{r} | n \rangle = \mathbf{a} \cdot \langle n', \mathbf{k} | \mathbf{r} | n \rangle \quad (13.111a)$$

We recall that in the dipole approximation the presence of $\mathbf{k}$ in the preceding matrix element is cosmetic. It merely reminds us that the element is relevant to spontaneous decay and includes a photon of wavevector $\mathbf{k}$ in the final state. It follows that the only vector in the matrix element on the right side of (13.111a) is $\mathbf{r}$. Consequently, this matrix element is likewise in the direction of $\mathbf{r}$ and we may conclude that the entire matrix element on the left side of (13.111a) is proportional to $\mathbf{a} \cdot \mathbf{r}$. With this observation, the sum in (13.110) is evaluated as depicted in Fig. 13.15. The propagation vector of the emitted photon, $\mathbf{k}$, makes an angle θ with the radius vector $\mathbf{r}$, which is held fixed and taken as the polar axis. The unit polarization vectors $\mathbf{a}_1, \mathbf{a}_2$ are normal to each other as well as to the propagation vector $\mathbf{k}$. To facilitate calculation, the vectors $\mathbf{k}, \mathbf{a}_1$, and $\mathbf{r}$ are taken to lie in the same plane (see Fig. 13.15). It follows that the vector $\mathbf{a}_2$ points out of the plane of the paper, so that $\mathbf{a}_1 \cdot \hat{\mathbf{r}} = \sin \theta$, $\mathbf{a}_2 \cdot \hat{\mathbf{r}} = 0$ ($\hat{\mathbf{r}}$ is a unit vector). Thus we obtain

$$\begin{aligned} \sum_{\mathbf{a}_i} |\langle n', \mathbf{k} | \mathbf{a}_i \cdot \mathbf{r} | n \rangle|^2 &= |\mathbf{a}_1 \cdot \langle n, \mathbf{k} | \mathbf{r} | n \rangle|^2 + |\mathbf{a}_2 \cdot \langle n', \mathbf{k} | \mathbf{r} | n \rangle|^2 \\ &= \sin^2 \theta |\langle n', \mathbf{k} | \mathbf{r} | n \rangle|^2 \end{aligned}$$

Inserting this result into (13.110) gives the following probability rate for spontaneous decay:

$$dw_{nn'} = \frac{e^2 \omega^3}{2\pi \hbar c^3} |\langle n', \mathbf{k} | \mathbf{r} | n \rangle|^2 \sin^2 \theta d\Omega \quad (13.111b)$$

So we reach the conclusion that the differential transition probability rate $dw_{nn'}$ depends only on the angle θ between $\mathbf{r}$ and $\mathbf{k}$. Consequently, the subsequent integration over $d\Omega$ is independent of the azimuthal angle, and with $\mathbf{r}$ still held fixed,

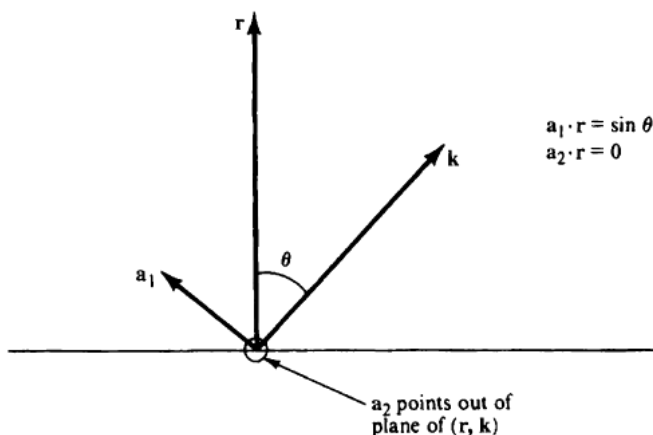


FIGURE 13.15 Configuration for summation over $\mathbf{a}_i$ in (13.110). The propagation vector $\mathbf{k}$ defines the direction of the differential of solid angle $d\Omega$.

we obtain

$$A_{nn'} = \int dw_{nn'} = \frac{4}{3} \frac{e^2 \omega^3}{\hbar c^3} |\langle n' | \mathbf{r} | n \rangle|^2 \quad (13.112)$$

Note that we are writing

$$|\langle n' | \mathbf{r} | n \rangle|^2 = |\langle n' | x | n \rangle|^2 + |\langle n' | y | n \rangle|^2 + |\langle n' | z | n \rangle|^2$$

The coefficient $A_{nn'}$ represents the probable rate of transition for spontaneous decay and may be identified with the Einstein A coefficient in (13.65). To obtain the corresponding expression for radiated power, one multiplies $A_{nn'}$ by $\hbar\omega$. There results

$$P_{nn'} = \frac{4}{3} \frac{e^2 \omega^4}{c^3} |\langle n' | \mathbf{r} | n \rangle|^2 \quad (13.113)$$

which agrees with our previous result (10.139).

The Einstein B coefficient may be found from our previously derived relation (see Problem 13.52):

$$\frac{A}{B} = \frac{\hbar\omega^3}{\pi^2 c^3}$$

which, with (13.112), returns our earlier finding (13.70).

It should be kept in mind that the preceding results are relevant to weak fields and the assumption that $\lambda \gg a_0$.

To apply the above formulas to an atom with Z electrons, one makes the replacement

$$\mathbf{r}_{nn'} = \langle n' | \mathbf{r} | n \rangle \rightarrow \langle n' | \sum_{i=1}^Z \mathbf{r}_i | n \rangle \quad (13.114)$$

in the preceding expressions for A and P . The sum in (13.114) runs over all the electrons in the atom. For such cases $e\mathbf{r}_{nn'}$ represents the total dipole matrix element of the atom.

The total probable rate of spontaneous decay of an atom in the n th energy state is given by

$$A_n = \sum_{E_{n'} < E_n} A_{nn'} \quad (13.115)$$

The summation runs over all states of lower energy than E_n . The corresponding mean lifetime of the n th excited state is then given by

$$\tau_n = \frac{1}{A_n} \quad (13.116)$$

Oscillator Strengths

An important parameter in radiation analysis is the oscillator strength, defined as the dimensionless form

$$f_{nn'} = \frac{2m\omega}{3\hbar} |\langle n' | \mathbf{r} | n \rangle|^2 = \frac{2m\omega}{3\hbar e^2} d_{nn'}^2 \quad (13.117)$$

It follows that

$$A_{nn'} = \frac{2e^2\omega^2}{mc^3} f_{nn'} \quad (13.118)$$

$$P_{nn'} = \frac{2e^2\hbar\omega^3}{mc^3} f_{nn'} \quad (13.119)$$

Oscillator strengths obey the so-called Thomas–Reiche–Kuhn sum rule,

$$\sum_{n'} f_{nn'} = 1 \quad (13.120)$$

from which, together with (13.117), it may be concluded that $f_{nn'} < 1$.

The sum rule (13.120) is simply derived with the aid of the basic commutator relation (5.61), according to which, in one dimension, one obtains

$$\langle n | [x, p_x] | n \rangle = i\hbar \quad (13.121)$$

Equivalently, we may write (see Problem 11.1)

$$\sum_{n'} [\langle n | x | n' \rangle \langle n' | p_x | n \rangle - \langle n | p_x | n' \rangle \langle n' | x | n \rangle] = i\hbar \quad (13.122)$$

With (13.105) we obtain

$$\begin{aligned} \sum_{n'} [im\omega_{n'n} |\langle n' | x | n \rangle|^2 - im\omega_{nn'} |\langle n' | x | n \rangle|^2] &= i\hbar \\ \sum_{n'} \frac{2m\omega_{n'n}}{\hbar} |\langle n' | x | n \rangle|^2 &= 1 \end{aligned}$$

The same result follows with x replaced by y or z . There results

$$\sum_{n'} \frac{2m\omega}{3\hbar} |\langle n' | \mathbf{r} | n \rangle|^2 = \sum_{n'} f_{nn'} = 1 \quad (13.123)$$

For an atom with Z electrons, with (13.114) one finds

$$\sum_{n'} f_{nn'} = Z \quad (13.124)$$

Values of the oscillator strengths of hydrogen corresponding to the first four $nP - 1S$ transitions are listed below.¹⁸

n	2	3	4	5
f_{nP-1S}	0.416	0.079	0.029	0.014

We conclude this section with an estimate of the lifetime of the excited states of one-electron atoms.¹⁹ With $|\mathbf{r}_{nn'}| \simeq a_0$, we rewrite $A_{nn'}$ (13.112) as

$$A_{nn'} \simeq \frac{4}{3} \frac{e^2 \omega^3 a_0^2}{\hbar c^3} \quad (13.125)$$

Introducing the effective nuclear charge $\bar{Z}$,

$$\hbar \omega \equiv \frac{\bar{Z} e^2}{a_0} \quad (13.126)$$

permits the preceding formula to be written

$$A_{nn'} \simeq \frac{4}{3} \alpha^3 \omega \bar{Z}^2 \quad (13.127)$$

where $\alpha = e^2/\hbar c$ is the fine-structure constant. With $\bar{Z} \simeq 1$, and $\hbar \omega \simeq \mathbb{R} = \alpha^2 m c^2/2$, we find

$$A_{nn'} \simeq \frac{\alpha^5}{\hbar} \frac{m c^2}{2} = \frac{\alpha^5 c}{2 \lambda_C} \quad (13.128)$$

In this expression, $\lambda_C = \hbar/mc$ is the Compton wavelength. Inverting $A_{nn'}$ gives the lifetime²⁰

$$\tau \simeq \alpha^{-5} \left(\frac{2 \lambda_C}{c} \right) \simeq 10^{-10} \text{ s} = 0.1 \text{ ns} \quad (13.129)$$

PROBLEMS

13.49 The radioactive isotope C^{11} decays through positron emission to B^{11-} . With the same assumptions holding as described in Problem 13.44, estimate the probability

¹⁸R. Loudon, *The Quantum Theory of Light*, Clarendon Press, Oxford, 1973.

¹⁹More detailed discussions of these topics may be found in H. A. Bethe and E. E. Salpeter, *Quantum Mechanics of One- and Two-Electron Atoms*, Plenum, New York, 1977; F. Constantinescu and E. Magyari, *Problems in Quantum Mechanics*, Pergamon Press, Elmsford, N.Y., 1971; S. Flügge, *Practical Quantum Mechanics*, Springer, New York, 1974.

²⁰Nanosecond (10^{-9}), picosecond (10^{-12}), and femtosecond (10^{-15}) measurements are presently commonplace in many laboratories. A review of these topics is given by P. W. Smith and A. M. Weiner, *IEEE Circuits and Devices*, **4**, 3 (May 1988). Note the powers of 10 in physics: speed of light, 3×10^{10} cm/s; decay time of an atom, 10^{-10} s; age of the universe, 10^{10} yr.

that B^{11-} is born in the ground state. (*Hint:* For your estimate, consult Problem 12.28 and pay attention to Z dependence of wavefunctions.)

Answer

As discussed in Problem 12.28, two-electron, ground-state wavefunctions (with $\hat{H}_{SO} = 0$) are given by

$$\varphi_Z(r_1, r_2) = \frac{1}{\pi a^3} \exp\left[\frac{-(r_1 + r_2)}{a}\right]$$

$$a \equiv \frac{a_0}{Z}$$

with normalization

$$\iint d\mathbf{r}_1 d\mathbf{r}_2 |\varphi_Z|^2 = (4\pi)^2 \int_0^\infty dr_1 r_1^2 \int_0^\infty dr_2 r_2^2 |\varphi_Z|^2 = 1$$

The transition probability to the ground state of B^{11-} is given by

$$\sqrt{P} = \langle \varphi_C | \varphi_B \rangle = \frac{Z_C^3 Z_B^3}{\pi^2 a_0^6} \iint \exp\left[\frac{-Z(r_1 + r_2)}{a_0}\right] d\mathbf{r}_1 d\mathbf{r}_2$$

where $Z \equiv Z_C + Z_B$. The preceding integral may be rewritten

$$\sqrt{P} = \frac{Z_C^3 Z_B^3}{\pi^2 a_0^6} I^2$$

$$I = a_0^3 \int_0^\infty dx x^2 e^{-Zx} = 2 \left(\frac{a_0}{Z}\right)^3$$

There results

$$\sqrt{P} = \left(\frac{4Z_B Z_C}{Z^2}\right)^3$$

With $Z_B = 5$, $Z_C = 6$, $Z = 11$, we obtain $P = 0.951$.

- 13.50 (a)** The density matrix (in a representation where $\hat{S}^2$ and $\hat{S}_z$ are diagonal) describing a beam of spinning electrons has the value

$$\hat{\rho} = \begin{pmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & 0 \end{pmatrix}$$

at $t < 0$. What are the values of $\langle S_z \rangle$, $\langle S_x \rangle$, and $\langle S_y \rangle$ for an electron in the beam?

- (b)** The beam interacts with a field which is turned on at $t = 0$. The corresponding interaction Hamiltonian has the matrix

$$\hat{H} = \hbar\omega_0 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

where ω_0 is a characteristic frequency. Estimate the value of the matrix $\hat{\rho}$ at the time Δt , where $0 < \Delta t \ll \omega_0^{-1}$.

- (c) What is the value of $\langle S_y \rangle$ at this value of time Δt ? [Hint: Recall (11.122).]

Answers

(a) $\langle S_z \rangle = \langle S_x \rangle = \hbar/2$, $\langle S_y \rangle = 0$.

(b) Set

$$\begin{aligned}\hat{\rho}(\Delta t) &= \hat{\rho}(0) + \left(\frac{\partial \hat{\rho}}{\partial t} \right)_0 \Delta t + \cdots \\ &= \hat{\rho}(0) + \frac{1}{i\hbar} [\hat{H}, \hat{\rho}]_0 \Delta t + \cdots\end{aligned}$$

Employing the given value of $\hat{\rho}(0)$ gives

$$\hat{\rho}(\Delta t) = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 0 \end{pmatrix} - i\omega_0 \Delta t \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & a \\ a^* & 0 \end{pmatrix}$$

where

$$a \equiv 0.5 + i\omega_0 \Delta t$$

- (c) At $t = \Delta t$

$$\begin{aligned}\langle S_y \rangle &= \text{Tr} \hat{\rho} \hat{S}_y = \text{Tr} \left[\hat{\rho} \frac{i\hbar}{2} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \right] \\ &= \frac{i\hbar}{2} \text{Tr} \left[\begin{pmatrix} 1 & a \\ a^* & 0 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \right] \\ &= \frac{i\hbar}{2} (a - a^*) = -\hbar\omega_0 \Delta t\end{aligned}$$

Thus, the perturbation causes a change in $\langle S_y \rangle$.

- 13.51** A one-dimensional harmonic oscillator of charge-to-mass ratio e/m , and spring constant K oscillates parallel to the x axis and is in its second excited state at $t < 0$, with energy

$$E_2 = \hbar\omega_0 \left(2 + \frac{1}{2} \right)$$

An oscillating, uniform electric field

$$\mathcal{E}(t) = 2\mathcal{E}_0 \cos \omega_0 t$$

is turned on at $t = 0$, parallel to the motion of the oscillator.

- (a) What is the new Hamiltonian of the oscillator at $t > 0$?
 (b) What are the matrix elements $\mathbb{H}'_{2n}$ for this system? You may leave your answer in terms of $\beta = \sqrt{m\omega_0/\hbar}$.
 (c) What are the probabilities P_{2n} that the oscillator undergoes a transition to the n th state at the end of $t < \omega_0^{-1}$ seconds? [Hint: Use harmonic perturbation theory and look at the short-time limit.]

(d) Using your answer to part (c), offer a technique for ‘pumping’ a harmonic oscillator to higher states.

13.52 In our discussion of Planck’s radiation law in Section 13.7, an expression for A/B was obtained as a function of frequency, ν . What is the corresponding expression for A/B in terms of angular frequency, ω ? [Hint: See Problem 2.43.]

13.53 Suppose φ_1 and φ_2 are two normalized eigenfunctions of an operator $\hat{A}$ with the same eigenvalue. Construct two new normalized, orthogonal functions, ψ_1 and ψ_2 , which are linear combinations of φ_1 and φ_2 . Offer a geometrical description of construction in the appropriate Hilbert space.

Answer (partial)

Let

$$\psi_1 = \varphi_1, \quad \psi_2 = \alpha\varphi_1 + \beta\varphi_2$$

and

$$\int \varphi_1^* \varphi_2 d\mathbf{r} = K \neq 0$$

Orthogonality of ψ_1 and ψ_2 gives

$$\alpha + \beta K = 0$$

whereas normalization of ψ_2 gives

$$|\beta|^2(1 - |K|^2) = 1$$

There results

$$\beta K = -\alpha = \frac{K}{\sqrt{1 - |K|^2}}$$

which determines ψ_2 . This construction gives the essentials of the *Schmidt orthogonalization procedure*.

13.54 (a) Show that the time-dependent Schrödinger equation may be written [compare with (11.144)]

$$\psi(t) = \psi(0) + \frac{1}{i\hbar} \int_0^t \hat{H}(t') \psi(t') dt'$$

(b) Show that for small $\hat{H}$ the preceding equation gives the series (*Neumann–Liouville expansion*) with $t > t'$, etc.

$$\psi(t) = \left[1 + \frac{1}{i\hbar} \int_0^t dt' \hat{H}(t') + \frac{1}{(i\hbar)^2} \int_0^t dt' \int_0^{t'} dt'' \hat{H}(t') \hat{H}(t'') + \dots \right] \psi(0)$$

(c) Show that the n th-order term in the preceding series may be written

$$\psi^{(n)}(t) = \frac{1}{n!} \frac{1}{(i\hbar)^n} \int_0^t \dots \int_0^t dt_1 \dots dt_n \hat{T}[\hat{H}(t_1) \dots \hat{H}(t_n)] \psi(0)$$

where the time-ordering operator

$$\hat{T} f(t_a) g(t_b) = \begin{cases} f(t_a) g(t_b) & \text{for } t_a > t_b \\ g(t_b) f(t_a) & \text{for } t_b > t_a \end{cases}$$

- 13.55** A particle of mass m is confined to a one-dimensional partitioned box with walls at $(-L/2, 0, L/2)$. The Hamiltonian for this configuration is labeled $\hat{H}_0$. See Fig. 13.16a. One of two perturbations, $\hat{H}'_{a,b}$, is applied as shown in Fig. 13.16b, c. The total Hamiltonians for the particle are

$$\hat{H}_{a,b} = \hat{H}_0 + \hat{H}'_{a,b}$$

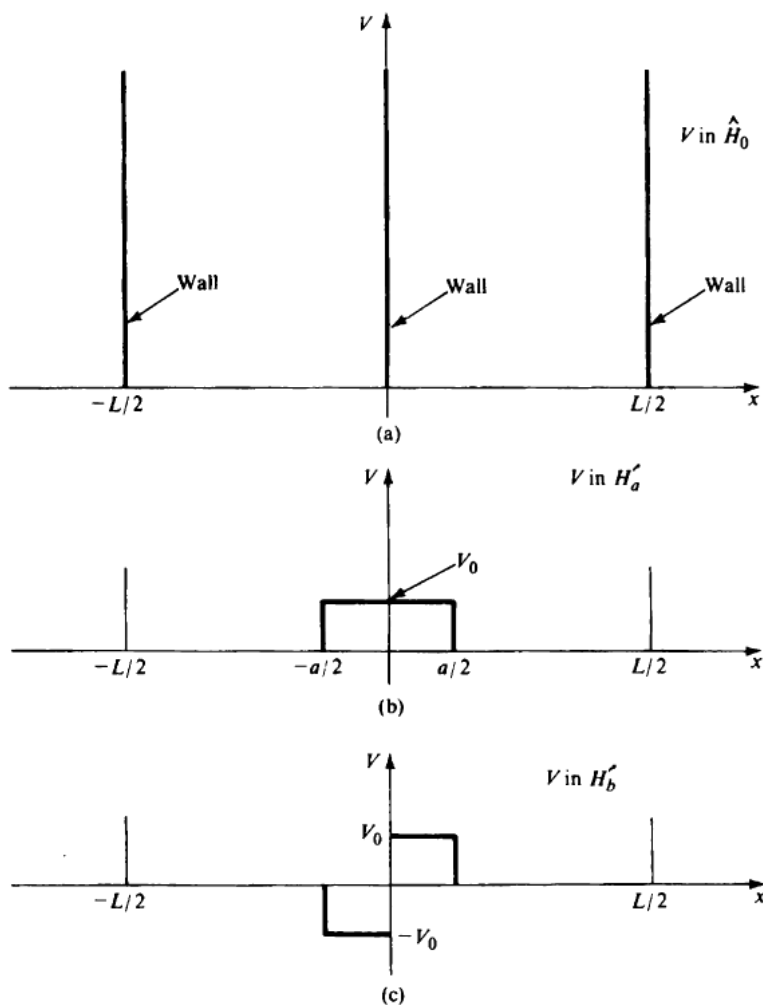


FIGURE 13.16 Potential configurations for Problem 13.55.

Let the perturbation potentials have magnitude $V_0 = 10^{-4}E_G$, where E_G is the ground state of $\hat{H}_0$, and let the normalized ground states of $\hat{H}_0$ be written $|r\rangle$ and $|l\rangle$ corresponding to the particle trapped in the right and left boxes, respectively.

- What are the coordinate representations of $|r\rangle$ and $|l\rangle$?
- What is the ground-state eigenenergy E_G of $\hat{H}_0$? What is the order of the degeneracy of E_G ?
- What is $\langle l|\hat{H}_0|r\rangle$?
- What are the values (zero or nonzero) of the commutators $[\hat{H}_0, \hat{\mathbb{P}}]$, $[\hat{H}_a, \hat{\mathbb{P}}]$, $[\hat{H}_b, \hat{\mathbb{P}}]$? Here we have written $\hat{\mathbb{P}}$ for the parity operator (6.90).
- Construct two simultaneous eigenstates of $\hat{H}_0$ and $\hat{\mathbb{P}}$ in terms of $|l\rangle$ and $|r\rangle$. Label these states $|S\rangle$ (symmetric) and $|A\rangle$ (antisymmetric). What are $\hat{H}_0|S\rangle$, $\hat{H}_0|A\rangle$, $\hat{\mathbb{P}}|S\rangle$, $\hat{\mathbb{P}}|A\rangle$?
- Using degenerate perturbation theory, obtain the splitting of E_G due to $\hat{H}'_a$ and $\hat{H}'_b$, respectively. Do both $\hat{H}'_a$ and $\hat{H}'_b$ remove the degeneracy of E_G ? Explain your answer. Call the new energies E_+ , E_- . What effect does the nonsplitting perturbation have on the ground-state energy?
- The trapped particle is an electron and $2a = L = 10 \text{ \AA}$. What are E_+ and E_- (in eV)?
- Can you suggest an alternative way to evaluate $E_{\pm}$ which does not employ degenerate perturbation theory?

Answers (partial)

- $\langle l|x\rangle = 0$ on $(0, L/2)$ and $\langle r|x\rangle = 0$ on $(-L/2, 0)$.

$$(e) \quad |S\rangle = \frac{1}{\sqrt{2}}[|l\rangle + |r\rangle], \quad |A\rangle = \frac{1}{\sqrt{2}}[|l\rangle - |r\rangle]$$

$$\hat{\mathbb{P}}|S\rangle = |S\rangle, \quad \hat{\mathbb{P}}|A\rangle = -|A\rangle$$

$$\hat{H}_0|S\rangle = E_G|S\rangle, \quad \hat{H}_0|A\rangle = E_G|A\rangle$$

- We must examine the secular equation

$$\begin{vmatrix} \langle l|H'_{a,b}|l\rangle - E' & \langle l|H'_{a,b}|r\rangle \\ \langle r|H'_{a,b}|l\rangle & \langle r|H'_{a,b}|r\rangle - E' \end{vmatrix} = 0$$

with

$$\mathcal{E}_a \equiv \langle l|H'_a|l\rangle = \langle r|H'_a|r\rangle$$

$$\mathcal{E}_b \equiv \langle l|H'_b|l\rangle = -\langle r|H'_b|r\rangle$$

$$\Delta\mathcal{E}_a \equiv \langle r|H'_a|l\rangle = \langle l|H'_a|r\rangle = 0$$

$$\Delta\mathcal{E}_b \equiv \langle r|H'_b|l\rangle = \langle r|H'_b|l\rangle = 0$$

We find that $\hat{H}'_b$ splits E_G but $\hat{H}'_a$ does not. The explanation of this result is that $\hat{H}'_a$ maintains the symmetry of $\hat{H}_0$. That is, for both $\hat{H}_0$ and $\hat{H}'_a$, there is no difference between right and left boxes. New energies (for $\hat{H}_b$) are given by

$$E_+ = E_G + \mathcal{E}_b$$

$$E_- = E_G - \mathcal{E}_b$$

- 13.56** An approximate form of the Hamiltonian of a positronium atom in the $1S$ state immersed in a weak magnetic field $\mathcal{B}$ is given by

$$\hat{H} = A\hat{\mathbf{S}}_1 \cdot \hat{\mathbf{S}}_2 + \Omega \cdot (\hat{\mathbf{S}}_1 - \hat{\mathbf{S}}_2) \equiv H_0 + \lambda H'$$

$$\Omega = \frac{e\mathcal{B}}{\mu c}$$

where H' is the magnetic field term. The subscripts 1 and 2 denote electron and positron, respectively, A is a constant, and μ is reduced mass.

- (a) Show that the coupled spin states ξ_S and ξ_A relevant to two spin- $\frac{1}{2}$ particles are eigenstates of $\hat{H}_0$. What is the ground-state energy of $\hat{H}_0$ and which eigenstate does this correspond to? [See Table 11.3 and recall (11.111).]
- (b) Employing ξ_S and ξ_A as basis states, use first-order perturbation theory to obtain eigenvalues and eigenstates of $\hat{H}$. Assume that $\mathcal{B}$ is aligned with the z axis.
- 13.57** In Section 11.12 we encountered an approximation scheme centered about the interaction picture. Again consider the Hamiltonian given in Problem 11.102, which we now write in more symbolic form,

$$\hat{H} = \hat{H}_0 + \lambda \hat{V}(t)$$

Let the system be in the initial state

$$|\psi(0)\rangle = |n\rangle$$

$$\hat{H}_0|n\rangle = E_n|n\rangle$$

where, we recall $\hat{H}_0$ is appropriate to an harmonic oscillator with natural frequency ω_0 .

- (a) Calculate the wavefunction $|\psi_I(t)\rangle$ to first order in λ .
- (b) For the problem at hand, show that

$$|\langle m|\psi(t)\rangle|^2 = |\langle m|\psi_I(t)\rangle|^2$$

- (c) Again, to $O(\lambda)$, obtain an integral expression for the time-dependent transition probabilities $P_{n \rightarrow m}$.
- (d) In the event that $\hat{V}$ is time-independent, show that your answer to part (c) reduces to (13.55) corresponding to the DC perturbation, $\omega = 0$.

Answers

(a)
$$|\psi_I(t)\rangle = |n\rangle + \sum_m \frac{\lambda}{i\hbar} \int_0^t e^{i\omega_{mn}t'} |m\rangle \langle n|\hat{V}(t')|n\rangle dt'$$

$$\hbar\omega_{nn} = E_m - E_n = \hbar\omega_0(m - n)$$

- (b) First, note that

$$|\psi(t)\rangle = |e^{-i\hat{H}_0 t/\hbar} \psi_I(t)\rangle$$

The desired equality follows since $\hat{H}_0|m\rangle = E_m|m\rangle$.

(c) With the result (b), we write

$$P_{n \rightarrow m} = |\langle m | \psi_I(t) \rangle|^2$$

where, for $m \neq n$,

$$\langle m | \psi_I(t) \rangle = \frac{\lambda}{i\hbar} \int_0^t e^{i\omega_{mn}t'} \langle m | \hat{V}(t') | n \rangle dt'$$

(d) In the event that $\hat{V}$ is time-independent, integration of the preceding finding gives

$$\begin{aligned} \langle m | \psi_I(t) \rangle &= \frac{V_{mn}}{\hbar\omega_{mn}} (1 - e^{i\omega_{mn}t}) \\ V_{mn} &\equiv \lambda \langle m | \hat{V} | n \rangle \end{aligned}$$

which yields

$$P_{n \rightarrow m} = \left| \frac{2V_{mn}}{\hbar\omega_{mn}} \right|^2 \sin^2 \frac{\omega_{mn}t}{2}$$

This result is seen to agree with (13.55) for the DC perturbation, $\omega = 0$.

13.58 Establish the following relations for the coefficients of the Fourier expansion (13.33) of the real potential function $V(x)$.

(a) $V_n^* = V_{-n}$.

(b) If $V(x)$ is even, then

$$V_n = V_{-n}$$

(c) If $V(x)$ is odd, then

$$V_n = -V_{-n}$$

(d) For $V(x)$ (even, odd) and with period $2a$, one writes

$$V(x) \begin{pmatrix} \text{even} \\ \text{odd} \end{pmatrix} = \sum_{n=1}^{\infty} \begin{pmatrix} a_n \cos \frac{n\pi x}{a} \\ b_n \sin \frac{n\pi x}{a} \end{pmatrix}$$

Show that (with $a_0 = 0$)

$$a_n = 2V_n = 2V_{-n}$$

$$b_n = 2iV_n = -2iV_{-n}$$

Answers (partial)

(a) As replacing n with $-n$ in the series (13.33) does not change the sum, we may set

$$\sum_{\forall n} V_n e^{i2\pi nx/a} = \sum_{\forall n} V_{-n} e^{-i2\pi nx/a}$$

For $V(x)$ real, $V(x) = V(x)^*$. With the preceding we then obtain

$$\sum_{\forall n} V_n^* e^{-i2\pi nx/a} = \sum_{\forall n} V_{-n} e^{-i2\pi nx/a}$$

whence $V_n^* = V_{-n}$.

(b) If $V(x)$ is even, then

$$\sum_{\forall n} V_n e^{i2\pi nx/a} = \sum_{\forall n} V_n e^{-i2\pi nx/a} = \sum_{\forall n} V_{-n} e^{i2\pi nx/a}$$

whence $V_n = V_{-n}$.

(c) If $V(x)$ is odd, then

$$\sum_{\forall n} V_n e^{i2\pi nx/a} = - \sum_{\forall n} V_n e^{-i2\pi nx/a} = - \sum_{\forall n} V_{-n} e^{i2\pi nx/a}$$

whence $V_n = -V_{-n}$.

13.59 (a) Evaluate the average dipole moment of hydrogen for the $3P \rightarrow 1S$ transition from the corresponding value of oscillator strength given in the text. Work in cgs units and state the dimensions of your answer.

(b) Compare your answer with the classical estimate, $d \simeq ea_0$.

13.60 Establish the commutator relation (13.103).

Answer

Since $\mathbf{r}$ commutes with $V(\mathbf{r})$, the relation reduces to

$$\hat{\mathbf{p}} = \frac{im}{\hbar} \left[\frac{\hat{\mathbf{p}}^2}{2m}, \mathbf{r} \right]$$

With reference to Problem 5.45, the preceding is rewritten

$$\hat{\mathbf{p}} = \frac{i}{2\hbar} \{ \hat{\mathbf{p}}[\hat{\mathbf{p}}, \hat{\mathbf{r}}] + [\hat{\mathbf{p}}, \hat{\mathbf{r}}]\hat{\mathbf{p}} \}$$

$$\hat{\mathbf{p}} = -\frac{i}{2\hbar} \{ \hat{\mathbf{p}}i\hbar\hat{I} + i\hbar\hat{I}\hat{\mathbf{p}} \}$$

where $\hat{I}$ is the identity operator,

$$\hat{\mathbf{p}}\hat{I} = \hat{I}\hat{\mathbf{p}} = \hat{\mathbf{p}}$$

which establishes the said relation.

13.61 A particle of mass m is in a three-dimensional, rigid-walled cubical box of edge length a . Edges of the box are aligned with the Cartesian axes, with one corner of the box at the origin.

(a) Write down the normalized ground state for this configuration.

(b) The face of the cube at $x = a$ is suddenly displaced to $x = 2a$. Obtain an expression for the probability that the particle remains in the ground state.

(c) Given that the particle is an electron and that $a = 2 \text{ \AA}$, what is the numerical value of this probability?

- 13.62** A particle of mass m is confined to the interior of a rigid-walled spherical cavity of radius a . The particle is in the ground state. At $t = 0$ the radius of the sphere begins to expand according to

$$a(t) = a_0 e^{-(t/\tau)^2}$$

where $\tau \gg ma_0^2/\hbar$. What are the wavefunction and energy of the particle at $t = \tau/2$?

- 13.63** At a given instant of time an harmonic oscillator undergoes a sudden change in spring constant from K to K' . Show that for energy to be conserved in the accompanying transition, $\sqrt{K/K'}$ must be the ratio of two odd numbers.
- 13.64** The same conditions as in Problem 13.26 apply. However, now the periodic potential is as shown in Fig. 13.17.

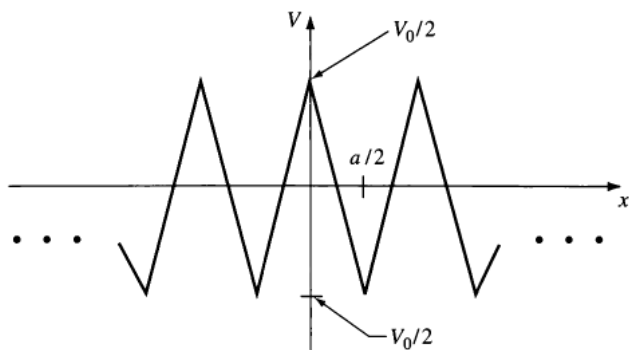


FIGURE 13.17 See Problem 13.64.

If $V_0 = 1.5$ eV and $a = 2.3$ Å, at what energy (eV) will photons incident on the crystal cause it to conduct? [Hint: Before you start, decide on the period of $V(x)$.]

Answer

You should obtain

$$E_g = 2V_1 = \frac{4V_0}{\pi^2} = 0.61 \text{ eV}$$

- 13.65** In our analysis of the *nearly free electron model* we set

(a) $H'_{k_j, k_j} = V_0 = 0$

(b) $H'_{k_j, -k_j} = V_n$

(c) $H'_{-k_j, k_j} = V_{-n}$

(d) $V_n = V_{-n}$

Establish the validity of these relations and state the relation between n and k_j .

- 13.66** The oscillator strength between the $4P$ and the $3S$ states of sodium is $f_{34} = 0.484$.²¹ Working in cgs units:
- What are the dimensions of f_{34} ?
 - What is the dipole moment, d_{34} , of the atom, corresponding to this transition? State the units of d_{34} (in cgs).
 - Compare your values for d_{34} with the value of the dipole moment, $|e| a_0$, where a_0 is the Bohr radius. [Note: Recall (10.138b).]
- 13.67** Derive the expression for time-averaged electromagnetic energy density, $\langle U \rangle$, as given by (13.94), carried in the plane-wave vector potential (13.91).
- 13.68** Show that the basis (13.11) comprises an orthogonal sequence.
- 13.69** (a) An electron with energy $E \gg V_0$ propagates through a one-dimensional crystal with potential

$$V(x) = 2V_0 \left[\cos\left(\frac{2\pi x}{a}\right) + \frac{1}{2} \cos\left(\frac{4\pi x}{a}\right) \right]$$

What are the values of the nonzero band gaps, δE_n , for this material? How many energy bands exist for this material?

- The second energy band is identified with the valence band of the material and the next-higher band is identified with the conduction band, which at $T = 0$ K is empty. Given that $V_0 = 1.5$ eV and that $a = 2$ Å, estimate the temperature at which this material will start to conduct.
- What are the new wavefunctions related to the first energy gap of this system?

Answers (partial)

- $T_{\text{cond}} = 0.17 \times 10^5$ K

13.10 ■ HARTREE–FOCK MODEL^{22,23}

In Section 10.8 we encountered the Thomas–Fermi statistical model for determining an effective atomic potential. The Hartree–Fock self-consistent model more realistically takes into account the effects of atomic electrons. In addition to an effective atomic potential, the model also determines eigenenergies and wavefunctions. Three primary elements of this model are: (1) Each electron moves in a central field equal to the nuclear potential and that due to the charge densities of the remaining atomic electrons. (2) The Schrödinger equation is solved for each electron in its own central field, and resulting wavefunctions are made self-consistent with the fields from which they are calculated. (3) The atomic

²¹R. D. Cowan, *The Theory of Atomic Spectra and Structure*, University of California Press, Berkeley, Calif., 1981, Section 14-14.

²²This analysis is due to D. R. Hartree, *Proc. Cambridge Soc.* **24**, 111 (1929).

²³Antisymmetrization of wavefunctions in this scheme was formulated by V. Fock, *Z. f. Phys.* **61**, 126 (1930).

wavefunction is a product of single-electron orthonormalized wavefunctions,

$$\psi(\mathbf{r}_1, \dots, \mathbf{r}_Z) = \varphi_1(\mathbf{r}_1) \dots \varphi_Z(\mathbf{r}_Z) \quad (13.130)$$

With statements (1) and (2) we note that each one-electron wavefunction satisfies the equation

$$\hat{H}_k \varphi_k(\mathbf{r}_k) = E_k \varphi_k(\mathbf{r}_k) \quad (13.131a)$$

$$\hat{H}_k \equiv \left[-\frac{\hbar^2 \nabla_k^2}{2m} + W(r_k) \right] \quad (13.131b)$$

$$W(r_k) \equiv V_C(r_k) + V_k(r_k) \quad (13.131c)$$

$$V_C(r_k) \equiv -\frac{Ze^2}{r_k} \quad (13.131d)$$

$$V_k(r_k) \equiv \int \sum_{j \neq k} |\varphi_j(\mathbf{r}_j)|^2 \frac{e^2}{r_{jk}} d\mathbf{r}_j \quad (13.131e)$$

$$r_{jk} \equiv |\mathbf{r}_j - \mathbf{r}_k| \quad (13.131f)$$

The term V_C represents electron-electron Coulomb energy. Summation over k (or j) runs from 1 to Z relevant to an atom with Z electrons. Note that $W(r_k)$ is a central potential and that spin-orbit effects are neglected in the preceding equations. As there are Z electrons in the atom, (13.131) constitutes Z simultaneous nonlinear integrodifferential equations for the Z functions. An iterative scheme is used to solve these equations. This approximation involves the following four steps:

1. An approximate central potential representing $W(r_k)$ is assumed, labeled $W^{(1)}(r_k)$. (Such a choice might be the Thomas–Fermi potential.)

$$W(r_k) \rightarrow W^{(1)}(r_k) \quad (13.132a)$$

2. Electron wavefunctions are computed employing this approximate potential.

$$\left[-\frac{\hbar^2 \nabla_k^2}{2m} + W^{(1)}(r_k) \right] \varphi_k^{(1)}(\mathbf{r}_k) = E_k^{(1)} \varphi_k^{(1)}(\mathbf{r}_k) \quad (13.132b)$$

3. With first-order wavefunctions, $\{\varphi_k^{(1)}(\mathbf{r}_k)\}$, at hand, charge densities, $e^2 |\varphi_k^{(1)}(\mathbf{r}_k)|^2$, are calculated.
4. Employing the preceding results, the second-order atomic potential, $W^{(2)}(r_k)$, is calculated.

The iterative scheme (steps 1 through 4) is repeated until

$$W^{(n+1)}(r_k) \simeq W^{(n)}(r_k) \quad (13.132c)$$

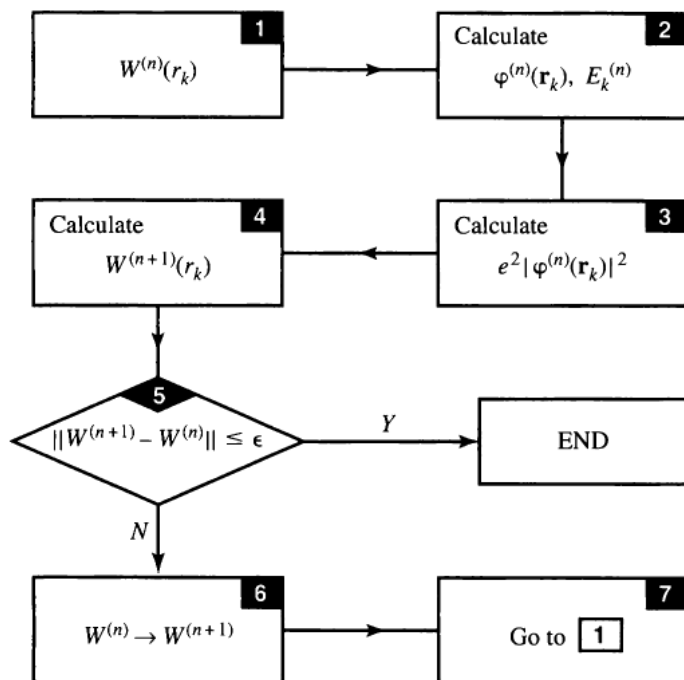


FIGURE 13.18 Flowchart illustrating the n th cycle of the iterative scheme for solution of the Hartree-Fock equation.

that is, until $W^{(n)}(r_k)$ does not change appreciably with the next iterative cycle. At the conclusion of this iteration one has a self-consistent effective atomic potential together with eigenfunctions and eigenenergies relevant to an atom with atomic number Z , to within a given order of accuracy. A flowchart illustrating this scheme is shown in Fig. 13.18. In step 5 of this chart we have written $\|W\|$ for the norm of W (see Section 4.4). The value of the parameter ϵ determines the accuracy of the solution.

PROBLEMS

- 13.70** (a) Neglecting spin-orbit effects, write down the Hamiltonian, $\hat{H}_Z$, of an atom with atomic number Z .
 (b) In what manner does $\hat{H}_Z$ differ from $\sum_k \hat{H}_k$ of the Hartree-Fock analysis?

Answers

$$(a) \quad \hat{H}_Z = \sum_k \left[-\frac{\hbar^2 \nabla_k^2}{2m} - \frac{Ze^2}{r_k} \right] + \sum_{j < k} \sum \frac{e^2}{r_{jk}}$$

The inequality $j < k$ restricts the double sum to distinct (j, k) pairs.

- (b) The difference between $\hat{H}_Z$ and $\sum_k H_k$ is due to the substitution of the effective potential $\sum_k V_k(r_k)$ for the exact double-sum electron-electron Coulomb energy in the preceding expression.

13.71 With reference to the preceding problem, show that the expectation $\langle H_Z \rangle$, in the state given by the wavefunction (13.130), may be written as a sum over the single-particle expectations $\langle H_k \rangle$.

Answer

With orthonormality of $\varphi_k(\mathbf{r}_k)$ wavefunctions we obtain

$$\begin{aligned} \langle H \rangle &= \int \cdots \int \psi^* \hat{H} \psi d\mathbf{r}_1 \cdots d\mathbf{r}_Z \\ &= \sum_k \iint \varphi_k^*(\mathbf{r}_k) \left(-\frac{\hbar^2 \nabla_k^2}{2m} - \frac{Ze^2}{r_k} \right) \varphi_k(\mathbf{r}_k) d\mathbf{r}_k \\ &\quad + \sum_{j \neq k} \iint \varphi_j^*(\mathbf{r}_j) \varphi_k^*(\mathbf{r}_k) \frac{e^2}{r_{jk}} \varphi_j(\mathbf{r}_j) \varphi_k(\mathbf{r}_k) d\mathbf{r}_k \end{aligned}$$

With reference to (13.131b) et seq., and the preceding result, $\langle H \rangle$ may be written

$$\langle H \rangle = \sum_k \langle H_k \rangle = \sum_k \int \varphi_k^*(\mathbf{r}_k) \hat{H}_k \varphi_k(\mathbf{r}_k) d\mathbf{r}_k$$

- 13.72** (a) Show that $\langle H \rangle$ may be made stationary through variation of individual one-electron atomic wavefunctions.
 (b) With reference to Problem 13.70, what can you conclude regarding the eigenstates of $\hat{H}_Z$?

Answer (partial)

- (b) In problem 4.30 we found that wavefunctions that make $\langle H \rangle$ stationary are eigenstates of $\hat{H}$ and, conversely, eigenstates of $\hat{H}$ render $\langle H \rangle$ stationary. With part (a) of this problem, we may conclude that ψ as given by (13.130) is an eigenstate of $\hat{H}_Z$, providing $\varphi_k(\mathbf{r}_k)$ are eigenstates of $\hat{H}_k$.

13.73 Obtain an expression for the eigenenergies of $\hat{H}_Z$ in terms of the eigenenergies of $\hat{H}_k$ and electron-electron Coulomb energy.

Answer

Employing expressions obtained in Problem 13.71, we find

$$E_Z = \langle H_Z \rangle = \sum_k E_k + \sum_{j < k} \iint |\varphi_j(\mathbf{r}_j)|^2 \frac{e^2}{r_{jk}} |\varphi_k(\mathbf{r}_k)|^2 d\mathbf{r}_j d\mathbf{r}_k$$

- 13.74** A particle of mass m is confined to the interior of a spherical cavity of radius a .
 (a) What is the ground-state wavefunction, $\psi_G(r, \theta, \phi)$, and energy, E_G , of this system?
 (b) A potential sphere of radius $a/10$ and height V_0 is placed concentric and interior to the original sphere. Employing nondegenerate perturbation theory, evaluate

the new first-order ground-state wavefunction, φ_G' , and new ground-state energy, E_G' .

- 13.75** An harmonic-oscillator Hamiltonian includes the perturbation field, $(C/a)x^3$ where C and a are constants. In the representation of (7.15), if dimensions of a are length, what are the dimensions of C ? What is an appropriate statement of smallness for the given perturbation? Calculate the first-order correction for the n th eigenstate energy for the given perturbed harmonic oscillator.

- 13.76** Throughout this text, it was stated that the Hamiltonian of a confined particle is that of a free neutral particle interior to the domain of confinement and infinite beyond the domain of confinement (cf., 4.2). Furthermore, it is assumed that the sudden expansion formalism is valid. Are these assumptions correct?

Answer

First note that we are discussing a particle with no internal degrees of freedom. To the degree that a perfectly rigid wall exists, the Hamiltonian form (4.2) is valid. Consider that the confining 'box' undergoes a sudden expansion. In this event, there is a finite probability for a transition of the system to a lower or higher energy. In a sudden expansion, no energy is lost to the particle in wall collisions. As the particle is neutral, there is no radiative mechanism to support either of these transitions, and the particle suffers no change in energy, in contradiction to standard perturbation theory. In a sudden compression, the particle can gain energy from the walls, but once in an excited state, it cannot radiatively decay as it is neutral.

If the particle is charged, and the confining walls are metallic or dielectric, then the simple Hamiltonian (4.2) is incorrect. The valid Hamiltonian is given by the electromagnetic form, (10.85). If it is the case that the confining walls are flat, then for either type confinement, the charged particle interacts with its image.

- 13.77** Consulting the periodic chart, we note that the elements palladium and platinum, in the solid state, have the respective electronic structures: $[\text{Kr}]4d^{10}$; $[\text{Xe}]5d^{10}$. We recall that the p shell can accommodate 10 electrons, so that the respective d shells in these atoms are closed. How is it that these solids are metals (i.e., have finite conductivity)?

Answer

There is overlap between the d shell and the s shell, so that electrons in the closed d shell can be promoted to the two-state s shell and contribute to conductivity. As noted previously, in this instance the material is said to be *degenerate* (Section 12.9). To denote this configuration in the periodic chart, the complete electronic structure is written: $[\text{Kr}]4d^{10}5s^0$; $[\text{Xe}]5d^{10}6s^0$.